



Review article

Reconstruction of interpretable causal network dynamics from time series data

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ABSTRACT

Reconstructing causal dynamic networks from multivariate time series is a foundational problem in complex systems science. Yet, the key scientific issue is not simply causality detection, but causal interpretability. Interpretable causality is the foundation for testable mechanistic hypotheses, transferable forecasting, and principled decision-making for intervention and control. In real-world complex systems, causal inference is often compromised by noise, missing data, high dimensionality, nonlinearity, time delays, heterogeneity, and partial observability. Classic approaches to interpretable causality yield explicit, inspectable quantities such as causal graphs, coefficients, and governing equations. However, this methodological shift has heightened expectations: AI-extended approaches are increasingly required to recover explicit causal mechanisms rather than opaque predictive dependencies, thereby preserving interpretability. This review summarizes four classic methods and their AI-extended counterparts based on time series data, including Granger frameworks, information-theoretic measures, nonlinear state-space/manifold reconstruction, and mechanistic differential-equation learning. Next, we elucidate their motivations, core principles, the origins of interpretability, and the assumptions required for meaningful conclusions. Finally, we highlight representative applications across climate studies, neuroscience, epidemiology, finance, social science, ecology and molecular biology, followed by a discussion of comparative analysis, open challenges, and future research directions.

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1. Introduction

1.1. Motivation

Reconstructing causal networks and their underlying interaction dynamics from observed time series constitutes a foundational inverse problem in complex systems science [1]. This challenge has far-reaching implications across scientific and engineering disciplines including climate system modeling [2], neuroscience [3], epidemiological forecasting [4], financial market interdependency analysis [5], social network evolution [6], and ecosystem stability assessment [7]. In these systems, collective behavior emerges from nonlinear interactions across multiple scales; consequently, system-level dynamics cannot be understood by analyzing components in isolation. Instead, such underlying interactions manifest as “spatiotemporal footprints” encoded within multivariate time series, which serve as the primary empirical window into unobservable causal interactions [8]. Extracting robust causal information from such noisy, high-dimensional observations, and subsequently reconstructing the intrinsic causal structure and dynamic rules governing the system, remains an enduring challenge for understanding and predicting complex systems [9].

Over the past decades, a diverse range of classic approaches have been developed for causal dynamics reconstruction from time series. These approaches span statistical-association and Granger causality frameworks [10], information-theoretic metrics (e.g., transfer entropy [11]), nonlinear state–space techniques [12], and mechanistic differential equation inference approaches (e.g., SINDy [13] and dynamic mode decomposition (DMD) [14,15]). A pivotal factor underpinning the enduring influence of these classic methods lies in their intrinsic interpretability, which is as critical to causal detection as statistical validity. In most scientific inquiries, causal inference extends beyond merely identifying “whether a causal link exists”; it demands answers to more nuanced, mechanistically relevant questions: What are the key driver variables, and in which directions and to what extent do they exert causal influence? Through which intermediate pathways do causal effects propagate? Over which temporal scales are these causal interactions operative? Under what assumptions do the inferred causality remain robust? A distinguishing strength of classic approaches is their ability to yield causal insights that are explicit, traceable, and falsifiable: they produce structured models or quantifiable metrics grounded in transparent, mathematically rigorous assumptions. These classic methodologies, rooted in established fields such as statistical physics, nonlinear dynamical systems theory, control theory, and econometrics, thus provide a rigorous benchmark for cross-dataset and cross-condition comparisons of causal evidence.

However, such interpretability is not universally guaranteed. It typically relies on simplifying assumptions or structural constraints that are frequently violated in real-world complex systems. Key limitations include a reliance on strong prior knowledge of system architecture, constraints of low dimensionality, and the requirement for sufficiently dense, noise-free observational data. If these conditions are unmet, as in systems with high dimensionality, structural heterogeneity, partial observability, and pervasive observational noise, classic methods are frequently characterized by substantial degradation in statistical power, computational tractability and out-of-sample reliability, and may become more susceptible to indirect effects and confounding.

In contrast, AI-driven black-box models, particularly deep learning architectures, exhibit unparalleled performance at fitting high-dimensional and strongly nonlinear time series with remarkable computational efficiency. Such data-driven approaches, despite their predictive power, are often built on statistical approximation rather than understanding of the

mechanism, resulting in a critical deficit of causal mechanism transparency. Due to this “black-box” nature, it is frequently difficult for them to support fundamental scientific questions centered on “why” causal relationships exist and “what if” counterfactual scenarios, which limits their utility for scientific explanations beyond predictive performance. This inherent tension between the flexibility of AI-driven methods and the interpretability of classic causal techniques has emerged as a central motivation for the burgeoning “AI for Science” paradigm, which seeks to integrate data-driven flexibility of machine learning with the theoretical grounding of classic causal methodologies.

Encouragingly, a promising frontier has emerged at the intersection of these two domains, focused on developing AI-driven methods for interpretable causal dynamics reconstruction that leverage the complementary strengths of classic and modern AI approaches. On the one hand, classic causal algorithms are being increasingly integrated with neural network modules or differentiable solvers, which enables them to overcome inherent limitations in handling large-scale systems, high noise levels, and strong nonlinearities. On the other hand, interpretability has evolved from a post hoc explanation tool to a built-in principle for model construction: contemporary AI models incorporate physical constraints, symbolic structures, and causal priors to enhance transparency and ensure consistency with fundamental scientific principles.

Representative advances such as sparse neural stochastic differential equations, physics-informed neural networks (PINNs) for causal inference, and symbolic causal discovery have demonstrated promising performance across complex systems, including climate pattern prediction, biomedical mechanism discovery, and financial risk propagation analysis. However, these successes have not yet closed the central scientific gap: in most current gray-box techniques, physical or domain-specific laws function primarily as constraints that regularize learning, rather than as objects of explicit identification or validation. As a result, these models retain constraints of the black-box type. They may output physics-consistent fragments of equations or sparse causal graphs, but the inferred structures are not sufficiently modular, falsifiable, or directly mappable to concrete physical, biological, or socioeconomic mechanisms. This motivates a sharper goal for interpretable causal dynamics reconstruction: to integrate scientific knowledge into the core causal representation so that the learning process can yield explicit, testable mechanisms, and ideally enable a transparent loop from prior laws to data-driven causal inference and back to improved understanding of the underlying mechanisms.

In this context, the present review focuses on interpretable causal dynamical reconstruction from time series, encompassing both classic methodologies and state-of-the-art AI-driven approaches. To transcend a mere methodological catalog, we organize these techniques around three foundational questions: What notion of causality is each method designed to detect? In what sense is the output interpretable? Which core assumptions (explicit or implicit) render the inferred causality meaningful and generalizable? By clarifying these foundations, we systematically discuss suitable application scenarios, inherent limitations, and potential challenges, aiming to move causal network reconstruction beyond simple detection toward rigorous scientific explanation. Ultimately, we argue that the field is maturing into a “gray-box” era, a paradigm of systematic fusion of classic mechanistic interpretability and neural flexibility, addressing the long-standing trade-off that has constrained progress in causal inference for complex systems.

1.2. Outline

This review is divided into eight sections. Section 1 introduces the problem and motivates the need for interpretable causal dynamics reconstruction from time series data. Section 2 establishes the foundations, covering the definitions of causality and interpretability, as well as the conceptual framework of network dynamics reconstruction. The core of the review (Sections 3–6) adopts a comparative structure; instead of treating classic and AI methods in isolation, we organize them into methodological families to highlight their evolution. In each section, we systematically pair established classic paradigms with their modern AI-driven extensions. Specifically, Section 3 focuses on classic and AI-driven statistical-association and Granger-causality methods; Section 4 reviews information-theoretic approaches; Section 5 examines nonlinear state–space reconstruction, and Section 6 covers techniques for discovering the governing equations. Following this methodological survey, Section 7 demonstrates the practical utility of these techniques through representative applications in climate science, neuroscience, epidemiology, economics, ecology and biology. Finally, Section 8 synthesizes a comparative analysis and outlines open challenges and future directions for the field.

2. Foundations

2.1. Causality

Causality is not uniquely defined in data analysis. Different methodological families adopt distinct operational notions of causality. Therefore, analyses of the same dataset may yield different causal conclusions under different frameworks without necessarily implying an error. The variation in outcomes reflects differences in the underlying causality definitions and assumptions. In what follows, we summarize several major definitions of causality, including causality in philosophy, causality based on counterfactual theories, Granger causality, and dynamical causality.

2.1.1. Causality in philosophy

In philosophy, causality is commonly understood as an influence by which one event, process, state or object contributes to the production of another event, process, state, or object [16]. Causality in time series refers explicitly to directional relationships where changes in one variable predictably influence changes in another.

2.1.2. Probabilistic causality

Probabilistic causality is based on the concept that a cause raises the probability of an effect [17]. Here we can give the formula for probabilistic causality [18]. C is a cause of E if $P(E | C) > P(E | \sim C)$, where $\sim$ represents negation.

2.1.3. Causality based on counterfactual theories

In 1973, David Lewis proposed the best known counterfactual theory of causation [19]. The basic idea of counterfactual causality is that the meaning of causality can be explained in terms of counterfactual conditionals of the form “if event c had not occurred, event e would not have occurred”. In 2001, Judea Pearl [20] proposed a new definition of actual cause to make this idea explicit. A structural equation framework has been developed for counterfactual causality.

Definition (Actual Cause). $\vec{X} = \vec{x}$ is an actual cause of φ in $(M, \vec{u})$ if the following three conditions hold:

AC1. $(M, \vec{u}) \models (\vec{X} = \vec{x}) \wedge \varphi$. (That is, both $\vec{X} = \vec{x}$ and φ are true in the actual world.)

AC2. There exists a partition $(\vec{Z}, \vec{W})$ of $\mathcal{V}$ with $\vec{X} \subseteq \vec{Z}$ and some setting $(\vec{x}', \vec{w}')$ of the variables in $(\vec{X}, \vec{W})$ such that if $(M, \vec{u}) \models Z = z^*$ for $Z \in \vec{Z}$, then

(a) $(M, \vec{u}) \models [\vec{X} \leftarrow \vec{x}', \vec{W} \leftarrow \vec{w}'] \neg \varphi$. In words, changing $(\vec{X}, \vec{W})$ from $(\vec{x}, \vec{w})$ to $(\vec{x}', \vec{w}')$ changes φ from true to false.

(b) $(M, \vec{u}) \models [\vec{X} \leftarrow \vec{x}, \vec{W} \leftarrow \vec{w}', \vec{Z}' \leftarrow \vec{z}^*] \varphi$ for all subsets $\vec{Z}'$ of $\vec{Z}$. In words, setting $\vec{W}$ to $\vec{w}'$ should have no effect on φ as long as $\vec{X}$ is kept at its current value $\vec{x}$, even if all the variables in an arbitrary subset of $\vec{Z}$ are set to their original values in the context $\vec{u}$.

AC3. $\vec{X}$ is minimal; no subset of $\vec{X}$ satisfies conditions AC1 and AC2. Minimality ensures that only those elements of the conjunction $\vec{X} = \vec{x}$ that are essential for changing φ in AC2(a) are considered part of a cause; inessential elements are pruned.

The three conditions capture a clear notion of causality. AC1 requires that both the cause and the effect be true. AC2 states that the effect counterfactually depends on $\vec{X}$ once we hold other variables under appropriate control. AC3 further imposes a minimality requirement, ensuring that no superfluous components are included in $\vec{X}$.

2.1.4. Interventional causality

Interventional causality defines causal relations in terms of how an outcome would change under suitable interventions. Woodward [21] defines the direct cause based on the interventionist theory of causation. The key idea underlying this theory is that, roughly, X is a cause of Y if intervening on X leads to changes in the value of Y .

Definition. A necessary and sufficient condition for X to cause Y or to figure in a causal explanation of Y is that the value of Y would change under some intervention on X in some background circumstances.

In modern causal inference, Pearl proposed a do-operator [22,23] to formalize interventions. The post-intervention distribution is defined as $P_M(y | do(x)) := P_{M_x}(y)$.

2.1.5. Granger causality

Granger causality [10] originates from the notion described by Wiener [24] and rests on the assumption that the future cannot cause the past. In practice, it can be tested by examining whether including the prior values of another time series improves the prediction of a target series' future values. Formally, variable X is said to cause Y if the historical information of X significantly improves the predictability of future values of Y beyond what can be achieved using Y 's own past information alone.

Definition (Granger Causality). If $\sigma^2(X | U) < \sigma^2(X | \overline{U - Y})$, we say that Y is causing X , denoted by $y_t \Rightarrow x_t$. We say that y_t is causing x_t if we are better able to predict x_t using all available information than if the information apart from y_t had been used.

2.1.6. Information-theoretic causality

Information-theoretic causality defines directional influence in terms of information flow between time series using concepts from information theory [25,26] such as entropy and mutual information. It measures causality in terms of the resolution of uncertainty. Information is quantified based on the statistical notion of how unlikely an event is to be observed [27]. Low-probability events provide more information than high probability events do. This perspective fundamentally measures causality by quantifying how much the uncertainty of a target variable's next state is uniquely reduced by knowing the past of a source variable, beyond the predictability already provided by the target's own history.

2.1.7. Dynamical causality

We can also define dynamical causality [28]. Consider a discrete dynamical system

$$\mathbf{x}(t+1) = F(\mathbf{x}(t), \mathbf{x}(t-1), \mathbf{x}(t-2), \dots, \mathbf{x}(t-p)) + \varepsilon_t \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ is an n -dimensional state vector, F is a smooth function that serves as a finite-dimensional approximation to the evolution operator, p is the temporal extent of memory, and ε_t is an error term. Within this framework, variable x_j is said to exert a dynamical causal influence on x_i if there exists a value $k \in \{1, 2, \dots, p\}$ such that $\partial F_i / \partial x_j(t-k) \neq 0$.

2.2. Network dynamics reconstruction

Networks are common models for data and visualization techniques representing relationships between entities in complex systems [29]. A network provides a representation for a complex system, in which nodes denote components and edges encode relationships or interactions among these components. For multivariate time series, network reconstruction aims to infer directed interaction structures that best explain how the underlying system evolves over time. The reconstruction target is a causal network in which the edges represent directed causal influences, potentially annotated by time delay, causal strength, and functional form reflecting mechanistic dependence.

In many real systems, interactions may change over time. Network dynamics refers to the temporal evolution of interaction structures among components within a complex system. When a system is represented as a network, its dynamics concern how interaction strengths and structure change over time in response to internal adaptation or external perturbations. Therefore, network reconstruction involves not only estimating a single static adjacency matrix, but also recovering a potential time-varying causal structure that can capture a system's evolving characteristics.

Here we introduce a general form of network dynamics [30] by splitting each variable's evolution into self-dynamics and interaction dynamics,

$$\frac{d\mathbf{x}_i(t)}{dt} = \mathbf{F}(\mathbf{x}_i(t)) + \sum_{j=1}^n A_{ij}(t) \mathbf{G}(\mathbf{x}_i(t), \mathbf{x}_j(t)), \quad (2)$$

where $\mathbf{F}$ captures self-dynamics of node i , and $\mathbf{G}$ represents the dynamical mechanism governing the influence of j on i . $A_{ij}(t)$ represents the existence and strength of time-varying causal interactions directed from j to i . These causal links may appear, disappear, strengthen, or weaken, leading to an evolving topology that reflects system states. Setting $A_{ij}(t)$ constant reduces this reconstruction problem to static ordinary differential equation (ODE) networks.

Network dynamics reconstruction involves inferring the directed adjacent matrix $A(t)$ and the functional components $\mathbf{F}$ and $\mathbf{G}$. Analyzing such temporal dynamics provides insight into how local interactions give rise to large-scale system behavior, such as feedback loops, cascading processes, or phase transitions [31]. This further enables the identification of transient regimes and structural instabilities.

2.3. Interpretability

Interpretability [32] refers to the extent and capacity to which a model's structure, outputs, internal logic and system's behavior can be understood and explained by human users [33,34]. In causal reconstruction, interpretability enables us to move beyond causal detection and toward understanding the underlying mechanisms that govern system behavior. This ensures that the causal findings are transparent, consistent, and understandable.

Here, we emphasize global rather than local interpretability. Global interpretability [35] refers to understanding all causal dynamics and system behaviors. This entails making sense of the full set of learned causal relationships, such as a discovered network of influences or the functional equations governing the system. We also emphasize parametric interpretability [36,37], which refers to the ability to understand the meaning of parameters (e.g., coefficients and weights) in our models, which can be regarded as the strengths and directions of influence. Classic methods typically offer high parametric and global interpretability but may lack flexibility or scalability. In contrast, AI-driven models often achieve high predictive performance but lack transparency. Hybrid approaches attempt to bridge this gap by incorporating causal priors, domain knowledge, and symbolic structures into causal learning, providing explanations that can promote trust, transparency, and accountability.

Interpretability is multidimensional. A useful causal model should have all three characteristics of transparency, stability, and fidelity. Transparency allows us to understand the underlying mechanism. Stability signifies robustness of causal conclusions under perturbations or resampling. Fidelity refers to consistency in the context of data and domain knowledge constraints.

Finally, a central challenge lies in the trade-off between model complexity and interpretability. Complex models, such as very high-dimensional or nonlinear models, might capture the patterns and interactions with high accuracy, but can become "black boxes" where the causal logic is opaque. Simpler models are far more interpretable, but might have lower predictive power. It is crucial to select appropriate approaches to balance this trade-off.

Fig. 1 provides a hierarchical methodological framework for the interpretable causal reconstruction approaches reviewed in Sections 3–6, with the aim of clarifying their underlying principles and conceptual relationships. This framework organizes these approaches according to their primary sources of causal information, including statistical dependence, temporal predictability, information flow, reconstructed state-space dynamics, graphical causal structure, and explicit governing equations. Rather than implying a rigid procedural sequence, this organization clarifies how different families of causal reconstruction methods infer causality from complementary perspectives. The arrows indicate conceptual connections among these method families, such as the transition from dependence-based measures to temporally directed causality and the use of graphical structures as priors for mechanistic equation discovery. Overall, it serves as a methodological map for interpreting the reviewed approaches and linking them to their typical outputs, including causal directionality, network topology, interaction strength, time delays, and mechanistic dynamics.

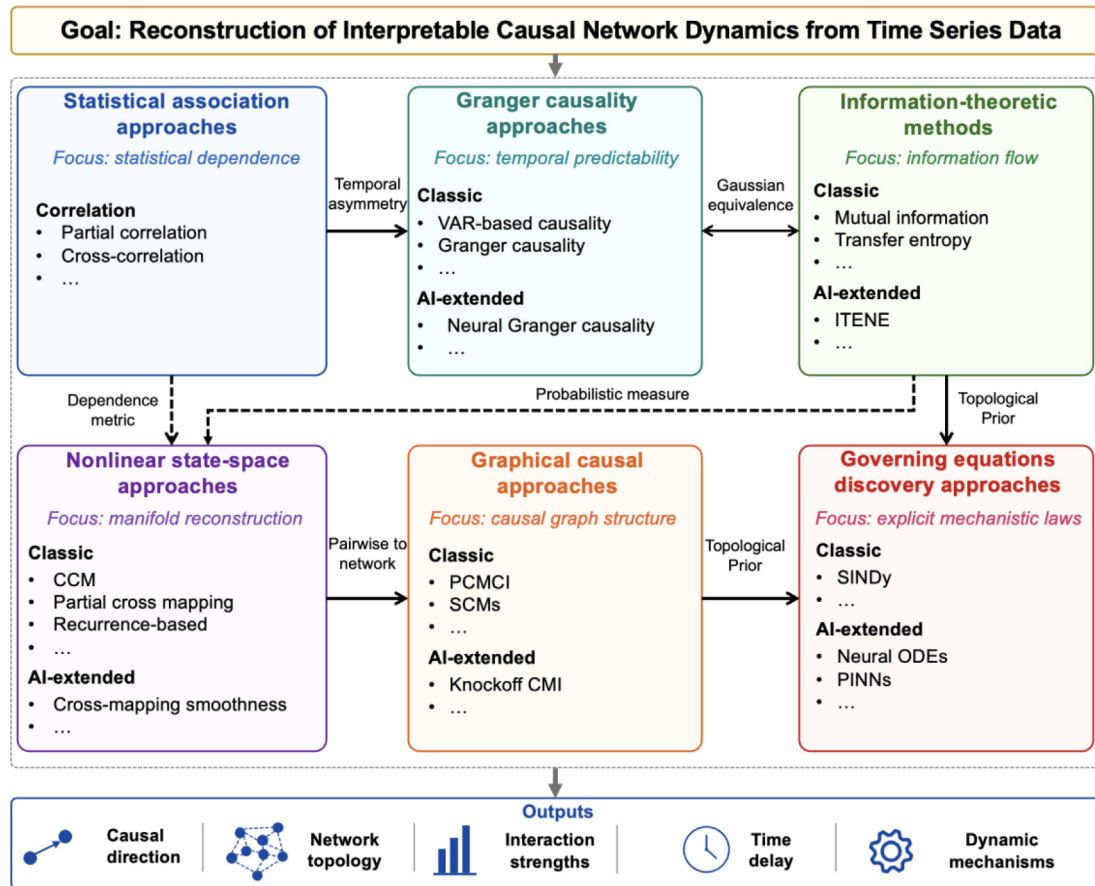


Fig. 1. Methodological landscape for reconstructing interpretable causal network dynamics. The arrows indicate conceptual connections among different approaches.

3. Granger causality and extensions

3.1. Statistical association approaches

In reconstructing causal network dynamics from time series, association-based approaches quantify inter-variable statistical dependence and encode it in a weighted graph. Historically, association-based network construction can be traced to the Pearson correlation coefficient (a product-moment correlation) as a measure of linear association between variables [38]. It has since expanded into a broad suite of dependence measures for multivariate time series, including correlation and covariance matrices and correlation-based network construction [39], spectral and wavelet coherence [40], lagged cross-correlation [41], partial correlation graphical models [42], event co-occurrence measures targeting extremes [43], recurrence analysis [44], gray correlation analysis [45] and directional spectral indices such as the phase-slope index [46]. All these methods map high-dimensional time series to weighted graphs that summarize “what fluctuates with what else” and the respective time scales or frequency bands.

The above statistical association approaches are computationally efficient and yield transparent summaries of inter-variable dependence. In finance, correlation and coherence networks reveal sectoral co-movement and market regimes; in climate science, they expose teleconnection patterns and coherent regions of variability; in neuroscience, they highlight functionally connected brain regions, and in epidemiology or social systems, they identify groups that tend to experience bursts or extremes together [42,47–49]. Thresholding these matrices or extracting their sparse representations yields community structures, hubs and backbones that can often be linked directly to domain knowledge such as circulation cells, economic sectors, or anatomical modules.

On this basis, Zhang et al. proposed the cross-validation predictability (CVP) framework as a statistical concept of causality that remained rooted in predictive association [50]. Given variables $\{X, Y, Z_1, \dots, Z_{n-2}\}$, let $\hat{Z} = \{Z_1, \dots, Z_{n-2}\}$ denote all variables except X and Y . CVP compares a model that predicts Y from $\hat{Z}$ alone with a model that additionally

includes X :

$$H_0: Y = \hat{f}(\hat{Z}) + \hat{\varepsilon} = \hat{f}(Z_1, Z_2, \dots, Z_{n-2}) + \hat{\varepsilon} \quad (3)$$

$$H_1: Y = f(X, \hat{Z}) + \varepsilon = f(X, Z_1, Z_2, \dots, Z_{n-2}) + \varepsilon \quad (4)$$

where f and $\hat{f}$ are typically linear regressions fitted on training folds of a k -fold cross-validation scheme. Let $\hat{\varepsilon}$ and ε be the summed squared prediction errors of H_0 and H_1 on the held-out data. The causal strength from X to Y named $CS_{X \rightarrow Y}$, is defined as

$$CS_{X \rightarrow Y} = \ln \frac{\hat{\varepsilon}}{\varepsilon} \quad (5)$$

where $\hat{\varepsilon}$ and ε are the total squared prediction errors on the held-out folds under H_0 and H_1 .

If $CS_{X \rightarrow Y} > 0$ and this improvement is statistically significant across folds, CVP infers a directed link $X \rightarrow Y$; otherwise, the edge is removed. In applications, one often starts from a correlation or partial-correlation network and then uses CVP to orient or prune neighboring pairs.

Even with such refinements, most statistical association measures are not causal and typically lack directionality; in contrast, CVP provides a directional notion of statistical causality based on predictive improvement. Classic measures such as correlation, covariance, coherence, and event co-occurrence and most recurrence indices are symmetric and do not encode temporal precedence. Directional variants such as lagged cross-correlation, the phase-slope index or CVP introduce arrows via lags, phase or predictive gains. However, they remain essentially pairwise and are only weakly conditioned on the rest of the system; thus, indirect pathways and common drivers cannot be excluded. Accordingly, these statistical association approaches (including directional schemes such as CVP) are best used as an interpretable, hypothesis-generating baseline: they provide transparent summaries of dependence and directional predictability that can guide a subsequent analysis, whereas stronger causal reconstruction typically requires additional assumptions and the more explicit predictive or structural models developed in the following sections.

3.2. Classic and generalized Granger causality

Granger's original causality concept [10] is defined predictively: if knowledge of the past of series X significantly improves the prediction of the future of series Y beyond the information in the past of Y alone, then X Granger-causes Y . Within a linear framework, this is operationalized by comparing restricted and unrestricted vector autoregressive (VAR) models [51]. In this section, we start from classic time-domain VAR-based Granger tests, review Geweke's decomposition and its spectral extension, and further discuss time-varying Granger causality, nonlinear Granger causality, and high-dimensional sparse VAR formulations. For these variants, interpretability stems from a shared mapping from model quantities to directed relationships of "something helps predict something else, on which lags and frequencies".

3.2.1. Classic VAR-based Granger causality in the time domain

The simplest question is whether the past of x_t helps predict y_t once the past of y_t has already been considered. In the linear case, VAR provides a system-level regression model for multivariate lagged interactions [52,53], so establishing the lack of Granger causality reduces to testing whether the corresponding lag coefficients are jointly zero.

Consider two zero-mean stationary time series $\{x_t\}$ and $\{y_t\}$. We fix a lag order p and compare restricted and unrestricted linear models for y_t . The restricted model M_{res} for y_t uses only own past of that series, while the unrestricted model M_{full} also includes lags of x_t .

$$M_{res}: y_t = \sum_{\ell=1}^p a_{\ell} y_{t-\ell} + \varepsilon_t^{(r)} \quad (6)$$

$$M_{full}: y_t = \sum_{\ell=1}^p a_{\ell} y_{t-\ell} + \sum_{\ell=1}^p b_{\ell} x_{t-\ell} + \varepsilon_t^{(u)} \quad (7)$$

where a_{ℓ} and b_{ℓ} are scalar autoregressive coefficients, p is the lag order, and $\varepsilon_t^{(r)}$ and $\varepsilon_t^{(u)}$ are residuals in the restricted and unrestricted models, respectively. The classic Granger F-statistic for testing the null hypothesis "X does not Granger-cause Y" is

$$F_{X \rightarrow Y} = \frac{(RSS_{res} - RSS_{full})/q}{RSS_{full}/(T - k)} \quad (8)$$

where RSS_{res} and RSS_{full} denote the residual sums of squares of the restricted and unrestricted models, respectively, q is the number of additional parameters (in this case, $q = p$), and k is the number of parameters in the unrestricted model. Under H_0 , this statistic $F_{X \rightarrow Y}$ follows an F-distribution with q and $T - k$ degrees of freedom, usually written as $F_{q, T-k}$. A significant positive $F_{X \rightarrow Y}$ implies that adding the past of X improves the prediction of Y , thus, we draw a directed edge $X \rightarrow Y$.

For a multivariate d -dimensional process $\mathbf{z}_t = (X_{1,t}, \dots, X_{d,t})^\top$, the full VAR(p) model is

$$\mathbf{z}_t = \sum_{\ell=1}^p A_\ell \mathbf{z}_{t-\ell} + \boldsymbol{\varepsilon}_t \quad (9)$$

with coefficient matrices $A_\ell \in \mathbb{R}^{d \times d}$ and residual covariance matrix $\Sigma = \text{cov}(\boldsymbol{\varepsilon}_t)$. The lack of Granger causality from node i to node j is tested by jointly setting to zero the coefficients $(A_\ell)_{ji}$ across all lags $\ell = 1, \dots, p$, and comparing restricted and unrestricted fits [54,55] via F-tests, likelihood-ratio tests, or log-variance ratios

In linear VARs, each directed edge $i \rightarrow j$ thus has a clear parametric meaning: conditionally on the past of all variables, knowledge of the history of node i significantly improves the prediction of node j . The corresponding coefficients and test statistics can be linked to impulse response functions and forecast error variance decompositions so that one can quantify how much a shock at node i contributes to the future variance of node j at different horizons [56]. Such time-domain VAR-based Granger causality serves as the baseline notion for all subsequent generalizations.

3.2.2. Geweke decomposition and spectral Granger–Geweke causality

Classic pairwise tests do not separate lagged influences from contemporaneous (instantaneous) coupling, nor do they fully account for the effects of other variables. Beyond such time-domain tests, the common next steps are to (i) explicitly partition linear dependence into directional versus instantaneous components, and (ii) further resolve the directional influence across frequencies (or time scales). Following this approach, Geweke's variance decomposition provides the canonical time-domain partition, its frequency-domain counterpart yields spectral Granger causality, and transfer function-based measures such as the directed transfer function (DTF) and partial directed coherence (PDC) offer alternative frequency-resolved connectivity indices derived from the same VAR representation. Geweke [57] proposed a decomposition of linear dependence within a multivariate VAR framework that partitioned the entirety of dependence into directional and instantaneous components.

Consider a stationary bivariate process $\mathbf{z}_t = (x_t, y_t)^\top$ with a joint VAR representation. Let Σ_1 and T_1 denote the residual variances of separate univariate autoregressions for x_t and y_t , respectively; let Σ_2 and T_2 be the residual variances of the corresponding conditional regressions in the joint VAR (i.e., for x_t given past X and Y , and for y_t given past X and Y), and let Σ be the residual covariance matrix of the full bivariate VAR. The entirety of the linear dependence between x_t and y_t , denoted $F_{X,Y}$, can be expressed in terms of these variances and covariances, and admits the following decomposition:

$$F_{X,Y} = F_{X \rightarrow Y} + F_{Y \rightarrow X} + F_{X,Y} \quad (10)$$

where $F_{X \rightarrow Y}$ measures the incremental predictive power of past x_t for y_t , $F_{Y \rightarrow X}$ is the reverse, and $F_{X,Y}$ captures contemporaneous coupling that cannot be attributed to lagged terms alone. In conditional settings, these quantities can be generalized to $F_{X \rightarrow Y|Z}$, $F_{Y \rightarrow X|Z}$ and $F_{X,Y|Z}$ by including additional conditioning variables z_t in the VAR. A simple summary of directionality is the net causal asymmetry

$$D_{X \rightarrow Y} = F_{X \rightarrow Y} - F_{Y \rightarrow X} \quad (11)$$

If $D_{X \rightarrow Y}$ is significantly positive (e.g., as assessed by bootstrap/permutation with FDR control), we infer a dominant direction $X \rightarrow Y$ and assign the edge weight using $D_{X \rightarrow Y}$ (or an equivalent score such as $-\log p_{X \rightarrow Y}$). Compared with pairwise F-tests, Geweke's decomposition makes explicit how much of the dependence between two series is due to lagged information flow in each direction and how much is due to instantaneous coupling, while retaining a clear probabilistic interpretation via log-variance ratios.

In multiscale systems, directed influence can concentrate in specific frequency bands. Spectral Granger causality [58] extends Geweke's framework into the frequency domain, asking not only “does X drive Y ?” but also “at which frequencies, corresponding to which time scales, is the influence $X \rightarrow Y$ strongest?”. The auto-spectrum $S_{yy}(\omega)$ of y_t and the spectrum $S_{yy|X}(\omega)$ of the residual obtained if the effect of x_t is removed via the VAR are derived from the VAR-based transfer function and the residual covariance matrix in the usual way. For a bivariate system, the spectral Granger causality from X to Y at frequency ω can be written as

$$F_{X \rightarrow Y}(\omega) = \log \frac{S_{yy}(\omega)}{S_{yy|X}(\omega)} \quad (12)$$

where $S_{yy}(\omega)$ is the auto spectrum of y_t and $S_{yy|X}(\omega)$ is the spectrum of the residual obtained if the effect of x_t is removed via the VAR. Integrating $F_{X \rightarrow Y}(\omega)$ over all frequencies recovers the time-domain Geweke index.

Related frequency-domain measures such as the directed transfer function (DTF) [59] and partial directed coherence (PDC) [60] derive directional connectivity indices directly from the transfer function $H(\omega)$, highlighting net “sources” and “sinks” of activity at different frequencies. In neuroscience, for example, spectral Granger causality, DTF and PDC are widely used to identify the brain regions that drive others in specific frequency bands (e.g., theta, alpha, and gamma), while in climate applications, these approaches can separate interannual from decadal modes of teleconnection. Spectral decomposition [61] thus yields frequency-resolved causal networks with edges that can be interpreted as directed information flows operating at distinct time scales.

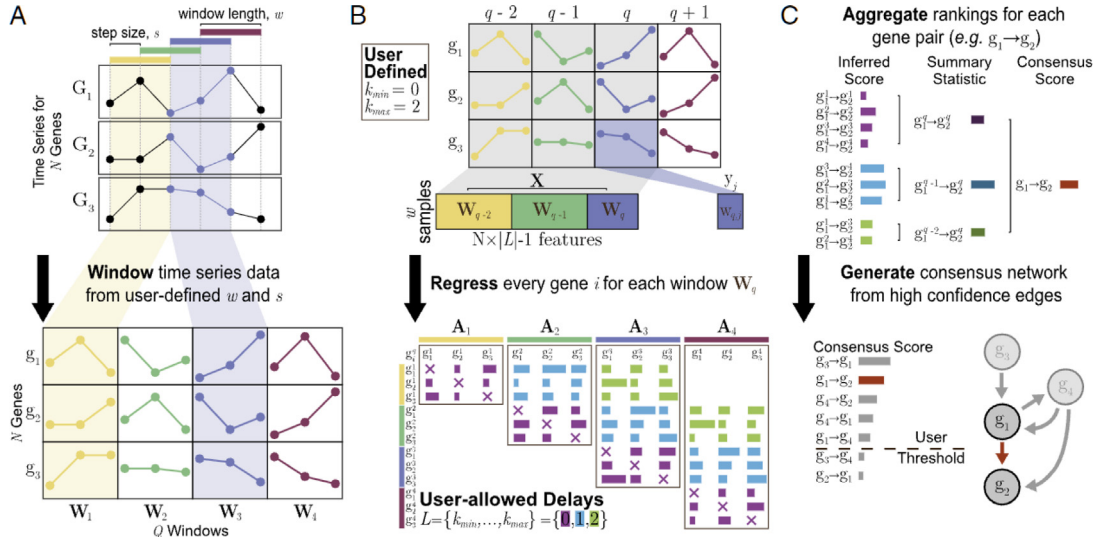


Fig. 2. Overview of the SWING framework. (A) Time-series data are divided into windows with a user-specified width, w . (B) For each window, inference is performed by iteratively selecting response and explanatory genes. The subset of available explanatory genes is defined by the minimum and maximum user-allowed time delays. (C) Edges from each window model are aggregated into a single network representation of the biological interactions between measured variables.

Source: Reprinted from Ref. [63].

3.2.3. Time-varying Granger causality

Many real-world systems are nonstationary, and directional dependencies may drift, undergo structural breaks, or switch regimes over time. Time-varying Granger causality is therefore typically approached either by (i) local-stationarity approximations, most commonly using a sliding-window estimation that yields a sequence of window-wise directed networks, or by (ii) models that explicitly evolve over time, such as state-space or time-varying-parameter VARs that impose smooth temporal dynamics on coefficients and produce continuously varying Granger measures. Focusing on the sliding-window route, one representative pipeline is the “dynamic causal network analysis” framework that applies sliding-window Granger modeling to multivariate time series and clusters the resulting connectivity patterns into a small number of recurring network states. A set of simultaneously observed time series is treated as a node set. For each ordered pair (x, y) , a bivariate VAR is built inside a window of length ω with overlap ρ between adjacent windows [62]. A restricted model with only y -lags yields a residual sum of squares RSS_{res} ; adding X -lags in the same window results in the full model with RSS_{full} . In each window, a standard Granger F-statistic or log-variance ratio, as described in Section 3.2.1, is computed to quantify the strength of $X \rightarrow Y$.

Doing this for all ordered pairs yields a directed weighted adjacency matrix for a given window; sliding the window generates a time series of adjacency matrices. To summarize typical patterns, these matrices can be vectorized and clustered (e.g., via k -means with the elbow method of selecting k) into a small number of causal states. Averaging the matrices within each cluster yields a representative network with edge weights $\bar{F}_{i \rightarrow j}$ that encode typical causal strengths in that regime.

Sliding-window Granger networks make regime switching explicit: bursts of connectivity patterns can be aligned with external covariates such as climate phases, monetary-policy regimes or market crises. In genetic regulatory network reconstruction, the sliding window inference for network generation (SWING) [63] fits multivariate Granger models in sliding windows (scanning window sizes and delays) and aggregates the resulting edges into a consensus gene-regulatory network in Fig. 2. Although window choices introduce a degree of arbitrariness and reduce the effective sample size, the resulting sequence of networks offers an intuitive picture of how information flow reorganizes over time.

A key modeling assumption is that coefficients are treated as constant within each window and may change abruptly at window boundaries. This brings intuitive time-localized networks, but window choices introduce arbitrariness, reduce the effective sample size, and can create boundary artifacts. When causal influences evolve more smoothly, time-varying-parameter VAR (TVP-VAR) models offer a more principled description by letting the VAR coefficients drift continuously over time within a state-space formulation. Let $\mathbf{z}_t \in \mathbb{R}^d$ follow a TVP-VAR(p) model,

$$\mathbf{z}_t = \sum_{\ell=1}^p A_{\ell,t} \mathbf{z}_{t-\ell} + \boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t) \quad (13)$$

where $A_{\ell,t}$ are time-varying coefficient matrices, and $\boldsymbol{\Sigma}_t$ is a (possibly time-varying) residual covariance.

A common specification assumes that the stacked coefficient vector $\alpha_t = \text{vec}(A_{1,t}, \dots, A_{p,t})$ evolves as a random walk,

$$\alpha_t = \alpha_{t-1} + \eta_t, \eta_t \sim \mathcal{N}(0, Q) \quad (14)$$

where $\text{vec}(\cdot)$ stacks the coefficient matrices into a state vector, and Q is the process-noise covariance controlling temporal smoothness [64,65]. The observation equation (the TVP-VAR model) and state equation (a random walk) together define a linear Gaussian state–space model, which can be estimated using Kalman filtering/smoothing or Bayesian simulation.

Given estimates $\{\hat{A}_{\ell,t}\}$ and $\hat{\Sigma}_t$, local Granger causality measures can be computed at each time t , including time-varying log-variance ratios, impulse response functions, and forecast error variance decompositions. These quantities together characterize a Granger causality network that evolves smoothly over time. Stochastic volatility components for Σ_t further improve robustness to time-varying noise scales, while shrinkage and sparsity priors (e.g., global–local shrinkage or spike-and-slab) stabilize estimates in high-dimensional, short-sample settings by adaptively shrinking small coefficients toward zero while preserving a few dominant edges [66]. In practice, temporal smoothness is tuned via a forgetting factor or by setting Q ; for very large networks, dynamic factor structures or local linear priors on $A_{\ell,t}$ further reduce dimensionality. TVP-VAR thus yields time-resolved Granger curves that can be mapped to slowly drifting mechanisms (e.g., gradual connectivity shifts under disease progression) in a way that remains parameter-level interpretable.

3.2.4. Granger causality in nonlinear and high-dimensional systems

In complex systems, linear Granger tests may miss nonlinear dependencies such as saturation, thresholds, or multiplicative effects. To retain the same restricted–unrestricted predictability test while allowing nonlinear predictors, kernel-based Granger causality [67,68] generalizes the linear prediction framework by embedding lagged vectors into a high-dimensional reproducing kernel Hilbert space (RKHS), where nonlinear relationships in terms of the original variables become linear in the feature space. Let $z_t \in \mathbb{R}^d$ be stationary and denote by u_t the concatenated lag vector up to order p , obtained by stacking past observations $(z_{t-1}, \dots, z_{t-p})$ into a single vector. For a given target component y_t and a candidate driver x_t , one defines two predictors in an RKHS with kernel $k(\cdot, \cdot)$: a restricted predictor that excludes the lags of x_t , and an unrestricted predictor that includes them. The nonlinear predictor is written as

$$\hat{y}_t = f(u_t) = \sum_s \alpha_s k(u_t, u_s) \quad (15)$$

where the coefficients $\{\alpha_s\}$ are estimated by kernel ridge regression, possibly with an additional regularization. Here, $u_t \in \mathbb{R}^{dp}$ is the concatenated lag vector. This kernel expansion predicts y_t as a weighted sum of kernel similarities between the current history u_t and past histories u_s , with weights α_s learned by a kernel ridge regression.

Let σ_r^2 and σ_u^2 denote the residual variances of the restricted and unrestricted kernel regressions, respectively. A kernel-based Granger index can then be defined as

$$F_{X \rightarrow Y}^{(\text{ker})} = \log \frac{\sigma_r^2}{\sigma_u^2} \quad (16)$$

analogously to the linear log-variance ratio, with significance assessed via permutation or bootstrap tests.

In a high-dimensional system, these approaches may overfit or deduce spurious causality. To address this high-dimensional issue, low-rank kernel regularization has been proposed to, for example, approximate the Gram matrix with a truncated eigendecomposition (Nyström- or spectral-truncation-style) before computing the kernel Granger statistic. To reduce overfitting and spurious causality in finite samples, a common strategy is to use a rank-reduced Gram matrix, keeping only the leading eigenvectors of the kernel matrix and constructing Granger indices from projections onto this low-dimensional subspace. Although the predictive model lives in an implicit, high-dimensional feature space, the graph structure remains interpretable in the Granger sense: a significant edge $X \rightarrow Y$ still asserts that including the nonlinear history of X improves the prediction of Y beyond all other variables, and the frequency or context in which such edges appear can be related to the underlying nonlinear mechanisms.

The number of nodes d is comparable to or larger than the sample size T in many applications, such as brain networks, gene regulation, and financial systems. A VAR(p) model then contains $O(d^2p)$ parameters, and an unrestricted estimation is infeasible and prone to severe overfitting. High-dimensional sparse VAR approaches address this by imposing sparsity on the coefficient matrices, effectively performing variable (and edge) selection while estimating Granger networks [69].

The VAR(p) model is written in a regression form,

$$Y = XB + E \quad (17)$$

where each row of Y collects the target series over time, X stacks lagged values of all variables, B is the matrix of regression coefficients encoding Granger edges, and E is the residual matrix. A standard approach estimates B via lasso or group-lasso penalized least squares,

$$\hat{B} = \arg\min_B \left\{ \|Y - XB\|_F^2 + \lambda \mathcal{P}(B) \right\} \quad (18)$$

where $\|\cdot\|_F$ is the Frobenius norm, $\mathcal{P}(B)$ is an ℓ_1 or group penalty that induces sparsity, and $\lambda > 0$ controls the overall level of shrinkage. Nonzero coefficients then indicate potential Granger-causal links.

The truncated lasso Granger approach [70] refines this idea by using a concave “truncated” penalty $\rho_\tau(\cdot)$ that behaves as the absolute value for small coefficients but flattens out beyond a threshold τ , strongly shrinking weak coefficients toward zero while leaving sufficiently large coefficients essentially unpenalized. This reduces the bias for large, truly nonzero coefficients while still eliminating weak and spurious links. Asymptotic or bootstrap-based tests can be layered on top of the penalized estimates to control type-I error rates.

Sparse VAR Granger networks thus highlight a small set of dominant predictive pathways in high-dimensional settings. Edges remain explicitly parameterized and interpretable as conditional linear Granger relationships, but now reflect a deliberate selection from many potential connections. In empirical studies, this approach has been used, for example, to identify core “hubs” and “brokers” in large-scale financial markets or brain networks, and to track how such sparse connectivity patterns reorganize over time. These high-dimensional sparse frameworks also provide a natural bridge to the neural Granger causality of Section 3.3, where sparsity constraints are built directly into the architectures of nonlinear predictors.

3.3. Neural Granger causality

Nonlinear dependencies and high-dimensional networks with complex interactions have so far only been captured crudely, even if one enforces sparsity in VAR coefficients. Neural Granger approaches address these gaps by replacing the linear predictor with a flexible function class (multilayer perceptrons, recurrent units, or reservoirs) while building sparsity or sensitivity constraints into the architecture so that the learned parameters can still be read as a Granger-causal graph [71,72]. Early research has already used feedforward networks to implement nonlinear Granger tests, either via non-uniform embedding of lagged inputs combined with explicit validation phases [73] or via multilayer perceptrons trained in parallel with linear VAR baselines [74]. More recent contributions systematize this into explicit “neural Granger causality” frameworks. Building on the classic as well as kernel and sparse Granger frameworks of Section 3.2, this section reviews neural extensions.

3.3.1. Sparse-input neural Granger predictors

A common nonlinear extension of Granger causality replaces the linear VAR predictor with a flexible nonlinear function class while preserving the predictive definition of Granger causality. Neural Granger causality extends the standard restricted and full model comparison to nonlinear predictors. In sparse-input variants, block sparsity enables causal edges to be identified from a single fitted predictor. This allows one to probe nonlinear predictability while retaining the familiar Granger hypothesis test.

Let $x_t = (x_{1,t}, \dots, x_{d,t})^\top$ be a d -dimensional stationary time series, and let u_t denote the lag vector obtained by concatenating all past observations up to order p into a single vector. Neural Granger models replace the linear VAR from Section 3.2 with nonlinear predictors of the form

$$\hat{x}_{i,t} = f_i(u_t; \theta_i) \quad (19)$$

where $f_i(\cdot; \theta_i)$ is a multilayer perceptron, recurrent unit or another differentiable model with parameters θ_i . To test whether series j Granger causes series i , one follows the same “restricted vs. full” strategy: the full predictor uses all coordinates of u_t , whereas the restricted predictor removes or clamps the components corresponding to the lags of x_j .

Let σ_r^2 and σ_u^2 denote the residual variances of the restricted and unrestricted neural predictors for target i . A nonlinear Granger index can then be defined as $F_{j \rightarrow i}^{(NN)} = \log \frac{\sigma_r^2}{\sigma_u^2}$ analogously to the linear log-variance ratio, with significance assessed via permutations or surrogate-data tests [75,76]. The meaning of an edge is unchanged in that adding the past of x_j significantly improves the prediction of x_i . However, the graph is not encoded directly in the network parameters, so one must refit restricted models for each candidate driver and target. This makes the procedure computationally intensive (one restricted model per ordered pair) and provides limited structural insight into how the network implements the dependence, motivating architectures in which sparsity patterns in the parameters themselves encode the Granger graph.

Refitting a separate restricted network for every ordered pair is expensive and obscures the global structure. To avoid such refitting, sparse-input neural Granger causality encodes the adjacency structure directly in the block organization of input weights of the predictor for each target node (as shown in Fig. 3). For each target series i , the first-layer weight matrix of f_i is partitioned into blocks $\{W_{ij}\}_{j=1}^d$, where each block W_{ij} collects the weights from all lags of series j into the hidden layer of the predictor for series i . Training proceeds by minimizing a squared-error loss with a group-lasso penalty on these blocks [71],

$$\mathcal{L}_i(\theta_i) = \frac{1}{T-p} \sum_{t=p+1}^T (x_{i,t} - \hat{x}_{i,t})^2 + \lambda \sum_{j=1}^d \|W_{ij}\|_F \quad (20)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. If, at the optimum, the block W_{ij} is driven to zero, then none of the lags of series j feed into the hidden representation for predicting x_i , and one declares that x_j does not Granger cause x_i . Conversely, a nonzero block W_{ij} indicates a directed edge $j \rightarrow i$. Component-wise recurrent variants replace the feedforward network with statistical recurrent units or other recurrent neural networks (RNN) to capture long-range temporal dependencies while retaining the same group-lasso structure on the input weights [77].

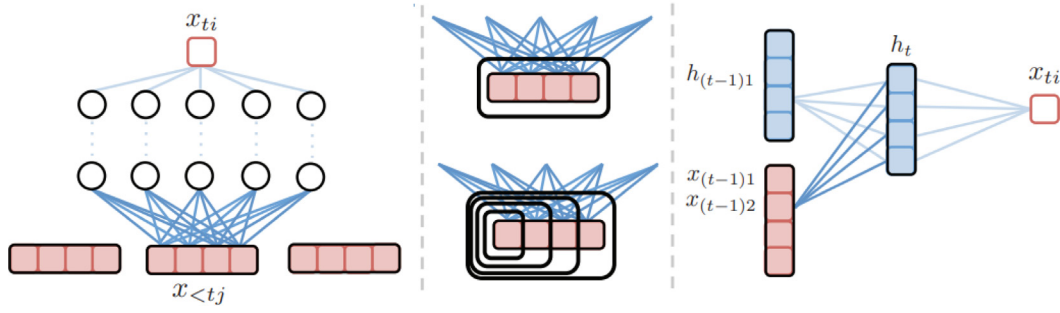


Fig. 3. Schematic of neural Granger causality with component-wise MLP/LSTM and structured sparsity. (left) Schematic for modeling Granger causality using cMLPs. If the outgoing weights for series j , shown in dark blue, are penalized to zero, then series j does not Granger-cause series i . (center) The group lasso penalty jointly penalizes the full set of outgoing weights while the hierarchical version penalizes the nested set of outgoing weights, penalizing higher lags more. (right) Schematic for modeling Granger causality using a cLSTM. If the dark blue outgoing weights to the hidden units from an input $x_{(t-1)j}$ are zero, then j does not Granger-cause series i .
Source: Reprinted from Ref. [71].

Block-level sparsity yields a one-to-one correspondence between nonzero input blocks and Granger edges. The adjacency matrix is read directly from blocks W_{ij} that survive regularization. This provides a clear structural interpretation that series j contributes nonlinearly to predicting series i , even though the internal hidden-layer nonlinearities remain opaque.

3.3.2. Decomposition-based neural Granger causality

Sparse-input networks identify the sources that matter, but their monolithic hidden representations still mix contributions from different series. A second family of neural Granger causality approaches therefore enforces an additive or concept-based structure so that each prediction can be decomposed into source- or concept-specific terms with explicit semantics. In this subsection, we group three such approaches: Neural Additive VAR (NAVAR) [78], which represents the conditional mean as a sum of source-specific subnetworks, self-explaining Granger framework, which expresses predictions as combinations of low-dimensional “concepts” tied to disjoint groups of inputs, and Latent-confounder recurrent models, which augment observable series with latent processes to account explicitly for hidden common drivers.

NAVAR approaches seek finer-grained interpretability by decomposing each prediction into a sum of source-specific components. The one-step conditional mean of $x_{i,t}$ is written as a sum of contributions from each potential cause,

$$\hat{x}_{i,t} = \sum_{j=1}^d f_{ij}(x_{j,t-1:t-p}) + \epsilon_{i,t} \quad (21)$$

where each f_{ij} is a small neural network acting only on the lags of series j . If $f_{ij} \equiv 0$ for all inputs, then series j has no Granger influence on i ; otherwise, the shape of f_{ij} encodes the magnitude, the sign and the nonlinearity of the Granger effect. Training these models involves minimizing

$$\mathcal{L}_i(f_{ij}) = \frac{1}{T-p} \sum_{t=p+1}^T \left[x_{i,t} - \sum_{j=1}^d f_{ij}(x_{j,t-1:t-p}) \right]^2 + \lambda \sum_{j=1}^d \Omega(f_{ij}) \quad (22)$$

where Ω is a functional penalty (e.g., a group norm on the parameters of f_{ij} , or a roughness penalty on its outputs) that encourages many of component functions to vanish. This additive structure underlies the NAVAR framework, which emphasizes interpretability and provides an explicit decomposition of the predicted signal into source-specific contributions.

NAVAR explicitly decomposes each prediction into a sum of source-specific terms: the partial dependence plots of f_{ij} show how lags of x_j affect x_i , and the penalty zeroes out unimportant components. Thus, both the existence of a Granger link (whether f_{ij} is nonzero) and its qualitative form (e.g., saturating vs. linear, or synergistic vs. antagonistic) are directly interpretable.

Even additive models may use high-dimensional hidden features with an unclear meaning. Self-explaining architectures aim to tie predictions to a low-dimensional set of “concepts” with explicit semantics and to produce time-localized effect sizes that can be traced back to specific input groups or latent factors.

Self-explaining neural Granger causality addresses interpretability by expressing each prediction as a weighted combination of a small set of explicit “concepts”, which enables time-local attributions and supports Granger edge

identification. In the self-explaining Granger framework [79], the prediction for series i at time t takes the form

$$\hat{x}_{i,t} = \sum_{m=1}^M \alpha_{im}(u_t) h_{im}(u_t) \quad (23)$$

where $h_{im}(u_t)$ are “concept” features depending simply, often linearly, on subsets of u_t (for instance, local linear projections of selected lags), and $\alpha_{im}(u_t)$ are attention-like coefficients describing how strongly each concept contributes at time t . By constraining the concepts to depend on disjoint groups of input series and imposing sparsity or stability constraints on the coefficients α_{im} , nonzero coefficients associated with the concept tied to series j can be interpreted as evidence that x_j Granger causes x_i . This yields both a Granger graph and signed, time-localized effect sizes, with better interpretability than that of sparse-input networks.

Neural Granger models can also incorporate latent confounding directly to mitigate spurious links induced by hidden common drivers and to obtain more reliable directed relationships. A deep recurrent Granger model has been proposed in which multiple RNNs parameterize predictive distributions for both observed and latent processes [80]. A dual-decoder setup is used to construct Granger tests that remain valid in the presence of latent confounders with different lags to cause and effect; in simulated nonlinear systems, this method reduces spurious edges that would be present under a naive application of neural Granger causality without confounder modeling.

3.3.3. Sensitivity-based and reservoir computing-based neural Granger causality

Sparse-input and additive models explicitly encode the adjacency structure in the predictor design, which can limit flexibility. A third family of neural Granger causality approaches adopts an attribution-based extraction strategy: it fits a flexible predictor and infers directed links from input–output sensitivities or from reservoir readouts. Specifically, we consider two subclasses: (i) sensitivity-based neural Granger causality, which defines edge strengths from Jacobian or gradient norms and regularizes these sensitivities toward sparsity, and (ii) reservoir computing-based Granger causality, which uses fixed nonlinear recurrent reservoirs (echo state networks and their extensions) and infers Granger edges from linear readouts on top of rich reservoir states.

Suppose that one trains a single multivariate predictor $\hat{x}_t = f(u_t; \theta) \in \mathbb{R}^d$ so that the i th component $\hat{x}_{i,t} = f_i(z_t; \theta)$ provides the one-step prediction for series i . At time t , the Jacobian

$$J_t = \frac{\partial f(u_t; \theta)}{\partial u_t} \quad (24)$$

collects the partial derivatives of all outputs with respect to all lagged inputs. For Granger analysis purposes, one groups the columns of J_t corresponding to the lags of series j and the rows corresponding to output i , and defines a local influence measure, for example,

$$\gamma_{j \rightarrow i}(t) = \|J_t^{(i,j)}\|_F^2 \quad (25)$$

where $J_t^{(i,j)}$ is the submatrix of J_t restricted to output i and the lagged inputs of series j . Averaging over time yields an overall neural Granger strength

$$\Gamma_{j \rightarrow i} = \frac{1}{T-p} \sum_{t=p+1}^T \gamma_{j \rightarrow i}(t) \quad (26)$$

where p is the maximum lag order, and T represents the length of the time series. If $\Gamma_{j \rightarrow i}$ is negligible, one concludes that x_j has no Granger influence on x_i ; if it is large, one infers a directed edge. This Jacobian-based definition has been used to construct a Granger test “using Jacobian in neural networks” for nonlinear dynamical systems [81]. Jacobian Regularizer-based Neural Granger Causality (JRNGC) goes further by adding a group-lasso penalty directly on time-averaged Jacobian blocks,

$$\mathcal{R}(\theta) = \lambda \sum_{i=1}^d \sum_{j=1}^d \left(\frac{1}{T-p} \sum_{t=p+1}^T \|J_t^{(i,j)}\|_F^2 \right)^{1/2} \quad (27)$$

where λ is the regularization strength, and d is the number of series, so that minimizing the sum of the prediction error and $\mathcal{R}(\theta)$ yields a sparse Granger matrix defined in terms of input–output sensitivities using a single multivariate network [76]. Gradient Regularization-based Neural Granger Causality (GRNGC) extends this idea by replacing Jacobian norms with L_1 -type penalties on stochastic gradients between inputs and outputs, which drives weak sensitivities toward zero to suppress spurious edges while allowing the Granger structure to be extracted from a single fitted predictor [82].

Here, edges are defined not by zero/nonzero input weights but by how sensitive each prediction is to small perturbations of each driver’s lags. The magnitude of $\Gamma_{j \rightarrow i}$ is directly interpretable as an average local effect size of x_j on x_i through the learned dynamics. Regularizing Jacobian or gradient norms pushes many sensitivities toward zero, producing a sparse, sensitivity-based Granger network that can be computed from any differentiable architecture.

Fully trainable RNNs are powerful but expensive to optimize and prone to overfitting in long or high-dimensional series. Reservoir computing offers a lighter alternative: a fixed nonlinear recurrent layer transforms inputs into a rich state, and only a linear readout is trained. Reservoir-based Granger causality exploits this structure to obtain nonlinear yet interpretable causal networks that scale well. Echo state networks (ESNs) are canonical reservoir models. Given an input vector x_t , the reservoir state $r_t \in \mathbb{R}^R$ evolves as

$$r_t = (1 - \alpha) r_{t-1} + \alpha \psi(W_r r_{t-1} + W_{in} x_t + b) \quad (28)$$

where W_r and W_{in} are fixed weight matrices satisfying the echo-state condition, ψ is a nonlinear activation (typically tanh), b is a bias vector and $\alpha \in (0, 1]$ is the leaking rate. The one-step-ahead prediction is then given by a linear readout

$$\hat{x}_{t+1} = W_{out} \begin{bmatrix} r_t \\ x_t \end{bmatrix} + c \quad (29)$$

where the output weight matrix W_{out} and offset c are learned by ridge regression. Echo-state-based nonlinear Granger causality uses this architecture as a nonlinear feature map, and defines a Granger index by comparing full and restricted readouts for each candidate edge [72].

Reservoir computing Granger causality (RCG) refines both the design of the reservoir and the statistical testing procedure [83], to better capture complex nonlinear dependencies and to improve robustness against noise. In RCG, the reservoir may be explicitly driven by delayed versions of the input, so that the state vector $r_t = \Psi(x_t, x_{t-1}, \dots, x_{t-p}; \xi)$ encodes a rich, nonlinear embedding of the joint past of all variables, controlled by hyperparameters ξ such as the spectral radius, input scaling and leakage. The Granger index $\mathcal{G}_{j \rightarrow i}^{RC}$ is evaluated over an ensemble of reservoirs to reduce variance, and permutation or surrogate-data tests are used to assign significance levels to each edge in the resulting causal network. Higher-order Granger reservoir computing (HoGRC) further extends RCG by directly targeting complex network structures, such as multistep pathways and higher-order interactions, while still providing accurate dynamical prediction [84]. In a simplified form, the readout for series i can be decomposed as

$$\hat{x}_{i,t+1} = \sum_j w_{ij}^{(1)\top} r_t^{(j)} + \sum_{j \neq k} w_{ijk}^{(2)\top} r_t^{(j,k)} + \dots \quad (30)$$

where $r_t^{(j)}$ are reservoir features primarily driven by series j , $r_t^{(j,k)}$ encode joint responses to series j and k , etc. Penalizing the norms of the coefficient vectors $w_{ij}^{(1)}$, $w_{ijk}^{(2)}$, etc., selects a sparse set of first-order and higher-order Granger interactions, thereby yielding a scalable representation of complex causal structures while maintaining multistep predictive accuracy.

Across these neural Granger causality variants, a unifying idea is to retain Granger's predictive definition while expanding the conditional-mean model class from linear VARs to neural networks and reservoir-based predictors. Compared with linear and kernel/sparse Granger methods, these neural approaches can capture richer nonlinear and higher-order dependencies, but typically require more careful optimization and hyperparameter tuning and demand additional interpretability checks to guard against spurious links.

4. Information-theoretic approaches

4.1. Classic information-theoretic approaches

Classic information-theoretic approaches quantify causal relationships by measuring the flow and structure of information between variables. These approaches rely on information-theoretic measures such as entropy, mutual information, and their extensions to capture dependencies. In this section, we review three representative families including mutual information and its extensions, information decomposition, and transfer entropy and its extensions.

4.1.1. Mutual information and its extensions

The concept of mutual information originates from information entropy theory [25,85] and is used to quantify the degree of dependency between two random variables. Its core extensions mainly include conditional mutual information and time-delayed mutual information.

Mutual information (MI), introduced by Shannon [25], quantifies the degree of dependency between two random variables [25,85]. For discrete variables, MI between X and Y is defined as

$$I(X; Y) = \sum_x \sum_y p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (31)$$

where $p(x)$ and $p(y)$ are marginal probability distributions and $p(x, y)$ represents the joint probability distribution. Commonly used methods for probability density estimation include k -nearest neighbor distances [86] and kernel density estimation [87]. For continuous random variables, summations are replaced by integrals [25]. Moreover, MI is symmetric, satisfying $I(X; Y) = I(Y; X)$.

However, MI cannot distinguish direct interactions from indirect interactions. To address this limitation, Conditional Mutual Information (CMI) was proposed. It quantifies the remaining dependency between variables X and Y given another variable Z . CMI is defined as [88]

$$I(X; Y|Z) = \sum_x \sum_y \sum_z p(x, y, z) \log \left(\frac{p(x, y|z)}{p(x|z)p(y|z)} \right) \quad (32)$$

where $p(x, y|z)$ is the joint conditional probability distribution of X and Y given Z ; $p(x, y, z)$ is the joint probability distribution of X , Y and Z ; and $p(x|z)$ and $p(y|z)$ are the marginal conditional probability distributions of X given Z and Y given Z , respectively.

By introducing a conditioning variable Z , CMI can effectively filter out the influence of confounding factors, allowing for a more accurate assessment of the true association between X and Y . If $I(X; Y|Z)$ is significantly smaller than $I(X; Y)$, it suggests that Z may be a confounding variable mediating the dependence between X and Y .

Fraser and Swinney further proposed time-delayed mutual information (TDMI) [89,90] to extend MI to causality involving temporal delays. For two time series X and Y , TDMI measures the mutual information between X delayed by τ time steps and Y , and is defined as

$$I(X_{t-\tau}; Y_t) = \sum_x \sum_y p(x_{t-\tau}, y_t) \log \left(\frac{p(x_{t-\tau}, y_t)}{p(x_{t-\tau})p(y_t)} \right) \quad (33)$$

where $p(x_{t-\tau}, y_t)$ is the joint probability distribution of Y_t and the series resulting from time-delaying by τ the time series X , named $X_{t-\tau}$, and $p(x_{t-\tau})$ and $p(y_t)$ are marginal probability distributions of $X_{t-\tau}$ and Y_t , respectively. In a study of how MI varies with the delay parameter τ , the delay at which TDMI peaks is interpreted as the information flow direction, extending information-theoretic dependence measures from static settings to the analysis of dynamic causal interactions.

4.1.2. Information decomposition

In multivariate systems, the informational contribution of multiple variables to a target or future state can be organized into structurally distinct and interpretable components, commonly characterized as unique, redundant, and synergistic information. Three representative approaches are partial information decomposition, integrated information decomposition and Synergistic-Unique-Redundant Decomposition.

While MI and CMI quantify the overall dependence, they cannot distinguish how different sources contribute. Partial information decomposition (PID) [91], introduced by Williams and Beer, addresses this by partitioning the mutual information that sources $[R_1, R_2, \dots, R_n]$ provide about a target S . In the three-variable case, PID is expressed as

$$I(S; R_1, R_2) = \text{Unq}(S; R_1) + \text{Unq}(S; R_2) + \text{Syn}(S; R_1, R_2) + \text{Rdn}(S; R_1, R_2) \quad (34)$$

where $\text{Unq}(S; R_1)$ represents the unique information, i.e., the information that R_1 individually provides about S that is not contained in R_2 . $\text{Syn}(S; R_1, R_2)$ is the synergistic information, the information that arises only if when R_1 and R_2 are considered jointly. $\text{Rdn}(S; R_1, R_2)$ denotes the redundant information, i.e., the information about S that is simultaneously provided by both R_1 and R_2 .

The interpretability of PID is reflected in providing a component-oriented view of information processing. By cleanly separating information into unique, redundant, and synergistic atoms, it offers an interpretable map of the information architecture within a system.

Chiang improves PID by establishing an explicit correspondence between PID components and a causal structure [92]. The cited study introduces information-theoretic signatures of causality, extending PID from a descriptive decomposition to a structural tool for causal discovery in Bayesian networks and hypergraphs. The core idea is to adopt a locally focused information-theoretic perspective, where the causal role of each variable can be inferred from its immediate informational footprint. Under suitable assumptions, unique information identifies direct causal neighbors, while synergistic information reveals collider and higher-order causal configurations, enabling the reconstruction of directed causal relations from local informational patterns without a global graph search.

PID resolves this only for many-to-one interactions and cannot handle the ubiquitous many-to-many structure of real dynamic systems. Integrated information decomposition (Φ ID) is a unified framework that decomposes all modes of information flow from the past to the future, enabling a principled taxonomy of information dynamics [93].

Φ ID extends PID to multiple targets, thereby capturing how information evolves among multiple sources and targets simultaneously. The core idea is to decompose TDMI into information atoms. Each information atom represents a specific transition pattern (redundant, unique, or synergistic) from the past to the future. For a two-variable system, TDMI is decomposed as

$$\text{TDMI} = I(X; Y) = \sum_{\alpha, \beta \in \mathcal{A}} I_{\alpha}^{\alpha \rightarrow \beta} \quad (35)$$

where $I_{\alpha}^{\alpha \rightarrow \beta}$ denotes a Φ ID atom, with α and β encoding the information patterns in the past and the future, respectively. Thus, TDMI is exactly partitioned into 16 nonoverlapping information atoms. These 16 atoms follow from decomposing

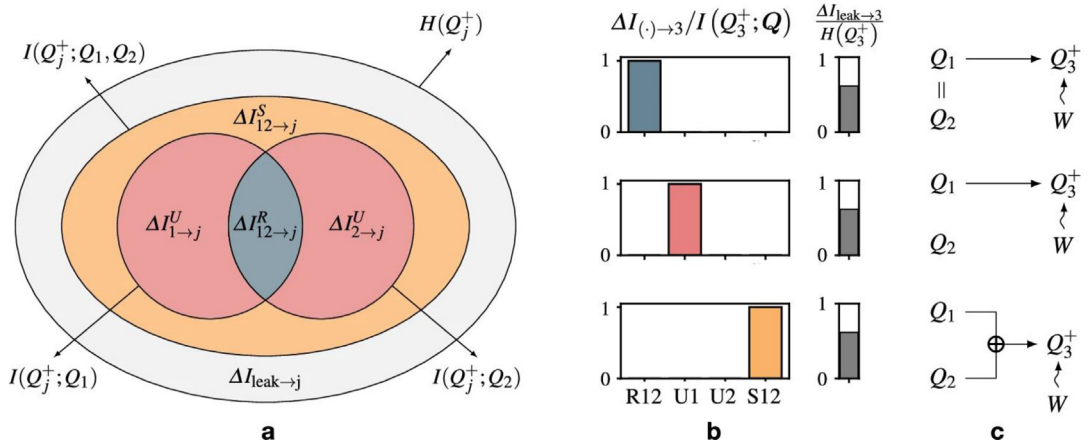


Fig. 4. SURD: Synergistic-Unique-Redundant Decomposition of causality. a Diagram of the decomposition of causal dependencies between a vector of observed variables $Q = [Q_1, Q_2]$ (past) and a target variable Q_j^+ (future) into their synergistic (S), unique (U) and redundant (R) components (in yellow, red, and blue, respectively) and contributions to the total, $I(Q_j^+; Q_1, Q_2)$, and individual, $I(Q_j^+; Q_i)$, mutual information. The causality leak is represented in gray. A version of this diagram for three variables is shown in the Supplementary Materials. b Redundant, unique, and synergistic causalities for the simple examples of (c) a duplicated input (top panel), an output equal to the first input (middle panel), and an exclusive-OR output (bottom panel). The notation used is such that $[R12, U1, U2, S12] \equiv [\Delta I_{12 \rightarrow 3}^R, \Delta I_{1 \rightarrow 3}^U, \Delta I_{2 \rightarrow 3}^U, \Delta I_{12 \rightarrow 3}^S]$. The target variable Q_3^+ is affected by external stochastic forcing W , which is independent of the observed variables in Q . The effect of W is measured by the causality leak, represented by the gray bar.

Source: Reprinted from Ref. [94].

TDMI between past and future states of a system into PID atoms at both time points. Each Φ ID atom corresponds to a mapping from a PID atom in the past to a PID atom in the future. Each atom captures a basic transformation from a past PID node α to a future PID node β .

The interpretability of Φ ID arises from extending PID into the temporal domain. By decomposing the entire past-to-future information flow into distinct atoms of redundant, unique, and synergistic transformations, Φ ID provides a transparently structured view of how information evolves across time.

Synergistic-Unique-Redundant Decomposition (SURD) [94] was introduced by Martínez-Sánchez in 2024. While PID decomposes static mutual information, SURD reformulates this decomposition in terms of entropy-based information increments associated with causal interactions. By decomposing these increments into unique, redundant, and synergistic causal contributions, SURD makes it possible to identify self-causation, disentangle redundant and synergistic collider effects, and quantify causality leaks from unobserved variables within a single framework. SURD quantifies causality by decomposing the information gained about a future state into redundant, unique, and synergistic components, providing a novel analytical perspective for causal discovery in multivariate systems (as shown in Fig. 4).

For a set of time-dependent variables $Q = [Q_1(t), Q_2(t), \dots, Q_N(t)]$, the goal is to quantify the causal influence of each variable on a future target variable Q_j . Specifically, the entropy of the future target is decomposed as

$$H(Q_j^+) = \sum_{i \in \mathcal{C}} \Delta I_{i \rightarrow j}^R + \sum_{i=1}^N \Delta I_{i \rightarrow j}^U + \sum_{i \in \mathcal{C}} \Delta I_{i \rightarrow j}^S + \Delta I_{\text{leak} \rightarrow j} \quad (36)$$

where $H(Q_j^+)$ denotes the Shannon entropy of the future target variable. $\Delta I_{i \rightarrow j}^R$ refers to the overlapping information shared among multiple variables. $\Delta I_{i \rightarrow j}^U$ captures the information contributed solely by a single variable, which cannot be obtained from any other individual variable, $\Delta I_{i \rightarrow j}^S$ is the causal contribution of the joint information between variables to the target variable, and $\Delta I_{\text{leak} \rightarrow j}$ represents causal leakage of unobserved variables, which is an information gap generated by unobserved variables in the system.

SURD enhances interpretability by explicitly separating causal influences into synergistic, unique, and redundant modes, and further quantifies the contribution of unobserved factors through the causality leak term.

Based on the idea of information decomposition, partial entropy decomposition (PED) has been proposed [95]. The motivation for exploring PED stems from the limitations of the standard network approach, which relies on pairwise links and is unable to assess higher-order interactions directly. The core idea of PED is to decompose the joint entropy of a system into a set of nonnegative atoms that describe redundant, unique, and synergistic interactions.

4.1.3. Transfer entropy and extensions

Information-theoretic approaches quantify causality by assessing whether knowledge of the past of one process reduces the uncertainty of prediction of another, thereby indicating a directed flow of information. On the basis of this idea, several

methods have been developed. Transfer entropy (TE) provides the foundational measure of directional information flow; its extensions, such as partial transfer entropy (PTE) and symbolic or rank-based variants, address indirect causality and non-stationarity, and the more recently proposed embedded entropy (EE) incorporates state–space geometry to capture nonlinear dynamical causality.

Transfer entropy (TE) [11], introduced by Schreiber, evaluates whether the future state of a target variable becomes more predictable if the source variable's past is incorporated [96]. Formally, for time series x_t and y_t , the transfer entropy from X to Y is defined as

$$T_{X \rightarrow Y} = \sum p(y_{t+1}, x_t, y_t) \log \frac{p(y_{t+1}|x_t, y_t)}{p(y_{t+1}|y_t)} \quad (37)$$

where $p(y_{t+1}, x_t, y_t)$ is the joint probability distribution of Y_{t+1} , X_t and Y_t , $p(y_{t+1}|x_t, y_t)$ is the conditional probability distribution of Y_{t+1} given X_t and Y_t , and $p(y_{t+1}|y_t)$ is the conditional probability distribution of Y_{t+1} given Y_t only.

A positive value of TE indicates directional causal influence, and, under Gaussian assumptions, TE is equivalent to Granger causality [97]. However, TE cannot distinguish direct from indirect influences, motivating the development of PTE. By conditioning on additional variables Z , PTE isolates direct information transfer [98], thereby removing common-cause and mediated effects,

$$PTE_{X \rightarrow Y|Z} = \sum p(y_{t+1}, x_t, y_t, z_t) \log \frac{p(y_{t+1}|x_t, y_t, z_t)}{p(y_{t+1}|y_t, z_t)} \quad (38)$$

where $p(y_{t+1}, x_t, y_t, z_t)$ is the joint probability distribution of Y_{t+1} , X_t , Y_t and Z_t , $p(y_{t+1}|x_t, y_t, z_t)$ is the conditional probability distribution of Y_{t+1} given X_t , Y_t and Z_t , and $p(y_{t+1}|y_t, z_t)$ is the conditional probability distribution of Y_{t+1} given Y_t and Z_t .

To improve robustness in noisy and nonstationary settings, symbolic extensions such as symbolic transfer entropy (STE) [99,100] and its multivariate form partial symbolic transfer entropy (PSTE) [101] replace amplitude values with ordinal patterns, while rank-based variants, such as transfer entropy on rank vectors (TERV) [102] and its multivariate extension called partial transfer entropy on ranks (PTERV) [101], achieve similar goals through rank transformations.

More recently, embedded entropy (EE) [28] has been proposed to measure the dynamical causality between two variables. EE measures the mutual information between the delay-embedded state of the effect variable and the nearest neighbors of the cause variable in the embedding space.

Formally, for variables X and Y , EE from Y to X is defined as

$$EE_{Y \rightarrow X} = MI(Y^*, X^{NN}) \quad (39)$$

where $Y^* = (y(t-1), \dots, y(t-p))^T$ is the delay-embedded state of Y , X^{NN} are the nearest neighbors of $X(t) = (x(t), \dots, x(t-p))^T$ in the embedding space, and $MI(\cdot, \cdot)$ denotes mutual information.

They also extended EE to conditional EE (cEE) [28] to detect conditional or direct causality,

$$cEE_{Y \rightarrow X|Z} = CMI(Y^*, X^{NN}|Z^{NN}) \quad (40)$$

where Z^{NN} represents the nearest neighbors of the conditioned variable $Z(t) = (z(t), \dots, z(t-p))^T$ in the embedding space, and $CMI(\cdot, \cdot)$ is conditional mutual information.

EE and cEE can be used not only to solve nonlinear causality problems but also to solve inseparability problems and reduce scale bias in numeric calculations.

4.1.4. Interventional dynamical causality

A novel framework named interventional dynamical causality (IntDC) [103] has been proposed to measure causality at the interventional level only from observational data without the need for additional interventions. The core idea is that without performing actual interventions, causality can be quantified by how perfectly a small change in the effect variable can reconstruct an instantaneous tiny perturbation that originated in the cause variable.

For two variables X and Y and their time-delay states $\mathbf{x}_t$ and $\mathbf{y}_t$, $\tilde{\mathbf{x}}_t$ and $\tilde{\mathbf{y}}_{t+1}$ are the states after the intervention. Here, we consider the intervention to be sufficiently small, and we can obtain $\delta \mathbf{x}_t = \tilde{\mathbf{x}}_t - \mathbf{x}_t$ and $\delta \mathbf{y}_{t+1} = \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_{t+1}$. If IntDC holds from the cause x to the effect y , the effect state $\delta \mathbf{y}_{t+1}$ contains complete information of $\delta \mathbf{x}_t$ in a neighborhood of $\mathbf{y}_{t+1}$. The interventional causal state $\delta \mathbf{x}_t$ should be reconstructed by $\delta \mathbf{y}_{t+1}$. It can be formulated by this smooth projection operator that $\delta \mathbf{x}_t = \nabla_{\mathbf{y}} \mathbf{F}(\mathbf{y}_{t+1}) \cdot \delta \mathbf{y}_{t+1}$. The interventional embedding entropy (IEE) has been proposed to measure the average information retention or reconstructability from the effect $\delta \mathbf{y}_{t+1}$ to the cause $\delta \mathbf{x}_t$,

$$IEE[X \rightarrow Y] := CMI(\delta \mathbf{x}_t, \delta \mathbf{y}_{t+1} | \mathbf{y}_{t+1}) \quad (41)$$

where CMI is conditional mutual information. The causal interpretability of IntDC arises from its adherence to the interventional criterion defined by the IEE principle, ensuring that the inferred causal relationships are consistent with the standards of interventional causality.

4.2. Neural information-theoretic approaches

Traditional information-theoretic estimators often struggle with high-dimensional data, non-stationarity, and the curse of dimensionality. Neural information-theoretic approaches have been proposed to enhance these estimators by leveraging the representational and optimization power of deep learning. In this section, we review such neural approaches, such as neural mutual information estimators, neural transfer entropy, information decomposition via machine learning, Dynamic Maxent across Entwined Scales, neighbor cross-mapping entropy and their extensions.

4.2.1. Neural mutual information estimators

To address the challenge of estimating MI in high-dimensional continuous data, neural mutual information estimators have been suggested. The core idea is to transform MI into an optimizable lower bound via neural networks, thereby enabling scalable, data-driven estimation of complex dependencies. The main approaches are Mutual Information Neural Estimation and InfoNCE.

The Mutual Information Neural Estimation (MINE) framework [104] parameterizes a function class using a deep neural network and constructs an optimizable lower bound for MI by leveraging dual representations of the Kullback–Leibler (KL) divergence (such as the Donsker–Varadhan representation) [105–108],

$$I_{\theta}(X, Z) = \sup_{\theta \in \Theta} \mathbb{E}_{\mathbb{P}_{XZ}} [T_{\theta}] - \log \left(\mathbb{E}_{\mathbb{P}_X \otimes \mathbb{P}_Z} [e^{T_{\theta}}] \right) \quad (42)$$

where $\sup$ represents the supremum, and T_{θ} is a neural network function. $\mathbb{P}_{XZ}$ and $\mathbb{P}_X \otimes \mathbb{P}_Z$ are the joint distribution and the product of marginal distributions, respectively. By maximizing this lower bound, MINE directly estimates MI from samples, avoiding the difficulties of high-dimensional density estimation encountered by traditional methods.

To further improve estimation stability and scalability, Oord [109] proposed the InfoNCE. Using the InfoNCE loss function, the model learns to distinguish positive samples drawn from the conditional distribution $p(x_{t+k}|c_t)$ from negative samples drawn from the marginal distribution $p(x_{t+k})$, thereby maximizing the mutual information between the latent representation and future signals. The loss function can be expressed as

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right] \quad (43)$$

where $f_k(x, c)$ is a scoring function used to measure compatibility between x and c , $f_k(x_{t+k}, c_t)$ is the score assigned to the true future sample x_{t+k} given the current context c_t , and $f_k(x_j, c_t)$ denotes the scores for all candidate samples in the contrastive set, including both the positive and negative examples.

We can evaluate the mutual information between the variables c_t and x_{t+k} as follows:

$$I(x_{t+k}, c_t) \geq \log(N) - \mathcal{L}_N \quad (44)$$

with the evaluation becoming tighter as the sample size N increases.

These methods not only significantly improve the feasibility of MI estimation in high-dimensional settings but also remain closely linked to classic information theory: their optimization objectives correspond to theoretical lower bounds of MI, ensuring interpretability and statistical consistency.

4.2.2. Neural transfer entropy

Transfer entropy is a classic technique for quantifying information flow and causal relationships. However, traditional estimation methods are hindered by the curse of dimensionality and non-stationary data. Neural transfer entropy provides more efficient solutions, relying on the powerful representational capacity of neural networks and a variational estimation framework. Two representative methods are the intrinsic transfer entropy neural estimator and transfer entropy estimation via transformers.

To distinguish between intrinsic and synergistic contributions in information transfer, the intrinsic transfer entropy neural estimator (ITENE) [110] decomposes transfer entropy [111]. First, transfer entropy is defined as conditional mutual information,

$$TE_{X \rightarrow Y}(m, n) \triangleq I(X_{t-m}^{t-1}, Y_t | Y_{t-n}^{t-1}) \quad (45)$$

where $X_{t-m}^{t-1} = (X_{t-m}, \dots, X_{t-1})$ and $Y_{t-n}^{t-1} = (Y_{t-n}, \dots, Y_{t-1})$ denote the past m and n samples of time sequences X_t and Y_t , respectively. An estimator of the intrinsic transfer entropy (ITE) can be defined as

$$\widehat{ITE}_{X \rightarrow Y}(m, n) = \inf_{\varnothing} \left(\hat{I}(X^-; Y^0, \bar{Y}^-) - \hat{I}(X^-; \bar{Y}^-) \right) \quad (46)$$

where $\inf$ represents minimization, and $\hat{I}$ represents mutual information estimation under the variational parameters $\varnothing$. $\hat{I}$ is computed based on MINE. $X^- = X_{t-m}^{t-1}$ represents the past m samples of X , and $Y^0 = Y_t$ is the present value of Y . $\bar{Y}^-$ approximates the past n samples of Y . The estimator uses a block coordinate descent approach that alternately optimizes the mutual estimation network and the variational parameters.

Another method is transfer entropy estimation via transformers (TREET) [112]. This method utilizes the Donsker–Varadhan variational representation and the attention mechanism for estimation. Transfer entropy is expressed as the difference of two conditional KL divergences,

$$TE_{X \rightarrow Y}(m) = D_{\text{KL}}\left(P_{Y_m|Y^{m-1}X^m} \parallel \tilde{P}_Y \middle| P_{Y^{m-1}X^m}\right) - D_{\text{KL}}\left(P_{Y_m|Y^{m-1}} \parallel \tilde{P}_Y \middle| P_{Y^{m-1}}\right) \quad (47)$$

where $D_{\text{KL}}(\cdot \parallel \cdot | \cdot)$ denotes the conditional KL divergence, Y_m represents the current values, and Y^{m-1} and X^m are the past $m-1$ and m samples of Y and X , respectively. $P_{Y_m|Y^{m-1}X^m}$ is the conditional probability of Y_m given Y^{m-1} and X^m . $\tilde{P}_Y$ is the absolutely continuous reference distribution. $P_{Y_m|Y^{m-1}}$ is the conditional probability of Y_m given Y^{m-1} . $P_{Y^{m-1}X^m}$ and $P_{Y^{m-1}}$ are the marginal probability distributions of $Y^{m-1}X^m$ and Y^{m-1} , respectively. By using a neural network g to approximate the variational function, the empirical estimator is obtained,

$$\hat{TE}_{X \rightarrow Y}(D_n; m) = \sup_{g_{xy} \in \mathcal{G}_{\text{ctf}}^{XY}} \hat{D}_{Y_m|Y^{m-1}X^m \parallel \tilde{Y}_m}(D_n, g_{xy}) - \sup_{g_{xy} \in \mathcal{G}_{\text{ctf}}^Y} \hat{D}_{Y_m|Y^{m-1} \parallel \tilde{Y}_m}(D_n, g_{xy}) \quad (48)$$

where D_n is the observed dataset, and $\mathcal{G}_{\text{ctf}}^{XY}$ and $\mathcal{G}_{\text{ctf}}^Y$ are sets of causal transformer architectures. $\hat{D}_{Y_m|Y^{m-1}X^m \parallel \tilde{Y}_m}(D_n, g_{xy})$ and $\hat{D}_{Y_m|Y^{m-1} \parallel \tilde{Y}_m}(D_n, g_{xy})$ are the estimates of $D_{\text{KL}}\left(P_{Y_m|Y^{m-1}X^m} \parallel \tilde{P}_Y \middle| P_{Y^{m-1}X^m}\right)$ and $D_{\text{KL}}\left(P_{Y_m|Y^{m-1}} \parallel \tilde{P}_Y \middle| P_{Y^{m-1}}\right)$, respectively. Therefore, TREET is ultimately computed as the difference between the solutions of two optimization problems.

4.2.3. Information decomposition via machine learning

Identifying the microscopic variations that drive macroscopic behavior is a central challenge. While mutual information links variables across scales, directly characterizing information distribution across many observables leads to a combinatorial explosion, rendering traditional methods infeasible for large-scale systems. Information decomposition via machine learning was proposed by Murphy and Bassett [113]. Based on the distributed information bottleneck (DIB) principle from information theory [114,115], the method decomposes information in complex systems by applying lossy compression to system measurements, preserving the information most relevant to macroscopic behavior while discarding the rest. The optimization is formalized by the minimization of the DIB Lagrangian,

$$\mathcal{L}_{\text{DIB}} = \beta \sum_{i=1}^N I(U_i; X_i) - I(\mathbf{U}; Y) \quad (49)$$

where X_i is the i th microscopic observable variable. U_i is a lossy compressed representation of X_i , Y is the target macroscopic behavior, and β is the bottleneck's strength.

The interpretability of this approach is embodied in the information allocation principle. The optimization process assigns an amount of information $I(U_i; X_i)$ to each observable variable X_i , and the magnitude of this value directly reflects the variable's contribution or relative importance in predicting the macroscopic behavior Y .

4.2.4. Dynamic Maxent across Entwined Scales

Traditional approaches have limitations: macro-constraints based on maximum entropy assume stationarity and fail to capture transients, while bottom-up mechanistic models often ignore the feedback from macro-states to micro-dynamics. Dynamic Maxent across Entwined Scales (DyMES) [4] provides a hybrid framework for modeling complex systems with bidirectional micro-macro causality, aiming to unify these two paradigms above. The core innovation of DyMES is in its hybrid architecture. It integrates top-down maximum entropy principles with bottom-up mechanistic dynamics, thereby explicitly describing the real-time, bidirectional constraints between macro-states and micro-distributions.

The probability distribution $R(x|X)$ over micro-variables x is derived at each moment by maximizing the Shannon information entropy, subject to constraints imposed by the instantaneous values of the macro-state variables X and their time derivatives dX/dt . This yields a time-dependent distribution,

$$R(x|X) = \frac{e^{-\sum_{\mu} \lambda_{\mu} f_{\mu}(x, X)}}{Z} \quad (50)$$

where $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_{2m})$ are Lagrange multipliers obtained by solving the constraint conditions, and Z is a normalization constant. A key technical advancement is the derivation of differential equations for λ_{μ} , such as $\sum_{v=1}^{2m} \text{Cov}(f_{m+i}, f_v) \frac{d\lambda_v}{dt} = 0$. This avoids numerical optimization at each time step and significantly improves computational efficiency.

The interpretability of DyMES is reflected in providing an explicit mathematical mapping for the abstract concept of bidirectional micro-macro causation. DyMES clearly separates and quantifies the two causal pathways. Downward causation is directly embodied and can be tested via the explicit dependence of transition functions $f_{\mu}(x, X)$ on the macro-state variables X . Upward causation is captured by the definition of state variables X themselves as averages over the micro-variable distribution.

4.2.5. Neighbor cross-mapping entropy and extensions

Neighbor cross-mapping entropy (NME) [116] was introduced by Li in 2023 as a novel information-theoretic criterion for inferring causal gene regulatory networks (GRNs) from both steady-state and time-series single-cell RNA-seq data.

The core idea of NME is to quantify continuous causality between variables by measuring the functional continuity of their neighbor-based mappings across varying neighborhood sizes. Specifically, if a causal relationship exists from variable X to Y , then the mapping from the nearest neighbors of X to the corresponding values in Y should exhibit continuity, which is captured by the neighbor cross-mapping mutual information.

Formally, for variables X and Y with n samples, NME from X to Y is defined as,

$$NME_{X \rightarrow Y} = MI(Y(X^{NN}); Y^{NN}) = \sum_{i=1}^n P(Y(x_i^{NN}), y_i^{NN}) \ln \left(\frac{P(Y(x_i^{NN}), y_i^{NN})}{P(Y(x_i^{NN})) P(y_i^{NN})} \right) \quad (51)$$

where $Y(X^{NN})$ denotes the cross-mapped values of the nearest neighbors of X in Y , and Y^{NN} represents the nearest neighbors of Y . This formulation allows NME to capture directional causality in a manner that is both mathematically rigorous and consistent with the concept of functional dependency.

The authors further extended NME to conditional NME (cNME) to distinguish direct from indirect causality in the presence of confounding variables.

$$cNME_{X \rightarrow Y|Z} = MI(Y^{NN}; Y(X^{NN})) - \frac{MI(Z^{NN}; Z(X^{NN}))}{H(Z^{NN})} * MI(Y^{NN}; Y(Z^{NN})) \quad (52)$$

NME and cNME are calculated based on mutual information of multiple variables, which can be estimated based on the kNN estimation. They also developed scNME [116], a single-cell-specific framework that infers GRNs, computes GRN activity via network entropy, and performs cell state identification and clustering based on the resulting activity matrix.

The interpretability of NME is based on the concept of continuous causality, where the magnitude of the NME value directly reflects the strength of causal influence between variables. The cNME approach eliminates indirect causal effects, enabling the identification of direct causal relationships between variables.

The interpretability of classic information-theoretic approaches is reflected in formulas with clear mathematical meanings and statistical significance of their magnitudes. In contrast, the interpretability of neural information-theoretic approaches is reflected in the in-depth analysis of the internal information structure of complex systems. Classic information-theoretic approaches focus on quantifying causality between variables, while neural approaches focus on the information composition and flow patterns within complex systems. The advantage of neural information-theoretic approaches is in their powerful ability to handle high-dimensional, nonlinear and complex systems, with better scalability. The disadvantage is that the computational complexity is usually higher, and interpretability depends to some extent on the model itself.

4.3. Classic graphical causal approaches

Classic graphical causal approaches essentially aim to explicitly represent causal mechanisms as graphical representations of multivariate time series, intervention mechanisms or other scenarios [8,117–119], thereby facilitating intuitive views of cause–effect relationships and enabling structure-based identification strategies. In this section, we review representative families, including PCMCi and its variants, and structural causal models (SCM). Note that “PC” is an abbreviation derived from the names of Peter and Clark [120] and that MCI is the momentary conditional independence test [8,48]. These representative families are particularly effective for synthesizing their theoretical foundations and methods to address the core challenges of graphical causal approaches.

4.3.1. PCMCi and its variants

PCMCi and its variants are constraint-based causal discovery methods tailored for large-scale time series. PCMCi, which was proposed by Runge [48], is a two-step causal discovery approach for multivariate time series. The main motivation of PCMCi is to overcome the challenges associated with autocorrelated, high dimensional and dimensional and nonlinear time series data. It combines a PC-style condition-selection step to prune unlikely parents with an MCI testing step [48] that evaluates the remaining candidate edges.

The core structure of the PC algorithm begins with a fully connected undirected graph. Subsequently, the graph is “thinned” by interactively removing links based on conditioning sets of growth. Note that casual directions for the contemporaneous links can be inferred through this process. Building on this, a two-stage procedure integrating the PC algorithm and the MCI test is followed. This algorithm first executes a condition-selection step (using the PC algorithm), and then the MCI test. (1) Specifically, in a time series graph $\hat{G}$, condition selection based on the PC algorithm can yield an estimate $\hat{P}(X_t^i)$ of parents $\mathcal{P}(X_t^i)$ for all variables $X_t^i \in (X_t^1, \dots, X_t^N)$. (2) Subsequently, these parents can be used as a condition in the MCI stage, which can assess all variables pairs $(X_{t-\tau}^i, X_t^j)$ with $i, j \in \{1, \dots, N\}$ and time lags $\tau \in \{1, \dots, \tau_{max}\}$, and determine whether a causal link $X_{t-\tau}^i \rightarrow X_t^j \in \hat{G}$ exists with the following formula:

$$MCI: X_{t-\tau}^i \not\perp\!\!\!\perp X_t^j | \hat{P}(X_t^i) \setminus \{X_{t-\tau}^i\}, \hat{P}_{P_X}(X_{t-\tau}^i) \quad (53)$$

where $\dagger$ denotes the absence of a conditional independence, and $\hat{p}_{P_X}(X_{t-\tau}^i) \subseteq \hat{p}(X_t^i)$ represents the P_X strongest parents related to the sorting operation in algorithm S1 [48]. This parameter is an optional choice. One may limit the maximum number of parents used for $\hat{p}(X_t^i)$; here, however, we impose no such restriction. If $\tau = 0$, one can also consider contemporaneous links as undirected.

Note that conditional independence tests can be implemented in both condition selection and MCI steps. These can use different test statistics; common choices include partial correlation (ParCorr) for linear–Gaussian relations, GP-based or distance-covariance approaches (e.g., GPCD), and nonparametric conditional mutual information estimators (CMI, e.g., CMlknn) [48].

PCMCI's strength is in accounting for serial dependence and yielding interpretable lagged causal graphs. However, the results remain conditional on the observed variables and assumptions. Several PCMCI variants have also been proposed, such as the WT-PCMCI algorithm [121] integrated wavelet transform (WT), and PCMCI+ [122], Regime-PCMCI [123] and Latent PCMCI [124] algorithms.

4.3.2. Structural causal models

Structural causal models (SCMs) were first described by Pearl [22,118], and are a foundational framework in causal inference that combines a graph model and structural equations. The motivation of SCMs is to overcome the inherent limitation of traditional statistical methods in distinguishing correlation ship from causation. It provides a formal and mathematical framework that supplies principles for representing interventions and counterfactuals. SCMs make the assumptions required for identification explicit, enabling rigorous causal reasoning and estimation in situations where randomized experiments are infeasible.

The core of the SCMs is in their integration of graphical modeling [125], the potential outcome framework (Neyman–Rubin) [126,127], decision analysis [118,128], and structural equation modeling approaches [118] to causation. Note that SCMs enable the evaluation of causal queries encompassing the effects of potential interventions and counterfactual reasoning [22].

Represented as a 4-tuple $\langle \mathcal{V}, \mathcal{U}, \mathcal{F}, P(\mathcal{U}) \rangle$, a typical (semi-Markovian) SCM [118,119] consists of four parts: a set $\mathcal{V}$ of observed (endogenous) variables with each V_i taking values in a domain $\mathcal{M}[V_i]$; the unobserved variable set $\mathcal{U}$ (each $U_i \in \mathcal{U}$ having a finite domain $\mathcal{M}[U_i]$); a set of modular, measurable functions $\mathcal{F}$ so that each endogenous variable is determined by $V_i = f_i(pa(V_i), U_i)$ where $pa(V_i) \in \mathcal{V} \setminus V_i$ represents are the endogenous direct causes of V_i ; and a probability distribution $P(\mathcal{U})$ over the exogenous variables.

Consequently, for any given SCM the modeler separates variables into endogenous and exogenous. Each V_i comes from a deterministic (or measurable) function f_i that combines the influences of its endogenous parents $pa(V_i)$ with an exogenous shock U_i to yield V_i . Importantly, an SCM corresponds to an underlying acyclic graph, where the set of vertices is given by the endogenous variables in $\mathcal{V}$ and the set of edges is given by the structural functions.

An SCM framework is typically described as having three layers, namely, associational, interventional, and counterfactual layers, which generally correlate to the common human behaviors of seeing, acting, and picturing, respectively [118,129]. Layer 1 deals with only “observational” factual data. Layer 2 encodes information about the hypothetical outcomes of activities or what might happen if an intervention were carried out. The final layer asks counterfactual questions about what would have happened if an intervention had been carried out, given that something else actually happened (perhaps at odds with the hypothetical intervention).

This section offers a comprehensive and rigorous overview of classic graphical causal approaches, including PCMCI and its variants, and SCMs, effectively synthesizing their theoretical foundations and methods in addressing the core challenges of causal inferences. By the representative algorithms offer a new view of field evaluation and the graphs that can support analysis of data characterized by non-linearity, non-stationarity, and high-dimensional features.

4.4. Neural graph-based approaches

While classic graph-based approaches to causality including PCMCI and its variants, and SCMs can help capture nonlinear causality under conditions such as large-scale and contemporaneous dependencies, performance of such techniques may present many challenges in terms of greater generalizability and robustness. Recent studies integrate machine learning models to propose graph-based approaches to causality with AI extensions, such as a causally informed neural network model combining PCMCI and a deep neural network, and a knockoff conditional mutual information model integrated with SCM and k-nearest neighbors (kNN) methods.

4.4.1. Causally-informed neural network model

Machine learning-based PCMCI framework has emerged as the most popular learning framework for analyzing graph-structured data in causal research. This section introduces a causally-informed neural network (CINN) model integrated PCMCI model and a deep learning model [130]. The aim of the proposed model that integrates causal domain insights is to address key limitations of conventional machine learning algorithms, predictions of which rely primarily on correlational patterns between predictors and predictands rather than causal mechanisms.

The core framework of the CINN model integrates existing subgrid parameterization models [131], a PCMCI model [48], and a deep neural network. The Superparameterized Community Atmosphere Model (SPCAM), composed of the climate

model CAM and the storm-resolving SP, is used to learn subgrid-scale targets (Y) from environmental conditions (X) provided by CAM. This study compares two parameterization strategies. In the causally-informed approach, PCMI is applied to the fully connected input space to remove spurious, nonphysical predictors and links; then 65 separate single-output neural networks are constructed and trained, each on the causal subset relevant to a specific output (resulting in varying input vector lengths). Importantly, the non-causal neural network always retains the entire 94-dimensional environmental input vector. Finally, both the non-causal and the causally-informed neural parameterizations of SPCAM physics are coupled into CAM.

This model, combining a causality discovery algorithm and a deep learning model, is a good framework for application in the climate field. The advantages of this approach are demonstrated: (a) causality discovery eliminates spurious correlations, (b) causality drivers are robust across colder and warmer climates [132], and (c) the CINN model faithfully reproduces the high-resolution model's climate and preserves a number of convection-associated features.

4.4.2. Knockoff conditional mutual information

From the perspective of intervention, knockoff conditional mutual information (KOCMI) [133] was proposed to capture interventional causality at the SCM framework's second level. This approach detects direct interventional causality between variable pairs from either time-independent data or time series data without needing prior network structure information, and it can infer a network with loops unlike many existing methods that require an acyclic structure.

A knockoff operation [134] is first carried out on the cause variable X within KOCMI, yielding the corresponding knockoff variable $\tilde{X}$. $\tilde{X}$ is generated to have the same distribution as X while maintaining correlations with all variables except for Y . As a result, the virtual intervention achieved by replacing X with $\tilde{X}$ does not change the inter-variable relationships involving X , apart from those with Y (Fig. 5A).

Second, KOCMI evaluates the difference between the original distribution of the effect variable Y , given the original cause variable X , and the alternative distribution of Y , given the on the knockoff variable Y (Fig. 5B). A notable difference signifies that the cause variable impacts the effect variable's distribution, thus denoting a causal link between the two. Conversely, if the difference is not significant, it shows that there is no causal relationship. Assume three random variables X , Y , and Z , with $X = x_i$, $Y = y_i$, and $Z = z_i$ for $i = 1, 2, \dots, n$. The calculation of the original conditional mutual information (CMI or I) proceeds as follows:

$$CMI(X, Y|Z) = \sum_z \sum_x \sum_y P(x, y, z) \log \frac{P(y|x, z)}{P(y|z)} \quad (54)$$

where $P(\cdot)$ represents the probability distribution associated with the corresponding random variable. By employing the knockoff approach, we can derive a knockoff variable $\tilde{X}$ from X , where $\tilde{x}_i$ signifies the knockoff data for the i th instance of X . The variable Z serve to block both the backdoor paths from X and the indirect path connecting X to Y ; and then, conditioning on Z measures that the estimated causality is direct. Consequently, the interventional direct causality from X to Y can be measured by the following formula:

$$D_{X \rightarrow Y} = CMI(\tilde{X}, Y|Z) - CMI(X, Y|Z) = \sum_z \sum_x \sum_y \sum_{\tilde{x}} P(x, \tilde{x}, y, z) \log \frac{P(y|\tilde{x}, z)}{P(y|x, z)} \quad (55)$$

If $D_{X \rightarrow Y}$ is not equal to zero, then we have $P(Y|\tilde{X}, Z) \neq P(Y|X, Z)$, suggesting a direct causal link from X to Y .

In addition, kNN is applied for estimating the conditional mutual information $\widehat{CMI}(X, Y|Z)$ or $\widehat{CMI}(\tilde{X}, Y|Z)$. The KOCMI algorithm is then utilized to determine the interventional direct causality from X to Y . This inference involves three main steps presented in Fig. 5C: (1) Generate M knockoffs $\tilde{X}^{(m)}$ for $m = 1$ to M using X as well as all other variables Z excluding Y , (2) Assess the difference between original CMI and that of the knockoff as follows $\hat{D}_{X \rightarrow Y} = \widehat{CMI}(\tilde{X}, Y|Z) - \widehat{CMI}(X, Y|Z)$ for all generated knockoffs, and (3) Check if $\hat{D}_{X \rightarrow Y}$ significantly deviates from zero by performing a permutation test with hypotheses $H_0: D_{X \rightarrow Y} = 0$ and $H_1: D_{X \rightarrow Y} \neq 0$ using M estimations, and estimate the causality strength as $cs = |\hat{D}_{X \rightarrow Y} / sd(\hat{D}_{X \rightarrow Y})|$. In small-sample settings, causal inference can be made by combining this measure with the criterion $cs > \text{threshold}$.

5. Nonlinear state-space theory approaches

5.1. Classic nonlinear state-space approaches

Classic nonlinear state-space approaches infer causality from the geometry of reconstructed dynamics. State-space methods for causal reconstruction do not rely on model fitting or parameter estimation. Instead, they assess causal influence by quantifying structural and geometric transformations between reconstructed dynamical trajectories. In this section, we review representative families, including neighborhood-based measures, cross-mapping criteria, interventional dynamical causality, and recurrence and symbolic dynamics approaches. These methods are particularly effective for deterministic systems in which system components interact through weak or moderate coupling.

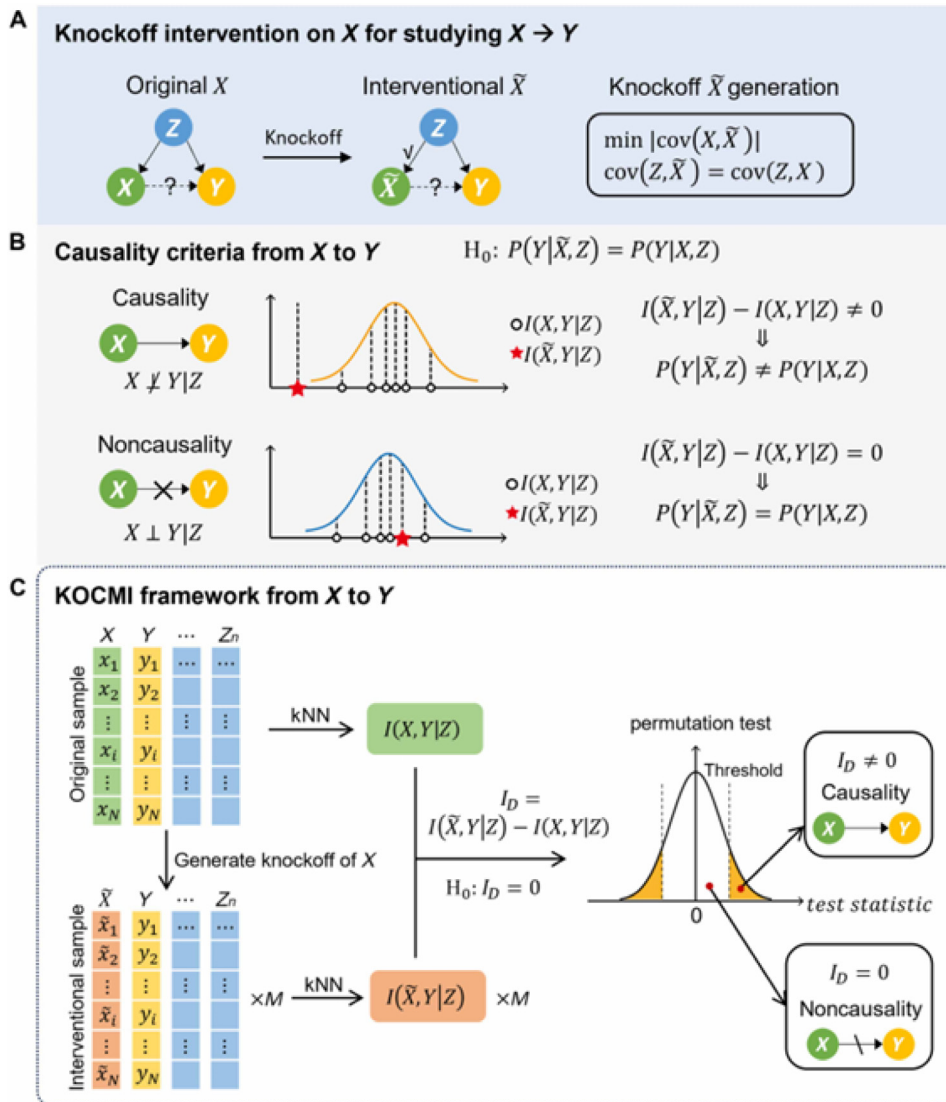


Fig. 5. The framework of KOCMI. (A) Knockoff intervention on X for studying $X \rightarrow Y$: generate a knockoff variable $\tilde{X}$ for the cause variable X . Knockoff method can ensure that correlation between the generated knockoff $\tilde{X}$ and other variables is the same as the correlation between the original variable X and other variables. B distribution invariance is used as the criterion for causal inference: if the conditional distribution of the outcome/effect variable changes after knockoff intervention, then a causal relationship exists; if the conditional distribution remains unchanged, then no causal relationship is present. C the computational framework of KOCMI: first, generate M knockoffs of the cause variable. Then, estimate the conditional mutual information (CMI or I) between the outcome variable and the original cause variable, as well as CMI between the outcome variable and the knockoff variable using the kNN method. A hypothesis test is used to determine whether the CMI changes before and after the intervention. If there is a change, then a causal relationship is inferred; if there is no change, then no causal relationship exists. note: I mean CMI, and I_D means $D_{X \rightarrow Y}$.

Source: Reprinted from Ref. [133].

5.1.1. Neighborhood-based measures

Early approaches were built on nearest-neighbor correspondences, such as topological causality (TC) [135] based on a local expansion of reconstructed state space mappings. Here we introduce two representative approaches, mutual nonlinear prediction and nonlinear interdependence. Importantly, these approaches are designed to distinguish dynamical causality from synchronization [136].

Schiff [137] proposed a mutual nonlinear prediction framework that captures both dynamical interdependence and coupling directionality among nonlinear systems. In a unidirectionally coupled system where information flows only from X to Y , the reconstructed attractor in Y represents the combined dynamics of $X \oplus Y$, while the reconstructed attractor in X represents only the dynamics of X alone. Based on the Takens' theorem, X will be predictable from Y since $X \oplus Y$ contains all dynamic properties of X .

Specifically, local zero-order maps are chosen as a nonlinear mutual predictor to detect the existence of a continuous map between X and Y . For each index point, the phase space vectors $\mathbf{x}_i$ and $\mathbf{y}_i$ have their respective k nearest neighbors, denoted by $\{\mathbf{x}_{nnx_i}^j\}_{j=1,\dots,k}$ and $\{\mathbf{y}_{nny_i}^j\}_{j=1,\dots,k}$. They also have mutual nearest neighbors, defined as points that share the same time index of the true nearest neighbor of the other system. With a prediction horizon of $H \gg 0$ time steps ahead, the prediction of $\mathbf{x}_i$ based on its k mutual nearest neighbors $\{\mathbf{x}_{nny_i}^j\}_{j=1,\dots,k}$ is the average translation given by $\langle v \rangle_{nnx(y)} = \frac{1}{k} \sum_{j=1}^k \mathbf{x}_{(nny_i+H)}^j$. Then, we can obtain the normalized prediction error $\Delta_{nnx(y)}$ between the actual $\mathbf{x}_{(i+H)}$ and the average predicted translation $\langle v \rangle_{nnx(y)}$ as the discriminating statistic for this method,

$$\Delta_{nnx(y)} \equiv \frac{\langle \varepsilon_{nnx(y)} \rangle_{\text{rms}}}{\langle \varepsilon_{\text{mean}} \rangle_{\text{rms}}} = \frac{|\mathbf{x}_{(i+H)} - \langle v \rangle_{nnx(y)}|}{|\mathbf{x}_{(i+H)} - \langle \mathbf{x} \rangle|} \quad (56)$$

where rms denotes root mean square, and $\varepsilon_{\text{mean}}$ represents the prediction error for the mean of the time series. A value of $\Delta_{nnx(y)} < 1$ indicates improved predictability of X from Y , implying the underlying unidirectional causal influence from X to Y . The interpretability of this method arises from linking causality to improved predictability, given mutual neighbors.

Arnhold [138] proposed a geometric measure of nonlinear interdependence by quantifying how nearest-neighbor distances in one reconstructed phase-space map to distances in another. We can obtain two reconstructed phase spaces $\mathbf{x}_n = (x_n, \dots, x_{n-(m-1)\tau})$ and $\mathbf{y}_n = (y_n, \dots, y_{n-(m-1)\tau})$ for variables X and Y . For each reconstructed state $\mathbf{x}_n$, the squared mean Euclidean distance to its k closest neighbors, $R_n^{(k)}(X)$, and the conditional mean squared Euclidean distance, conditioned on the closest neighbor times in Y , $R_n^{(k)}(X|Y)$, are defined as

$$R_n^{(k)}(X) = \frac{1}{k} \sum_{j=1}^k (\mathbf{x}_n - \mathbf{x}_{r_{n,j}})^2 \quad (57)$$

$$R_n^{(k)}(X|Y) = \frac{1}{k} \sum_{j=1}^k (\mathbf{x}_n - \mathbf{x}_{s_{n,j}})^2 \quad (58)$$

where $r_{n,j}$ and $s_{n,j}$ are the time indices of the k neighbors of $\mathbf{x}_n$ and $\mathbf{y}_n$, respectively. If X and Y are completely independent, then $R_n^{(k)}(X|Y) \gg R_n^{(k)}(X)$. Accordingly, the local and global interdependence measures $S_n^{(k)}(X|Y)$ and $S^{(k)}(X|Y)$ are introduced as

$$S_n^{(k)}(X|Y) \equiv \frac{R_n^{(k)}(X)}{R_n^{(k)}(X|Y)} \quad (59)$$

$$S^{(k)}(X|Y) \equiv \frac{1}{N} \sum_{n=1}^N S_n^{(k)}(X|Y) = \frac{1}{N} \sum_{n=1}^N \frac{R_n^{(k)}(X)}{R_n^{(k)}(X|Y)} \quad (60)$$

We can deduce that X and Y are independent within the limits of accuracy if $S^{(k)}(X|Y) \approx (k/N)^{2/D} \ll 1$, where D denotes the effective dimensionality. In addition, $S^{(k)}(X|Y) \gg (k/N)^{2/D}$ indicates a statistical dependence of X on Y , although this does not necessarily establish a causal link. The interpretability of this method arises from linking causality to the systematic contraction of local neighbor distances when conditioned on the other system.

5.1.2. Convergent cross mapping

Convergent cross mapping methods are based on the idea that causal influence leaves an informational imprint on the response dynamics. A range of extensions have been developed to accommodate various situations. Representative variants include Extended Convergent cross mapping, Multiview Convergent cross mapping, Segment convergent cross mapping, and modified formulations of irregular spike training.

Sugihara [12] introduced Convergent cross mapping (CCM) as a framework for detecting causal interactions in deterministic dynamical systems. CCM is based on the principle that if two variables X and Y originate from the same deterministic dynamical system, their lag-embedded reconstructions should share information about the other. Specifically, we consider the shadow manifolds M_x and M_y formed by the reconstruction of X and Y . The causality from X to Y is inferred by evaluating whether and to what extent the nearest neighbors in the shadow manifold M_y reliably identify the corresponding nearest neighbors in M_x . Here, we show the cross-mapped estimation of $Y(t)$ by identifying its $E+1$ neighbors.

$$\hat{Y}(t) | M_x = \sum w_i Y(t_i), i = 1 \dots E+1 \quad (61)$$

$$w_i = u_i / \sum u_j, j = 1, \dots, E+1 \quad (62)$$

$$u_i = \exp \{-d[\mathbf{x}(t), \mathbf{x}(t_i)] / d[\mathbf{x}(t), \mathbf{x}(t_1)]\} \quad (63)$$

where w_i is a weight based on the Euclidean distance between $\mathbf{x}(t)$ and its i th nearest neighborhood in M_x , $Y(t_i)$ is the contemporaneous observed value of Y at time index t_i , and $\hat{Y}(t)$ is the cross-mapped estimate of Y at time t .

An essential feature of CCM is its convergence criterion, reflecting the notion that as the length of a time series increases, the attractor trajectory gradually becomes denser. Consequently, the nearest neighbors in the reconstructed

phase space draw closer, reducing estimation errors. By examining how predictive skill evolves with increasing library sizes, CCM effectively discriminates true causal relationships from spurious correlations. The accuracy of this mapping from one manifold to another quantifies the strength and directionality of the causal relationship. However, the detected causal relationships are scale-dependent. Different scales and data resolutions will lead to different causal results [139].

Moreover, extensions have been proposed to evaluate the cross-map between two variables. Ma [140] proposed a CME score to replace Pearson correlation in CCM. To explicitly account for temporal delays, Ye [141] introduced extended convergent cross mapping (ECCM), which incorporates time delays into the cross-mapping framework to identify delayed interactions among time series. It relies on the principle that true causal directions manifest as negative lags because the response variable can better predict the past values of the driving variable rather than future values. Hence, we should cross map from the current state of the system $\mathbf{y}(t)$ to another observation variable $\mathbf{x}(t + l)$, where l indicates the lag value.

To handle high-dimensional systems effectively, Ye [142] proposed a multiview embedding approach. By constructing multiple low-dimensional shadow manifolds from different subsets of variables and aggregating their cross-map skills, one can reduce the required data length and improve robustness to noise. The basic idea is straightforward: each reconstruction derived from a specific subset of variables yields a distinct perspective of the system, capturing complementary information. Distinct prediction errors are observed from each view for short and noisy time series. By integrating multiple complementary views, a more complete representation of system dynamics can be obtained to improve predictive performance because such a representation covers the full range of system behavior.

Duan [143] reported that CCM might produce misleading results in systems that exhibit twofold symmetry. Segment convergent cross mapping (sCCM) solved this problem by projecting the inversion-symmetric shadow manifold onto its fundamental domain. The influence of symmetry was eliminated by splitting the symmetric shadow manifolds into two sub-manifolds. In addition, a modified CCM method [144] was proposed to detect causality in the point processes with irregular sampling intervals. The basic idea was to reconstruct the state space from interspike intervals and then apply the CCM principle.

5.1.3. Partial cross mapping

The causality transition effect may lead to a failure to distinguish direct causality from indirect causality. To solve this problem, Leng [145] proposed the partial cross mapping (PCM) method to isolate the true direct causal relationships from indirect influence. It extends the CCM framework by integrating partial correlation coefficients into the cross-mapping process. PCM can effectively filter out indirect causal influences and achieve high accuracy in detecting the direct causal interactions within multivariate time series. PCM assesses the correspondence between a variable and its cross-map estimate while conditioning on information from an additional variable, as illustrated in Fig. 6.

In the simplest case involving three variables X , Y and Z , the mutual cross mapping (MCM) index ϱ_C is used to measure the correlation coefficient. The correlation index ϱ_D is introduced to measure the directional causal relationships from X to Y conditioned on the indirect causation through Z .

$$\varrho_C = \left| \text{Corr} \left(X, \hat{X}^Y \right) \right| \quad (64)$$

$$\varrho_D = \left| \text{Pcc} \left(X, \hat{X}^Y | \hat{X}^{\hat{Z}^Y} \right) \right| \quad (65)$$

where $\hat{X}^Y$ represents the mapping from Y and $\hat{X}^{\hat{Z}^Y}$ represents the mapping from $\hat{Z}^Y$. Corr denotes the full correlation, and $\text{Pcc}(\cdot, \cdot | \cdot)$ is the partial correlation coefficient between the first two variables conditioned on the third. The indirect causal influence can be eliminated because we can regard ϱ_D as the projection of ϱ_C onto the information space orthogonal to indirect information $\hat{X}^{\hat{Z}^Y}$. The causal links among three variables X , Y and Z can be detected by the order of the correlation index among ϱ_C , ϱ_D and the empirical threshold T . The conditions $\varrho_C \geq \varrho_D \geq T$, $\varrho_C \geq T \gg \varrho_D$, and $T > \varrho_C \geq \varrho_D$ correspond to three cases: a direct causal link from X to Y , a single indirect causal link, and the absence of a causal link.

The PCM framework naturally extends to networked systems comprising an arbitrary number of interacting variables, denoted by $X, Y, Z^1, \dots, Z^s$, where $s \geq 2$. Their partial correlation coefficient can be formalized as

$$\varrho_{D_1} = \left| \text{Pcc} \left(X, \hat{X}^Y | \{ \hat{X}^{\hat{Z}^{iY}} | i = 1, \dots, s \} \right) \right| \quad (66)$$

where ϱ_{D_1} is a first-order measure for distinguishing a direct link from an indirect causal link from X to Y .

5.1.4. CCM extensions across domains

Beyond the standard time-domain setting, CCM has been generalized to spatial data and frequency domains in these two directions. First, it has been generalized from purely temporal trajectories to spatial or spatiotemporal data. Representative examples include multispatial CCM and geographic CCM. Second, CCM has been extended from the time domain to the frequency domain, with Frequency-Domain CCM being an example.

In recent years, causal inference methodologies have evolved beyond purely temporal frameworks to encompass spatially cross-sectional datasets. A significant challenge in this domain arises if individual time series are remarkably short, even though spatial replication is abundant. To address this, multispatial CCM [146] has been proposed, combining the principles of CCM and dewdrop regression. Its core innovation entails leveraging the information from spatial

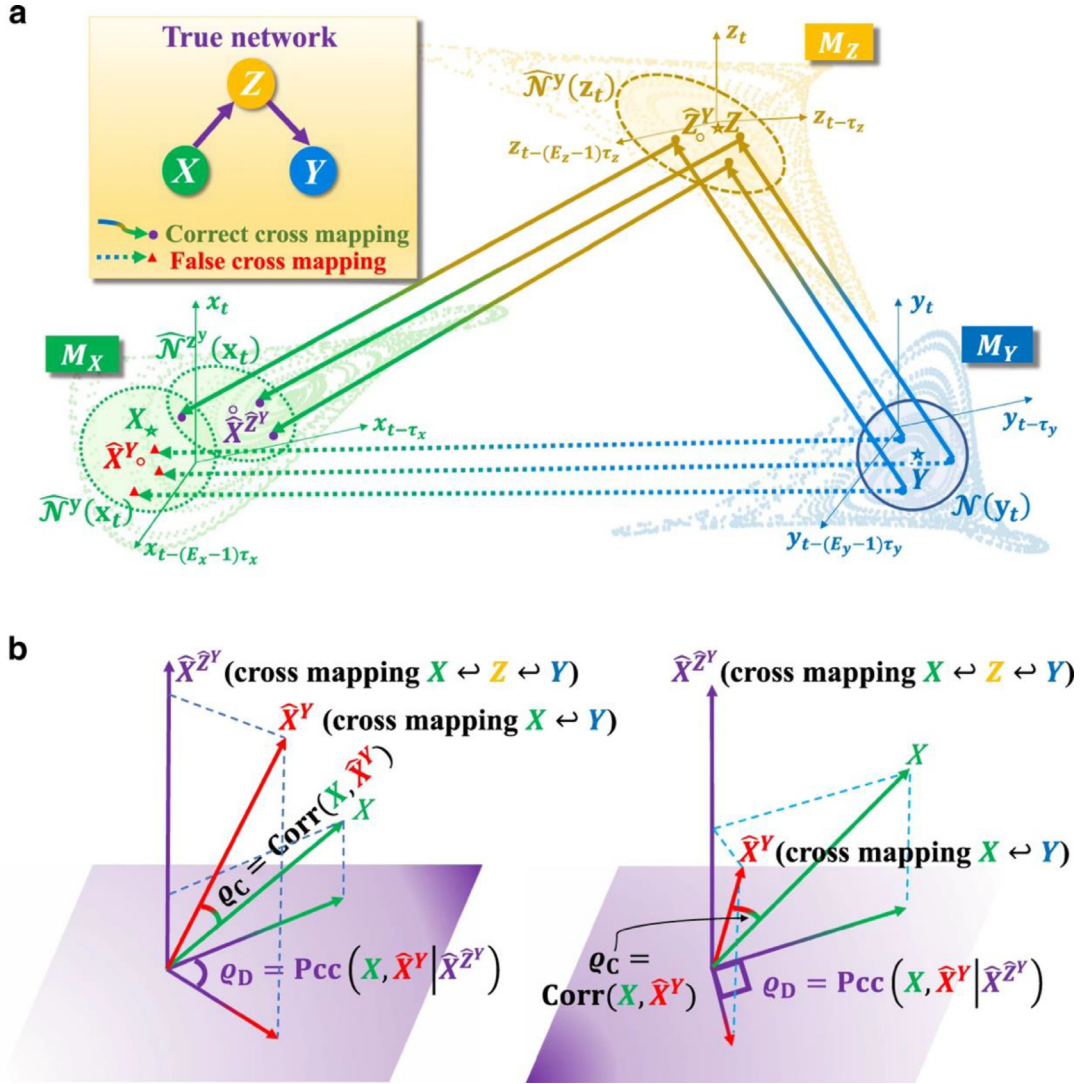


Fig. 6. Basic principles of the PCM framework. a For the illustrative setting of three variables interacting in a unidirectional causal chain, the MCM method maps $\mathcal{N}(y_t)$ to the left circled region $\hat{\mathcal{N}}^y(x_t)$ in M_X , where the estimated $\hat{X}^y$ is close to the true X , denoting full causal information from X to Y and leading to erroneous identification of the indirect causal link as the direct link. b For our proposed PCM method, partial correlation coefficient between X and $\hat{X}^y$ is calculated by conditioning on the information about $\hat{X}^z^y$, which is mapped from $\mathcal{N}(y_t)$ through M_Z and then to the right circled region $\hat{\mathcal{N}}^{z^y}(x_t)$ in M_X in a, denoting indirect information flow. Geometrically, q_C corresponds to the cosine of the angle between X and $\hat{X}^y$ in the entire space, while q_D is the projection of q_C onto the subspace orthogonal to $\hat{X}^z^y$. Because $q_C \geq q_D$, the example in a corresponds to the sketch on the right side of b, where the projection is close to the right angle, leading to a near-zero value of q_D . Source: Reprinted from Ref. [145].

replicates to compensate for the limited length of individual time series. Sampling from spatially replicated observations allows the identification of states that exhibit comparable historical dynamics. The expected dynamics over subsequent time steps can be estimated by a weighted average of them. This approach has been proven to successfully detect causality even when only five observations are available.

Unlike spatial replication, the geographical convergent cross mapping (GCCM) model [147] was specifically proposed to bridge the Earth's spatial cross-sectional data with the methodologies of complex systems. It enables effective identification and quantification of causal spatial interactions, significantly extending the applicability of CCM-based approaches from a spatial perspective. Spatial distribution and cross-sectional data encode their spatial interactions, can also be treated as snapshots of the dynamical system. For spatial datasets, we construct an embedding $\psi(x, s)$ by taking the values at one specific unit and its neighbors as a set of observation functions:

$$\psi(x, s) = \langle h_s(x), h_{s(1)}(x), \dots, h_{s(L-1)}(x) \rangle \quad (67)$$

where s is the focal unit currently under investigation, $s(i)$ is its i th order of spatial lag, and $h_s(x)$ and $h_{s(i)}(x)$ are their observation functions. Spatial lag refers to a shift to spatial neighbors from the focal spatial unit.

Based on the generalized embedding theorem, the shadow manifold M_X and M_Y for two spatial variables X and Y can be reconstructed by $\psi(x \text{ or } y, s)$. Setting one point $\psi(x, s)$ in M_X as a focal state, we can identify its close states and their corresponding states in M_Y based on the mutual spatial locations. For each state x , the value of y can be inferred from the neighborhood states identified in M_X . The cross-mapping prediction can be defined as

$$\hat{Y}_s|M_X = \sum_{i=1}^{L+1} (w_{si} Y_{si}|M_X) \quad (68)$$

where $\hat{Y}_s|M_X$ is the prediction value of y , s represents a spatial unit at which the value of Y needs to be predicted. L is the number of embedding dimensions, si indicates the spatial unit used in this prediction, and Y_{si} is the real observation value at si .

CCM relies on temporal dynamics and does not explicitly exploit spectral information. To bridge this gap, several extensions have been proposed to adapt CCM to the frequency domain. One such approach is the spectral measure named Frequency-Domain CCM (FDCCM) [148], which replaces the time-delayed embedding in CCM with spectrogram representations. Specifically, each state in the attractor manifold M_X is defined as $\mathbf{x}_t = \{x_t^{f_1}, x_t^{f_2}, \dots, x_t^{f_D}\}$, where the superscripts represent D different frequency bands of equal bandwidth. The basic idea is to extend CCM to the frequency domain by applying coherence metrics for evaluation.

Another extension is the Cross-Mapping Coherence (CMC) method [149], designed to detect frequency-specific causal relationships. CMC evaluates causality by calculating the coherence between the predicted and real time series instead of the coefficient in CCM. The coherence at a specific frequency f signifies the magnitude of the average cross-spectral density between the real x and predicted values $\hat{x}$. It can be defined as

$$\text{coh}_{x\hat{x}}(f) = \frac{|S_{x\hat{x}}(f)|}{\sqrt{S_{xx}(f) S_{\hat{x}\hat{x}}(f)}} \quad (69)$$

where $S_{xx}(f)$ and $S_{\hat{x}\hat{x}}(f)$ represent spectral power densities of the respective signals.

To detect the causality between different frequencies, the CEEMDAN-based cross-frequency symbolic convergent cross mapping (CEEMDAN CF-SCCM) [150] has been proposed to identify the phase–amplitude interactions of irregular oscillation signals in complex systems. This method can effectively reduce the sensitivity to noise.

5.1.5. CCM extensions for causality quantification

Here, we mainly introduce two techniques for detecting causality: cross-mapping cardinality (CMC) and Continuity Scaling.

CMC is a model-free approach grounded in the concavity of intersectional cardinality (IC) [151] and proposed to detect nonlinear causal interactions among variables. This method quantifies the dynamical causality by evaluating the concave-down characteristic of the IC curve against the neighbor size. It avoids the linear operation adopted by existing methods and is capable of handling nonlinear and indirect interactions while simultaneously addressing the issue of non-separability. To detect a causal link from Y to X , we define IC between the nearest neighbors $Y^{rNN}(t)$ and the corresponding states after cross-mapping $Y(X^{kNN}(t))$ as $IC(r) = |Y(X^{kNN}(t)) \cap Y^{rNN}(t)|$ to quantify the continuity of a dependent function from X to Y . It follows that

$$Y(X^{kNN}(t)) = \{Y(t_i^x), i = 1, 2, \dots, k\} \quad (70)$$

$$Y^{rNN}(t) = \{Y(t_i^y), i = 1, 2, \dots, r\} \quad (71)$$

Based on this concept, IC continuity is defined to detect dynamical causality. Specifically, given a integer $k > 0$, one can find an integer $r \in [k, N]$ such that $Y(X^{kNN}(t)) \subseteq Y^{rNN}(t)$ or $IC(r) = k$. IC continuity is defined as $1/r$. We consider two contradictory hypotheses, H_1 which assumes a causal influence from y to x , and H_0 which assumes no such causal relationship. Under H_1 , the k nearest neighbors of $X(t)$, denoted by $X^{kNN}(t)$, have been mapped onto M_Y . If there exists a causal link, the corresponding points of them, denoted by the set $Y(X^{kNN}(t))$, are expected to lie in the neighborhood of $Y(t)$. As the size r increases from 0 to N , $IC(r)$ converges quickly to k , and we can find that $AUC(H_1) > 0.5kN$. Based on the difference between H_1 and H_0 , the CMC framework quantifies causal strength (dividing by kN) through a normalized strength measure, defined as $2(AUC(H_1) - 0.5)$. The stronger the IC continuity is, the closer the cross-mapping neighbors $Y(X^{kNN}(t))$ are to $Y(t)$.

Further geometric perspectives have also been explored in the reconstructed space. The recognition that genuine causal interactions manifest as a coherent geometric expansion in the reconstructed space led to the Continuity Scaling method [152] being proposed to quantify causality in complex systems. Instead of measuring cross-map smoothness or prediction skill, it directly quantifies causality by examining the scaling behavior associated with the continuity properties of the underlying dynamical system itself.

The core principle is based on the epsilon-delta definition of continuity: if variable Y causes X , a small change in $\ln \varepsilon_x$ will lead to the correspondingly direct change in δ_y , where ε_x and δ_y denote the neighborhood sizes of the resulting variable

X and the causal variable Y , respectively. This causality detection process can be described as a “first- ε -then- δ ” operation. By systematically varying the size of the neighborhood ε_x and identifying the necessary corresponding neighborhood size δ_y , the process reveals a linear scaling law between the average δ and the logarithm of ε . The slope of this linear scaling law serves as a quantitative index of causal strength.

Here we consider discrete-time dynamical systems $\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t, \mathbf{y}_t)$. If variations in Y induce changes in the value of $\mathbf{f}(\mathbf{x}, \mathbf{y})$ for any given X , there exists a directional causal influence from Y to X . For a given point $\mathbf{x}_g \in \mathcal{M}$, we define $\mathbf{x}_l \triangleq \mathbf{f}_{\mathbf{x}_g}(\mathbf{y}_p)$ as its image under the given function. Applying the logic statement of a continuous sufficiently smooth map $\mathbf{f}_{\mathbf{x}_g}(\cdot)$, for any neighborhood $\mathcal{O}(\mathbf{x}_l, \varepsilon)$ centered at $\mathbf{x}_l$ and of radius $\varepsilon > 0$, there exists a neighborhood $\mathcal{O}(\mathbf{y}_p, \delta)$ centered at $\mathbf{y}_p$ of radius $\delta > 0$, such that $\mathbf{f}_{\mathbf{x}_g}(\mathcal{O}(\mathbf{y}_p, \delta)) \subset \mathcal{O}(\mathbf{x}_l, \varepsilon)$. The neighborhood and its radius are defined as

$$\mathcal{O}(\mathbf{p}, h) = \{\mathbf{s} \in \ell | \text{dist}_\ell(\mathbf{s}, \mathbf{p}) < h\}, \mathbf{p} \in \ell, h > 0 \quad (72)$$

where $\text{dist}_\ell(\cdot, \cdot)$ represents a measure used to quantify the distance between two states in a specified manifold $\ell = \mathcal{M}$ or $\mathcal{N}$. Thus, once a neighborhood of the resulting variable $\mathbf{x}_l$ is specified, a corresponding neighborhood of the causal variable $\mathbf{y}_p$ can be identified to satisfy the above mapping.

5.1.6. Recurrence-based approaches

Initial recurrence-based approaches used recurrence plots to characterize a univariate recurrence structure and perform recurrence quantification analyses. They then were naturally extended to bivariate relationships, such as cross recurrence plots and joint recurrence plots. More recently, studies have moved toward examining multivariate and conditional relationships, such as conditional joint recurrence plots, the recurrence measure of conditional dependence, and shared-recurrence methods.

The concept of recurrence, first introduced by Poincaré in 1890 [153], describes a fundamental property of dynamical systems in which trajectories eventually return arbitrarily close to a previous state. A significant advance in visualizing recurrences from time series data occurred when Eckmann [154] proposed the concept of recurrence plots (RPs) in 1987. The RPs provide a two-dimensional graphical representation of the recurring states and visualizes the time-dependent behavior of a system's trajectory in the phase space. It does not depend on assumptions about the distributional properties or stationarity of the data and can be effectively applied to short datasets [44].

Following Takens' embedding theorem, the trajectory $\{\mathbf{x}_i\}_{i=1}^N$ of a time series can be reconstructed in its phase space. An RP is formally represented by a recurrence matrix $\mathbf{R}(\varepsilon)$ defined as

$$\mathbf{R}_{i,j}(\varepsilon) = \Theta(\varepsilon - \|\mathbf{x}_i - \mathbf{x}_j\|) \quad (73)$$

where $\|\cdot\|$ can be any norm, such as Manhattan, Euclidean, or maximum norm, $\Theta(\cdot)$ is the Heaviside function, and ε is a predefined threshold distance that determines whether the two states are close. $\mathbf{R}_{i,j}$ is a binary element where a value of 1 denotes a recurrence, and 0 denotes a lack of recurrence; these values are often visualized as black and white pixels, respectively.

In recent years, variations of RPs have been developed, such as global recurrence plot, corridor thresholded recurrence plot [155], perpendicular recurrence plot [156], iso-directional recurrence plot [157], and fuzzy recurrence plot [158]. Recurrence Quantification Analysis (RQA), first introduced by Zbilut and Webber [159,160], quantifies small-scale structures within RPs by analyzing their recurrence points, diagonal lines, and vertical lines.

Cross recurrence plots (CRPs) [161,162] compare the phase space trajectories of two different systems in the same phase space. The cross-recurrence matrix $\mathbf{CR}$ is defined based on the distance between $\mathbf{x}_i$ and $\mathbf{y}_j$,

$$\mathbf{CR}_{i,j} = \Theta(\varepsilon - \|\mathbf{x}_i - \mathbf{y}_j\|) \quad (74)$$

To identify the interrelationships between two physically different quantities, a geometric method based on the inter-system recurrence networks (IRN) [163] has been proposed to determine the coupling direction between two dynamical systems. The inter-system recurrence matrix is formulated as

$$\mathbf{IR}(\varepsilon) = \begin{pmatrix} \mathbf{R}^X(\varepsilon^X) & \mathbf{CR}^{XY}(\varepsilon^{XY}) \\ [\mathbf{CR}^{XY}(\varepsilon^{XY})]^T & \mathbf{R}^Y(\varepsilon^Y) \end{pmatrix} \text{ with } \varepsilon = \begin{pmatrix} \varepsilon^X & \varepsilon^{XY} \\ \varepsilon^{XY} & \varepsilon^Y \end{pmatrix} \quad (75)$$

$$\mathbf{A}(\varepsilon) = \mathbf{IR}(\varepsilon) - \mathbb{I}_N \quad (76)$$

where ε denotes the set of threshold distances, $\mathbb{I}_N$ is the N -dimensional identity matrix, $\mathbf{R}^X$ and $\mathbf{R}^Y$ are the recurrence matrices of X and Y , respectively, and $\mathbf{CR}^{XY}$ denotes the cross-recurrence matrix between dynamical systems X and Y .

The global cross-clustering coefficient c^{XY} and the cross-transitivity $\mathcal{T}^{XY}$ can then be defined,

$$c^{XY} = \left\langle \frac{1}{k_v^{XY}(k_v^{XY} - 1)} \sum_{p,q \in V_Y} A_{vp} A_{pq} A_{qv} \right\rangle_{v \in V_X} \quad (77)$$

$$\mathcal{T}^{XY} = \frac{\sum_{v \in V_X: p,q \in V_Y} A_{vp} A_{pq} A_{qv}}{\sum_{v \in V_X: p,q \in V_Y} A_{vp} A_{vq}} \quad (78)$$

The coupling direction can be inferred by comparing the qualitative behavior of $\mathcal{C}^{XY}$ vs. $\mathcal{C}^{YX}$ and $\mathcal{T}^{XY}$ vs. $\mathcal{T}^{YX}$.

Order recurrence plot (ORP) [164] has been proposed to visualize the dynamics of different dynamical systems by introducing symbolic dynamics. Here, the neighborhood is defined by the local order structure. ORP is a plot of a Boolean matrix of size $M \times N$, where M and N are the lengths of the order pattern series:

$$\mathbf{ORP}(t, t') = \begin{cases} 1, & \pi_x(t) = \pi_y(t') \\ 0, & \text{otherwise} \end{cases} \quad (79)$$

where $\pi_x(t)$ and $\pi_y(t')$ are the order patterns. If these two order patterns coincide, the phase space vectors $\mathbf{x}(t)$ and $\mathbf{y}(t')$ are defined as neighbors.

Joint recurrence plots (JRPs) [165] are a bivariate or multivariate extension of the recurrence plot, considering the simultaneous recurrence of states in multiple systems. It can be used to determine the weak coupling strengths between two different systems. The JRP matrix is defined by the joint probability that recurrences occur simultaneously in the respective phase spaces of the considered systems. The joint recurrence matrix is as follows:

$$\mathbf{JRP}_{i,j}^{X,Y} = \Theta(\varepsilon_x - \|\mathbf{x}_i - \mathbf{x}_j\|) \times \Theta(\varepsilon_y - \|\mathbf{y}_i - \mathbf{y}_j\|), i, j = 1, \dots, N \quad (80)$$

A recent study proposed Conditional Joint Recurrence plots (CJRP) [166] to identify delayed causal relationships between a driver and a target, formulated as follows:

$$\mathbf{CJRP}_{XY|Z} = \mathbf{JRP}_{XY} \odot (1 - \mathbf{JRP}_{ZY}) \quad (81)$$

Hence, the causal influence between X and Y is detected by filtering out the influence of Z on Y . This CJRP approach can be easily extended to more than three quantities by simply adding terms to the Hadamard product. CJRP has been proven to more effectively mitigate the influence of confounding factors compared to conditional transfer entropy (CTE) [167].

On the basis of JRP, joint order pattern recurrence plots (JOPRPs) [168] have been proposed to evaluate the relationships between different dynamical systems. Its core principle is to replace the RPs in JRPs with the ORP. The JOPRP matrix is defined as

$$\mathbf{JOPRP}_{i,j} = \mathbf{ORP}_{i,j}(m_X) \odot \mathbf{ORP}_{i,j}(m_Y) \quad (82)$$

where m_X and m_Y represent the embedding dimension of each system.

Considering the failure to identify the driver and the response of JRPs, the mean conditional probabilities of recurrence (MCR) [169] between two systems X and Y have been introduced,

$$\mathbf{MCR}(Y|X) = \frac{1}{N} \sum_{i=1}^N p(\mathbf{y}_i|\mathbf{x}_i) = \frac{1}{N} \sum_{i=1}^N \frac{\sum_{j=1}^N \mathbf{JRP}_{i,j}^{X,Y}}{\sum_{j=1}^N \mathbf{R}_{i,j}^X} \quad (83)$$

where $p(\mathbf{y}_i|\mathbf{x}_i)$ is an estimated conditional probability quantifying the likelihood that the trajectory of Y recurs to the neighborhood of $\mathbf{y}_i$, given that the trajectory of X simultaneously recurs to the neighborhood of $\mathbf{x}_i$.

The basic idea is that if X drives Y , the evolution of Y depends on the states of both X and Y , and the dimension of Y will be larger than that of X . Then, the probability $p(\mathbf{y}_i)$ that the trajectory of Y recurs to the neighborhood of $\mathbf{y}_i$ will decrease. Hence, $p(\mathbf{y}_i) < p(\mathbf{x}_i)$, and then $M_{CR}(Y|X) < M_{CR}(X|Y)$.

Inspired by the conditional mutual information, the researchers proposed the recurrence measure of conditional dependence (RMCD) [170] that could be applied to detect the causal dependencies among different variables. The index $\mathcal{J}_{RMCD}(X, Y|Z)$ can quantify the recurrence relation between X and Y given Z .

$$\mathcal{J}_{RMCD}(X, Y|Z) = \frac{1}{N} \sum_i \frac{1}{N} \sum_j \mathbf{R}_{ij}^{X,Y,Z} \times \log \left(\frac{\sum_j \mathbf{R}_{ij}^{X,Y,Z} \sum_j \mathbf{R}_{ij}^Z}{\sum_j \mathbf{R}_{ij}^{X,Z} \sum_j \mathbf{R}_{ij}^{Y,Z}} \right) \quad (84)$$

However, this measure's detection accuracy relies heavily on the chosen threshold parameter and is limited to analyzing only two variables. Thus, a symbolic approach to recurrence quantification analysis [171] was proposed to address these limitations and better detect causal coupling among multiple variables.

A physics-based unsupervised learning algorithm called shared recurrences (SHREC) [172] was suggested to address the fundamental problem of reconstructing an unobserved causal driving signal from observed time series. It is based on the core principle that simultaneous or shared recurrence events across multiple response time series contain sufficient information to reveal the structure of their common unobserved driving signal. Specifically, SHREC introduces a higher-order network embedding method to construct a more robust connectivity matrix and find the consensus recurrence graph. The spectral decomposition of the graph Laplacian is subsequently applied to identify the driving states.

5.1.7. Symbolic dynamics techniques

Symbolic dynamics extend state-space ideas by replacing continuous trajectories with discrete symbol sequences (e.g., ranks or ordinal patterns) so that causal influence can be assessed through pattern consistency. Following this perspective, a representative study proposes a rank-based statistic to test causality by the correspondence of neighbors. A complementary development is pattern causality that encodes local dynamics into symbolic signatures.

A measure based on rank statistics [173] was proposed in 2009 to detect directional couplings among time series. It detects directional causality by evaluating how consistently close states in the driven dynamics correspond to close states in the driving dynamics. Here, we denote by $\mathbf{x}_i$ and $\mathbf{y}_i$ the reconstructed states of time series X and Y at time i , respectively. The non-synchronizing couplings can be tested by several rank-based measures. Here, we present only one representative statistic defined as

$$G_i^k(X | Y) = \frac{1}{k} \sum_{j=1}^k g_{i, w_{i,j}} \quad (85)$$

$$A(X | Y) = \frac{1}{N} \sum_{i=1}^N \frac{G_i(X) - G_i^k(X | Y)}{G_i(X) - G_i^k(X)} \quad (86)$$

$$\Delta A = A(X | Y) - A(Y | X) \quad (87)$$

where, for each reconstructed state $\mathbf{x}_i$, $g_{i,j}$ represents the rank of the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ in the list of distances from $\mathbf{x}_i$ to all $\mathbf{x}_{j \neq i}$ sorted in the ascending order, and k is the number of nearest neighbors. $G_i(X) = \frac{N}{2}$ and $G_i^k(X) = \frac{k+1}{2}$ denote the mean and minimal mean rank, respectively. $G_i^k(X | Y)$ represents the Y -conditioned mean rank. The criterion is that $\Delta A > 0$ indicates a directional coupling from X to Y .

Stavros [174] developed a pattern causality (PC) method based on symbolic dynamics to detect hidden interdependencies in complex systems. The basic idea is that the influence from X to Y is quantified by comparing patterns from M_X 's current neighborhood to predict M_Y 's future pattern $\hat{P}_{Y_{t+h}}$. We can test X 's ability to predict Y 's values by encoding each time series into discrete symbols. We use pattern information to enable robust estimation of three categories, namely, positive, negative, and hidden causality.

For each state $\mathbf{x}_t$ in M_X , we find its $E + 1$ nearest neighbors with the projected time indices $t_{\mathbf{x}_1}, t_{\mathbf{x}_2}, \dots, t_{\mathbf{x}_{E+1}}$.

$$S_{t_n}^y = \left(\frac{Y(t_{\mathbf{x}_2}) - Y(t_{\mathbf{x}_1})}{Y(t_{\mathbf{x}_1})}, \dots, \frac{Y(t_{\mathbf{x}_{E+1}}) - Y(t_{\mathbf{x}_E})}{Y(t_{\mathbf{x}_E})} \right) \quad (88)$$

$$\hat{S}_{y_{t+h}} = \sum_{t_n=t_{\mathbf{x}_1}}^{t_{\mathbf{x}_{E+1}}} w_{t_n}^x S_{t_n}^y \quad (89)$$

$$P_{\mathbf{x}_t} = \text{signature}(S_{\mathbf{x}_t}) \quad (90)$$

where $Y(t_{\mathbf{x}_1})$ is the value at time $t_{\mathbf{x}_1}$ of time series Y , $w_{t_n}^x$ is the weight, $\hat{S}_{y_{t+h}}$ represents the vector of successive percentage changes of $\mathbf{y}_{t+h}$, and $P_{\mathbf{x}_t}$ is the current pattern of $\mathbf{x}_t$.

To test the hypothesis of causality, the predicted pattern $\hat{P}_{y_{t+h}}$ should be the same as the real pattern $P_{y_{t+h}}$. We should repeat this procedure for every state of M_X , and we can obtain a PC matrix, which is a 3D array that models the causal influence strength. The pattern's strength can be calculated as

$$PC[P_X, P_Y, t] = \frac{\|S_{y_{t+h}}\|}{\|S_{\mathbf{x}_t}\|} \quad (91)$$

We can obtain three types of causal influences, namely, positive, negative and dark, labeled $P(t)$, $N(t)$ and $D(t)$. These series indicate the influence strength at each time step.

5.2. AI-extended state-space approaches

While classic state-space approaches rely on time-delay embedding and neighbor mappings, their performance may present many challenges in cases with short or noisy observations, heterogeneity, temporal delays, missing values, irregular sampling or high-dimensional interactions. Recent developments integrate machine learning to propose extensions, such as cross-mapping smoothness, latent CCM, continuity scaling, cross-mapping cardinality, and RC-based dynamical causality.

5.2.1. Cross-mapping smoothness

Ma [175] developed the cross-mapping smoothness (CMS) algorithm specifically to improve performance in cases of limited data. By training a neural network to approximate the cross-mapping relationship from a minimal time series length, CMS measures the smoothness of the learned mapping between two observed variables as a proxy for causal strength, effectively extracting reliable signals even if CCM fails.

With the CCM method, the geometric property of mutual neighbors can be used to detect causality. To test the causality from X to Y , the map from nearest neighbors to mutual neighbors is defined as the cross-map $\Phi_{YX}: M_Y \rightarrow M_X$. Here, we observe that finding mutual nearest neighbors is equivalent to measuring the smoothness of the cross-map. If X causally influences Y , the cross-map Φ_{YX} is locally smooth around $\mathbf{y}_t$ so that it maps a neighborhood of $\mathbf{y}_t$ to a neighborhood of $\mathbf{x}_t$.

The smoothness of Φ_{YX} indicates the strength of the causality. Instead of finding nearest neighbors, we can approximate this smoothness by a neural network method.

Specifically, we can train the neural network N to estimate Φ_{YX} , where the input is the entirety of states of $\mathbf{y}_t$ on M_Y , and the output is the entirety of states of $\mathbf{x}_t$ on M_X .

5.2.2. Causality under invisible confounders

Here, we consider the possibility that many unobserved variables exist in complex systems that may mislead our causality detection results. To overcome this challenge, a causality under invisible confounders (CIC) framework [176] based on the orthogonal decomposition theorem was proposed, detecting causality by eliminating spurious links. The core idea is that the invisible confounder can be reconstructed by the shared latent subspace, and if all information of X is contained in Y , then we can determine that X has a causal effect on Y .

For two observed variables x_t and y_t , we can obtain their delayed embedding reconstructed states $\mathbf{x}_t$ and $\mathbf{y}_t$, and we can perform orthogonal decomposition mappings G_x and G_y for $\mathbf{x}_{t-1}$ and $\mathbf{y}_t$, respectively.

$$G_x(\mathbf{x}_{t-1}) = \{z_{x_{t-1}}, z_{xy_{t-1}} | z_{x_{t-1}} \perp z_{xy_{t-1}}\} \text{ and } G_y(\mathbf{y}_t) = \{z_{y_t}, z_{xy_{t-1}} | z_{y_t} \perp z_{xy_{t-1}}\} \quad (92)$$

where $\perp$ is the orthogonal symbol, and $z_{xy_{t-1}}$ is the shared space between $\mathbf{x}_{t-1}$ and $\mathbf{y}_t$.

If there exists a causal effect from X to Y , then the following criterion is satisfied:

$$\mathbf{x}_{t-1} = F(z_{xy}) \text{ or } \|z_x\| = 0 \Leftrightarrow X \rightarrow Y \quad (93)$$

The causality index is reconstructed by using the two subspaces $\|z_x\|$ and $\|z_{xy}\|$ as follows:

$$CIC_{X \rightarrow Y} = \frac{\|z_{xy}\|}{\|z_x\| + \|z_{xy}\|} \quad (94)$$

where $CIC_{X \rightarrow Y}$ is the dynamical causality from X to Y . If $CIC_{X \rightarrow Y} = 1$, there is causality from X to Y . If $CIC_{X \rightarrow Y} \in (0, 1)$, there exists a confounder of X and Y .

5.2.3. RC-based dynamical causality

In recent years, reservoir computing (RC) has proven highly effective for causality reconstruction. Duan [177] showed that a general reservoir network could rigorously serve as a high-dimensional embedding of the original input nonlinear dynamical system. From this point of view, RC provides an alternative to explicit phase space reconstruction, and causality can be detected by whether information about the driver is encoded in the response dynamics. Building on this principle, reservoir computing causality, reservoir cross mapping and intervened reservoir computing have been proposed to detect causal influence from many perspectives, such as prediction improvement, cross mapping learning, and counterfactual intervention.

To avoid the estimation accuracy sensitivity induced by the phase space parameters estimated by phase space reconstruction, the Reservoir Computing Causality (RCC) model [178] was proposed based on the ECCM principle to detect causality between time series. Specifically, RCC assumes that the predictability achieved by reservoir computing and traditional phase-space prediction models originates fundamentally from the forward transmission of information through time. Within this framework, we propose replacing “the phase space prediction model” with “the RC prediction model” (RCC), and we test whether RCC can also identify causal relationships. RCC is robust to noise and offers a higher computational efficiency.

The core idea entails transforming a geometric topology problem into a machine learning prediction problem to detect causality. The underlying reasoning is that if causal influence exists from X to Y , the future state of Y is encoded in the information of X . Then, we can estimate the future values of Y using the current value of X . Specifically, we can estimate the value of $Y(t + \tau)$ from $X(t)$. The RC computing framework used to accomplish this includes three layers: the input-to-reservoir layer, the reservoir neuron layer, and the reservoir-to-output layer. In addition, the prediction skill measures the intensity of causality.

Reservoir cross mapping method(RCM) [179] was proposed to replace linear estimation by integrating reservoir computing with mutual cross mapping. Instead of treating RC as a predictor, this method uses RC to fit the cross mapping and reconstruct the system's state space directly under the generalized embedding theorem. RCM achieves more precise identification and mitigates the effects of indirect causality. The causation from X to Y can be identified in two steps. First, time series x_t and y_t are fed into the reservoir, yielding generalized embeddings r_t^x and r_t^y of variables X and Y . Second, the embedding r_t^y is supplied to another reservoir in order to estimate the output r_t^x , i.e., defining a cross-mapping from the state space M_y to M_x , denoted as $\Phi_{yx}: M_y \rightarrow M_x$. The training loss is quantified using the normalized mean squared error and can indicate the causal strength. If causal interaction occurs from X to Y , the training loss is expected to be sufficiently low, indicating that the estimated $\hat{r}_t^x$ closely match r_t^x and that the cross-mapping Φ_{yx} exhibits smooth.

The intervened reservoir computing method [180] was proposed to detect the dynamical causality in time series. The basic idea is that if a causal link exists, eliminating the influence of one node should induce a change in the subsequent evolution of another node. Here, an RC neural network is used to construct a high-dimensional embedding of the target system (as shown in Fig. 7). The reservoir computing model is trained using observations of the target system and

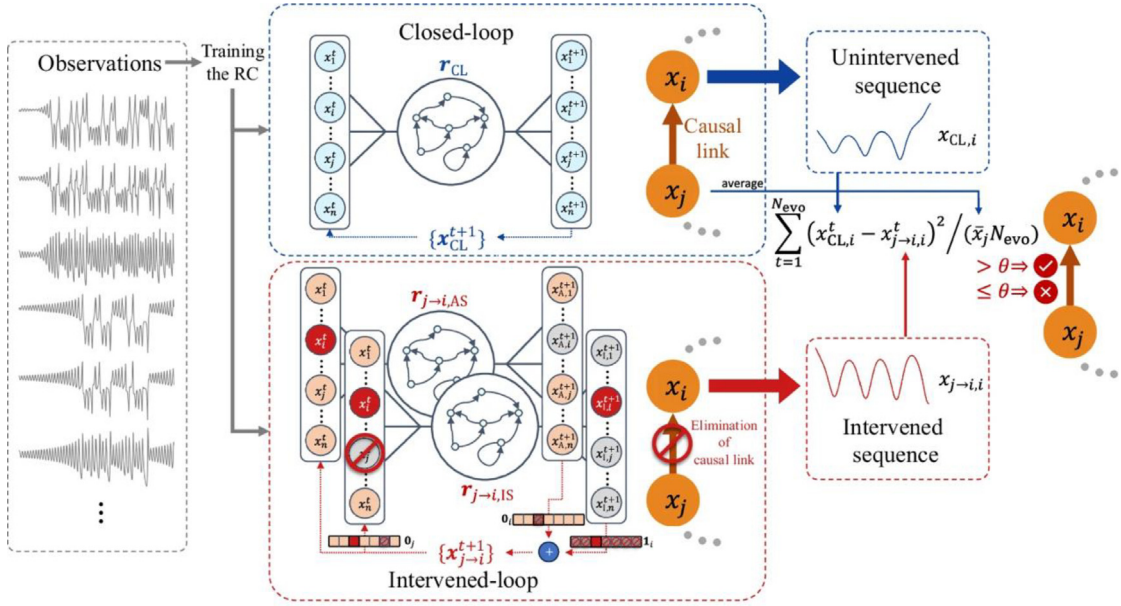


Fig. 7. The overall architecture of intervened reservoir computing. The high-dimensional observations of the target system are used to train the reservoir computing, which generates predicted trajectories through two different modes (closed-loop and intervened-loop), respectively. The causal links within the target system can be identified by comparing the differences in the predicted trajectories. The trajectory deviation indices calculated using Eq. (10) measures these differences. For better viewing, $x_{A,k}^t$ and $x_{i,k}^t$ are used here to denote the k -th dimension of the output of aiding structure and intervened structure, respectively.

Source: Reprinted from Ref. [180].

subsequently used to generate predicted trajectories through a self-iterative process referred to as the closed loop, together with sequences generated under intervened inputs using an intervened loop. If we test the causal influence from x_j to x_i , the intervened-loop is presented to simulate the elimination of the causal effect from x_j to x_i and to obtain the intervened sequences. Specifically, we generate the unintervened series $\{x_{CL}^t\}_{t=0}^{N_{evo}}$ using the closed-loop and the intervened series $\{x_{j \rightarrow i}^t\}_{t=0}^{N_{evo}}$ through the intervened-loop. We can identify the causal links by comparing the differences in the predicted trajectories, which we refer to as trajectory deviation indices, $\mathbf{TDI} = [\mathbf{TDI}_{j \rightarrow i}]$. $\mathbf{TDI}$ is calculated to minimize the influence of varying x_j values for all pairs of nodes (x_i, x_j) , $i, j = 1, 2, \dots, N$, $i \neq j$.

$$\bar{x}_j = \frac{1}{N_{evo}} \sum_{t=1}^{N_{evo}} x_{CL,j}^t \quad (95)$$

$$\mathbf{TDI}_{j \rightarrow i} = \frac{1}{\bar{x}_j N_{evo}} \sum_{t=1}^{N_{evo}} (x_{CL,i}^t - x_{j \rightarrow i,i}^t)^2 \quad (96)$$

where $x_{CL,k}^t$ and $x_{j \rightarrow i,k}^t$ denote the k th dimensions of x_{CL}^t and $x_{j \rightarrow i}^t$, respectively. We repeat the entire procedure for multiple RCs with different initial states.

Beyond reservoir computing-based approaches, recent AI extensions based on the state-space theory have been proposed to infer causality by learning latent dynamical representations directly. A latent CCM method [181] has been introduced to infer causal structures in short time series with missing or noisy values. It mainly focuses on learning a neural ODE latent process model, instead of relying on traditional time-delay embedding reconstruction, without the need to explicitly specify embedding dimensions and time delays. Causality is then inferred by examining the existence of a continuous map between these learned latent processes. For two time series x_t and y_t , we denote the resulting space of latent vectors from time series X on all segments by $\mathcal{H}_X$. If a continuous map exists between $\mathcal{H}_Y$ and $\mathcal{H}_X$, we conclude that X causes Y .

6. Governing equations discovery approaches

6.1. Differential equation-based causal network reconstruction

6.1.1. Classic mechanistic differential equation forms

Empirical dynamic data can be sampled in continuous or discrete time and may be well approximated by linear dynamics in some regimes while exhibiting strongly nonlinear behavior in others; thus, various classic mechanistic differential equation forms have been developed for causal network reconstruction. In this section we focus on three representative model families: continuous-time linear network models obtained by linearizing the dynamics around a working point, generalized Lotka–Volterra interaction networks widely used in ecological and microbiome studies, and ODE-based schemes such as gradient matching and modular response analysis, including improved MRA and retrograde insulation, which infer local response coefficients from steady-state or perturbed data.

A natural way to endow causal networks with mechanistic meaning is to start from explicit differential equations for node dynamics and treat network inference as an inverse problem: recover the coupling structure and parameters in $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \theta)$ from the observed trajectories. Here, θ collects reaction rates, interaction strengths and other physical parameters, and directed edges in the network correspond to nonzero partial derivatives $\partial f_i / \partial x_j$ or entries of the Jacobian matrix $J(\mathbf{x}; \theta)$ so that each edge can be interpreted directly in physical, biochemical or ecological terms. Compared with purely statistical association or Granger-type models, these mechanistic reconstructions provide greater interpretability and closer links to established domain equations, but do so at the cost of assuming reasonably correct functional forms and carefully addressing parameter identifiability and experimental design.

If the dynamics are close to a fixed point or operate in a narrow regime, it is often reasonable to approximate them by a linear system. This yields a simple, globally interpretable model in which every edge corresponds to an element of a coupling matrix.

A minimal example is a continuous-time linear network model, which approximates the dynamics near a working point as

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (97)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ encodes directed couplings among n nodes and $\mathbf{B}$ represents how external inputs drive the system. If time derivatives $\dot{\mathbf{x}}(t_k)$ can be estimated from the data, for example via finite differences, smoothing splines or Gaussian processes, observations at times $\{t_k\}_{k=1}^T$ may be assembled into matrices $\dot{\mathbf{X}}$, $\mathbf{X}$ and $\mathbf{U}$, collecting $\dot{\mathbf{x}}(t_k)$, $\mathbf{x}(t_k)$ and $\mathbf{u}(t_k)$, respectively, this yields a regularized regression formulation

$$\min_{\mathbf{A}, \mathbf{B}} \|\dot{\mathbf{X}} - \mathbf{A}\mathbf{X} - \mathbf{B}\mathbf{U}\|_F^2 + \lambda_A \|\mathbf{A}\|_1 + \lambda_B \|\mathbf{B}\|_1 \quad (98)$$

where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|_1$ promotes sparsity in the inferred coupling matrices. Nonzero entries A_{ij} then define directed edges $j \rightarrow i$, and their signs and magnitudes quantify effective strengths of linear influence. Variants of this continuous-time VAR/linearization approach have been used to reconstruct interaction graphs in systems biological and ecological systems, often with additional constraints such as stability, sign structure or conservation laws to regularize the inverse problem [182,183].

In ecological and microbiome communities, interactions are often approximately proportional to current abundances and can be decomposed into sums of self-growth terms and pairwise influences. Generalized Lotka–Volterra models encode this structure directly and thus provide a mechanistically meaningful template for network reconstruction.

The abundance $x_i(t)$ of species or taxon i evolves according to

$$\dot{x}_i(t) = x_i(t) \left(r_i + \sum_{j=1}^n a_{ij} x_j(t) \right) \quad (99)$$

where r_i is an intrinsic growth rate and a_{ij} quantifies how species j promotes ($a_{ij} > 0$) or inhibits ($a_{ij} < 0$) species i . Dividing both sides by $x_i(t)$ yields a regression form

$$\frac{\dot{x}_i(t)}{x_i(t)} = r_i + \sum_{j=1}^n a_{ij} x_j(t) + \varepsilon_i(t) \quad (100)$$

so that, after smoothing or differentiating the time series to obtain $\dot{x}_i(t)$, each row $(r_i, \{a_{ij}\}_j)$ can be estimated by a (possibly sparse) linear regression of $\frac{\dot{x}_i}{x_i}$ on the current state $\{x_j\}_j$. This strategy underlies many recent microbiome network studies, which integrate gLV dynamics with LASSO-type penalties, stepwise selection and bootstrap aggregation to obtain directed, signed interaction networks together with uncertainty estimates [184,185]. Structural identifiability analyses have clarified the conditions under which such time series contain enough information to uniquely determine the gLV parameters, and the circumstances when only combinations of interaction coefficients can be recovered [186].

Each coefficient has an immediate ecological meaning: r_i is the net growth rate of species i in isolation, while a_{ij} describes facilitation or competition between species j and i . Signs, magnitudes and even simple derived quantities such

as $-r_i/a_{ii}$ (the single-species carrying capacity) can be linked to ecological theory and independent experiments, making the resulting network highly interpretable.

A direct regression on numerically estimated derivatives can, however, be unstable under noise and irregular sampling because numerical differentiation amplifies measurement errors. Gradient-matching methods therefore introduce an intermediate smooth representation of the trajectories. The central idea is to interpolate each component $x_i(t)$ with a flexible smoother, most commonly based on splines or Gaussian processes, so that both $x_i(t)$ and an approximate derivative $\tilde{x}_i(t)$ are available in a closed form. The mechanistic model is then estimated by minimizing the discrepancy between the model-implied derivatives and the smoothed derivatives. It can be expressed as

$$\min_{\theta} \sum_{k=1}^T \|\tilde{\mathbf{x}}(t_k) - \mathbf{f}(\tilde{\mathbf{x}}(t_k), \mathbf{u}(t_k), \theta)\|^2 \quad (101)$$

where θ denotes the parameters of the dynamical system. This formulation can be extended to a Bayesian framework, in which θ and the smoothing hyperparameters are inferred jointly [187,188]. This approach avoids repeatedly solving ODEs inside an inner optimization loop. It has proven particularly effective in applications such as biochemical signaling networks, where the interaction structure is known a priori from biochemistry, while θ includes dozens of reaction rates. Extensions aimed specifically at network reconstruction adopt a comparative strategy, in which parametric gradient-matching models defined on a prescribed interaction topology are evaluated against nonparametric surrogate models. Model selection across candidate graphs is then employed to identify the edges required to explain the data [189].

Because gradient matching still fits an explicit ODE model, the inferred parameters retain their mechanistic meanings (reaction rates, interaction coefficients, etc.), while the smoothing step improves numerical robustness. This method is particularly appealing if the network structure is unknown but the functional form of f is reasonably specified, as it provides a compromise between fully numeric inverse problems and purely statistical black-box fits.

Modular response analysis (MRA) infers direct regulatory couplings between modules in biochemical networks from small steady-state perturbations without fitting the full dynamical system. The key idea is to treat each module as a black box characterized by local elasticities and control coefficients and to relate these local quantities to global steady-state responses that can be measured experimentally.

Consider a network with metabolite concentrations $s \in \mathbb{R}^m$ and reaction rates $v(s, p)$ driven by external parameters p . The mass-balance dynamics are

$$\dot{s}(t, p) = N v(s(t, p), p) \quad (102)$$

where $N \in \mathbb{R}^{m \times r}$ is the stoichiometric matrix. At a steady state s^* with $N v(s^*, p) = 0$, small parameter perturbations $p \rightarrow p + \delta p$ induce steady-state changes δs that can be summarized by global (logarithmic) response coefficients

$$R_{ij} = \frac{\partial \ln x_i}{\partial p_j} \quad (103)$$

where x denotes a suitable set of independent state variables obtained, if necessary, by eliminating conserved moieties.

The local logarithmic connections between modules are described by $r_{ij} = \frac{\partial \ln x_i}{\partial \ln x_j}$ with $r_{ii} = -1$ by convention. The classic MRA [190] shows that global and local quantities obey linear constraints

$$\sum_k r_{ik} R_{kj} = 0 \text{ for all } i, j \quad (104)$$

which can be solved for r , given measurements of R and mild structural assumptions (for example, that each communication species is directly controlled by one independent parameter). In this way, one reconstructs an effective Jacobian between modules directly from steady-state perturbation data.

In large networks only a small subset of “communication species” transmit signals between modules. Using the notation $x = (x^{\text{com}}, x^{\text{int}})$, the connection matrix r adopts a block structure in which the sub-block $r_{x^{\text{com}}, x^{\text{com}}}$ encodes direct inter-module regulation. The linear relations above then reduce to a small system involving only communication species,

$$R_{x^{\text{com}}}^p = -(r_{x^{\text{com}}, x^{\text{com}}})^{-1} r_{x^{\text{com}}}^p \quad (105)$$

so that the effective inter-module Jacobian $r_{x^{\text{com}}, x^{\text{com}}}$ can be reconstructed from steady-state responses of communication species to small perturbations. Each entry r_{ij} thus quantifies how a small relative change in the activity of j 's activity alters the steady-state activity of module i , after marginalizing over intramodular details; the resulting graph is mechanistic at the level of modules, even though many microscopic rate constants remain unidentifiable.

The classic MRA assumes that modules are insulated: perturbations propagate “downstream” through communication species but do not provide feedback upstream. In enzyme cascades this assumption is often violated. If a communication species is a free enzyme that forms complexes with downstream substrates, retrograde flux through enzyme–substrate sequestration can produce apparent upstream regulation and mutually inconsistent estimates of r across different perturbations.

To address this issue, retrograde-insulated MRA [191] redefines communication variables as weighted combinations of free enzymes and complexes, $x_i(a) = x_i^{\text{free}} + ac$. This method analyzes how their steady-state responses depend on the weight a . For the correct choice $a = a_{\text{opt}}$ determined by the underlying association and dissociation rates, changes in free enzymes and complexes cancel for purely retrograde effects, restoring the insulation condition and yielding a unique, physically consistent connection matrix. Practically, one measures the steady-state responses of x_i^{free} and c , reconstructs $r(a)$ from the same linear constraints $\sum_k r_{ik} R_{kj} = 0$, and scans a over a range. Spurious edges induced by sequestration change their sign and vanish near a_{opt} , whereas genuine regulatory edges maintain a stable sign. Taking $a = a_{\text{opt}}$ thus yields a retrograde-insulated connection matrix r^* that reflects true inter-module regulation.

Retrograde-insulated MRA therefore exposes how subtle biochemical mechanisms can bias naïve perturbation-based network reconstruction and provides a systematic correction that recovers the “true” inter-module regulatory graph from steady-state data alone, with each edge again interpretable as an effective logarithmic control coefficient between modules.

6.1.2. Sparse identification of nonlinear dynamics (SINDy)

In many applications, we still wish to work with continuous-time differential equations, but the data are strongly nonlinear and do not obviously fit into any of the classic templates in Section 6.1.1; thus, specifying an explicit functional form in advance is difficult. The mechanistic ODE models in Section 6.1.1 assume that the functional form $f(x, p)$ of the dynamics is known up to a moderate number of parameters. In many complex systems this assumption is unrealistic: we may trust that the dynamics are low-dimensional and smooth, but we do not know a priori which nonlinearities are present. Sparse identification of nonlinear dynamics (SINDy) addresses this gap by learning both the structure and the parameters of explicit ODEs directly from time series [13,192]. This section reviews the SINDy formulations most relevant to network reconstruction. We first outline regression-based SINDy in a generic nonlinear function library (as shown in Fig. 8), then discuss implicit extensions for rational dynamics, and finally consider mass-action, neural and Bayesian formulations that encode chemical structure and uncertainty while preserving reaction-level interpretability.

The central assumption is that, in an appropriate function space, the true vector field uses only a small subset of possible terms. From a network perspective, each component equation $\dot{x}_i = f_i(x)$ is expressed as a sparse linear combination of candidate functions (library terms) built from the state variables. Nonzero coefficients indicate the variables and interaction monomials that actually contribute to $\dot{x}_i$; thus, the sparsity pattern of the coefficient matrix encodes a directed network, possibly with higher-order (hypergraph) interactions.

If one believes that the true dynamics are simple in some function basis (e.g., low-order polynomials, or trigonometric functions), then discovering the governing equations reduces to selecting a small subset of basic functions that fit the observed time derivatives. This provides explicit equations and a direct mapping from terms to causal interactions.

Let the observed time series be arranged in a matrix $X \in \mathbb{R}^{T \times S}$, where each row corresponds to one time point and each column to one of the S state variables, and let $\dot{X}$ be the corresponding derivative matrix, obtained, for instance, by spline smoothing and finite-difference differentiation. In the function space of the state vectors, construct an overcomplete candidate library

$$\Theta(X) = [1, X, X^{(2)}, \dots, X^{(d)}, \sin X, \dots] \quad (106)$$

where $X^{(d)}$ denotes the column vectors of all d th order polynomial terms of the time series. Regress the corresponding columns of $\dot{X}$ onto $\Theta(X)$ to obtain the overdetermined linear system $\dot{X} = \Theta(X) \mathcal{E}$, $\mathcal{E} = [\xi_1, \dots, \xi_S]$. Because the true dynamics are typically sparse in this function space, the coefficients are estimated via least squares with L_1 regularization

$$\xi_i^* = \operatorname{argmin}_{\xi} \|\dot{x}_i - \Theta(X) \xi\|_2^2 + \lambda \|\xi\|_1 \quad (107)$$

where $\dot{x}_i$ is the i th column of $\dot{X}$. A nonzero entry $\xi_{j,i}$ indicates that library term j contributes to the dynamics of variable i ; if the term depends on x_k , this defines a directed edge $k \rightarrow i$ with the sign and magnitude that encode direction (excitatory vs. inhibitory) and strength. When monomials such as $x_j x_k$ are included in the library, nonzero coefficients of such terms can be interpreted as higher-order candidate interactions $(j, k) \rightarrow i$.

In practice, standardization of columns of $\Theta(X)$ and simple sparse solvers (e.g., LASSO or sequentially thresholded least squares) are sufficient. From the compressed-sensing viewpoint, if the dynamics are exactly sparse in the chosen library and certain incoherence and sample-complexity conditions hold, ℓ_1 -based SINDy can probably recover the true coefficient pattern, clarifying the trade-off between sparsity and residual error while retaining an explicit, simulatable ODE network [193].

Basic SINDy struggles with rational nonlinearities (e.g., Michaelis–Menten or Hill functions), which cannot be represented sparsely in a purely polynomial or trigonometric library. Implicit SINDy addresses this by rewriting the dynamics in an implicit form where the numerator and denominator terms appear jointly.

Suppose that the dynamics of variable x_k include rational terms. One [13] can rewrite its equation in implicit form $f_N(x) - f_D(x) \dot{x}_k = 0$ and expand the library to

$$\Theta(X, \dot{x}_k) = [\Theta_N(X) \quad \operatorname{diag}(\dot{x}_k) \Theta_D(X)] \quad (108)$$

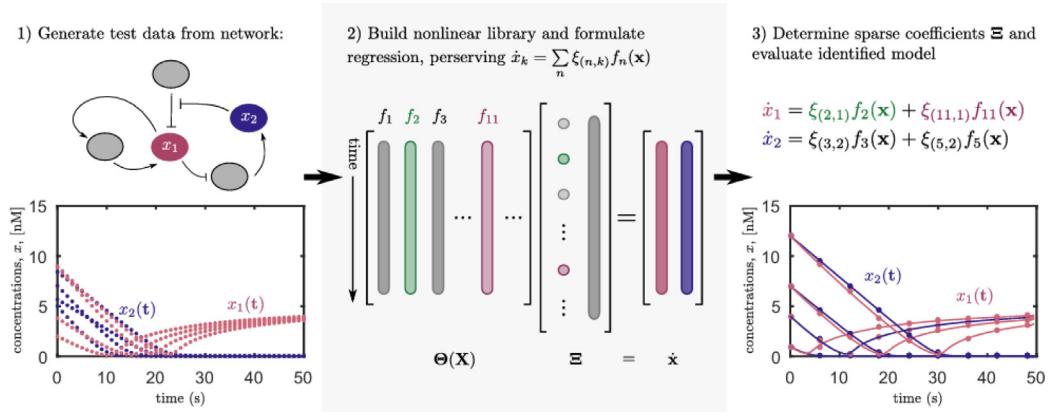


Fig. 8. Methodology for sparse identification of nonlinear dynamics (SINDy) from data. First, data is generated from a dynamical system, in this case a biological network. The time series of data is synthesized into a nonlinear function library, and the terms in this library are related to the time derivative by an overdetermined linear regression problem. Enforcing sparsity ensures that only a small number of coefficients are nonzero, identifying the few active terms in the dynamics that are needed to model the system.

Source: Reprinted from Ref. [13].

where $\Theta_N(X)$ and $\Theta_D(X)$ are candidate libraries for the numerator and denominator functions, respectively, and $\text{diag}(\dot{x}_k)$ multiplies each denominator column by the derivative. For each variable, one then solves $\Theta(X, \dot{x}_k) \xi_k = 0$ and searches for the sparsest nonzero ξ_k in the null space. This nonconvex problem is approached by alternating-direction methods with sparsity enforced via a threshold λ . As λ increases, the model's dimensionality decreases monotonically; plotting residuals $\|\Theta \xi_k\|$ versus complexity yields a Pareto front with the “elbow” representing the simplest sufficient model. Aggregating the supports of ξ_k across windows or the full record defines a sparse differential-equation-based network, in which nonzero entries again correspond to directed edges.

The support of ξ_k identifies the numerator and denominator terms that are needed to represent the dynamics of x_k , making rational structures (e.g., saturation and inhibition) explicit. Aggregating these supports across variables defines a sparse differential equation-based network where nonzero entries correspond to directed edges and where the functional form of each interaction (e.g. saturating versus linear) is directly visible.

Generic SINDy automatically selects driving terms from a large nonlinear library but does not explicitly enforce the law-of-mass-action structure or stoichiometric constraints that characterize chemical reaction systems. Reactive SINDy addresses this by lifting the candidate library from abstract function terms to concrete reaction channels so that a sparse, nonnegative regression simultaneously identifies the active reactions and estimates their rate constants [194].

Given the concentration time series for S species arranged in $X \in \mathbb{R}^{T \times S}$ and approximate derivatives $\dot{X}$, one enumerates R candidate reactions as vector-valued basis functions $y_r(x) \in \mathbb{R}^S$, each corresponding to a zero-, first- or higher-order mass-action term with fixed stoichiometry. Evaluating these at all time points yields “reaction trajectories” $\theta_r(X)$, which are stacked into a design matrix $\Theta(X)$. The dynamics can then be written in reaction-linear form

$$\dot{X} = \Theta(X) \Xi \quad (109)$$

where $\Xi = (\xi_1, \dots, \xi_R)^\top$ represents the unknown macroscopic rate constants. A nonnegative elastic-net problem

$$\min_{\Xi \geq 0} \frac{1}{2T} \|\dot{X} - \Theta(X) \Xi\|_F^2 + \alpha \lambda \|\Xi\|_1 + \alpha (1 - \lambda) \|\Xi\|_2^2 \quad (110)$$

encourages both sparsity (few active reactions) and robustness while enforcing nonnegative rates. Coefficients below a threshold are pruned; the remaining reactions define a sparse mass-action network with interpretable stoichiometry and positive rate constants. Multiple trajectories with different initial conditions can be concatenated in X and $\dot{X}$ to reduce noise sensitivity and improve the detection of weak reactions.

At larger scales and under pronounced multiscale effects, SINDy-style ideas can be embedded into neural network architectures that enforce mass-action structure while learning stoichiometry and rate constants jointly [195]. A single-hidden-layer “chemical reaction neural network” $N(\mathbf{x})$ is constructed whose output has the form,

$$N_k(\mathbf{x}) = \sum_r S_{k,r} \left(e^{\epsilon_r^f} \prod_j x_j^{-\min(0, S_{j,r})} - e^{\epsilon_r^r} \prod_j x_j^{\max(0, S_{j,r})} \right) \quad (111)$$

where the trainable integer matrix S encodes stoichiometry (negative entries for reactants, positive for products) and the exponents $\epsilon_r^{f,r} = \ln k_r^{f,r}$ enforce nonnegative forward and reverse rates. A mean-square-error loss penalizes the mismatch between $N(\mathbf{x}(t_n))$ and the estimated derivatives $\dot{x}_k(t_n)$, together with regularization promoting sparsity and

near-integer stoichiometric coefficients. To escape poor local minima and recover discrete stoichiometry, partial parameter freezing (PPF) rounds rows of S that are close to integer vectors and keeps them fixed while re-optimizing the remaining parameters; once all entries of S have converged to integers, $\theta_r^{f,r}$ are refined with S being fixed. The result is an automatically discovered multiscale reaction network with integer stoichiometry and positive rate constants, sitting at the boundary between sparse equation discovery and neural dynamics learning.

To simultaneously satisfy the triad of sparse discovery, physical consistency, and uncertainty quantification, reaction identification can be recast as a probabilistic inference problem. This is achieved by combining a hierarchical ODE formulation with regularized horseshoe priors, giving rise to Bayesian Reactive SINDy [196]. First, concentration observations $\hat{X}(t_j) \in \mathbb{R}^N$ ($j = 0, \dots, T$) are collated. Mass-action reactions are automatically enumerated up to the second order to obtain D candidates with rate functions $f_i(X)$ and an integer stoichiometry matrix $S_c \in \mathbb{Z}^{N \times D}$. Under mass action, $dX/dt = S_c[k_1 f_1(X), \dots, k_D f_D(X)]^\top$, $k_i > 0$, and a regularized horseshoe prior on the rate constants promotes sparsity while retaining heavy tails. To avoid bias from noisy numerical differentiation, the observed concentrations $\hat{X}(t_j)$ are treated as noisy observations of a latent ODE trajectory $Z(t)$ satisfying the same mass-action dynamics. Posterior sampling (e.g., via NUTS–HMC) yields distributions over trajectories and parameters; reactions with coefficients that strongly shrink toward zero are pruned, and the remaining rate constants and stoichiometric columns form the inferred reaction network with credible intervals.

Across these variants, SINDy-style approaches exhibit a shared interpretability strength. By producing explicit differential equations, these methods expose individual dynamical terms that are amenable to mechanistic interpretation, while the resulting sparsity structure directly encodes a directed graph or hypergraph of causal interactions. If the library is enlarged to include higher-degree monomials, the same framework can infer hypergraphs in which terms such as $x_j x_k$ are interpreted as genuinely ternary hyperedges $(j, k) \rightarrow i$ rather than two independent pairwise edges, allowing one to distinguish collective synergies from mere superpositions of pairwise interactions [197]. In these methods, performance hinges on several strong assumptions: derivatives must be estimated reliably from noisy and possibly irregular time series, the chosen function or reaction library must be rich enough to contain the true dynamics but not so large as to trigger severe multicollinearity, and sparsity thresholds can produce unstable edge sets, especially in high dimensions or under multiscale dynamics.

6.1.3. Dynamic mode decomposition (DMD)

In situations where the dynamics are observed only through time-resolved snapshots and the underlying governing equations are unknown or too complex to specify, it is natural to directly approximate the linear evolution operator that propagates the state from one time step to the next. Dynamic mode decomposition (DMD) provides such a data-driven way to approximate the linear evolution operator of a nonlinear dynamical system from time-resolved snapshots, and can be viewed as a finite-dimensional approximation to the Koopman operator [14,15]. In the context of network reconstruction, the best-fit linear map between successive snapshots can be interpreted as a directed, weighted adjacency matrix, while the associated eigenvalues and modes provide a spectral decomposition of dominant, coevolving patterns and their time scales. Compared with SINDy, which seeks sparse, explicitly parametric equations of motion, DMD emphasizes a linear Koopman surrogate with matrix entries that can be read as one-step predictive couplings and with modes and spectrum that decompose the dynamics into spatial patterns and temporal scales. This section reviews basic DMD and its Koopman interpretation, DMD with control for input-driven systems, sliding-window and graph DMD formulations for time-varying networks, and delay-embedded and kernel DMD variants that construct similarity networks in the mode domain.

Given a set of N -dimensional time series observation vectors at M discrete points in time, assemble the snapshots columnwise into

$$X = [x(t_1) \ x(t_2) \ \dots \ x(t_M)] \in \mathbb{R}^{N \times M} \quad (112)$$

Form the one-step shifted pairs

$$X_1 = [x(t_1) \ \dots \ x(t_{M-1})], \ X_2 = [x(t_2) \ \dots \ x(t_M)] \quad (113)$$

and assume that there exists a constant-coefficient linear operator A such that $X_2 = AX_1$. Minimizing the Frobenius norm $\|X_2 - AX_1\|_F$ yields the analytic solution $A^* = X_2 X_1^\dagger$ ($\dagger$ denotes the Moore–Penrose pseudoinverse); to reduce dimensionality and suppress noise, perform a truncated singular value decomposition on $X_1 \approx U_r \Sigma_r V_r^*$ and obtain the low-rank operator $\tilde{A} = U_r^* X_2 V_r \Sigma_r^{-1}$. Solving $\tilde{A} y_k = \mu_k y_k$ and mapping the eigenvectors back to the original space yields the DMD modes $\psi_k = U_r y_k$; the continuous spectrum corresponding to the discrete eigenvalue μ_k is $\omega_k = \frac{\ln \mu_k}{\Delta t}$ (where Δt is the sampling interval). Hence the state at any time can be reconstructed as

$$x(t) = \sum_{k=1}^r b_k \psi_k e^{\omega_k t}, \quad b = \Psi^\dagger x(t_1), \quad \Psi = [\psi_1 \ \dots \ \psi_r] \quad (114)$$

The matrix A^* can be interpreted as the adjacency matrix of a directed, weighted network; element A_{ij} quantifies the one-step linear influence of variable j on variable i , while the modes ψ_k and spectrum ω_k reveal dominant, coevolving structures and their growth, decay, or oscillation rates. Repeating the procedure in sliding windows produces a time-evolving network with edge weights and modes that change with the window location, enabling mechanism-agnostic network construction and spatiotemporal pattern decomposition in a fully data-driven framework [198].

In many applications, the observed dynamics are driven not only by intrinsic couplings but also by known exogenous inputs. Fig. 9 shows the three major components of applying dynamic mode decomposition with control (DMDc) [199]. DMDc augments the state equation to $x_{k+1} = Ax_k + Bu_k$ which leads to the matrix equation $X_2 \approx AX_1 + BU$. To estimate both mapping matrices in one least-squares step, observations and controls are combined into an augmented matrix

$$\Omega = [X_1 U] \in \mathbb{R}^{(N+L) \times (M-1)}, X_2 \approx G\Omega, G = [AB] \quad (115)$$

A truncated SVD is performed on Ω : $\Omega \approx \tilde{U} \tilde{\Sigma} \tilde{V}^*$; after the rank- p approximation, one sets

$$G \simeq X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}^* \quad (116)$$

Splitting $\tilde{U}^*$ by rows as $(\tilde{U}_1^*, \tilde{U}_2^*)$ leads to the extraction of the two high-dimensional estimates $A \approx \bar{A} = X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}_1^*$ and $B \approx \bar{B} = X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}_2^*$. If the observation dimension greatly exceeds the effective dynamic dimension, one can to reduce noise and accelerate the subsequent spectral decomposition by performing a second truncated SVD on X_2 : $X_2 \approx \hat{U} \hat{\Sigma} \hat{V}^*$ (to be truncated to rank $r \ll N$). Under the basis change $x = \hat{U} \hat{x}$ obtain the low-rank operators are obtained,

$$\tilde{A} = \hat{U}^* \bar{A} \hat{U} = \hat{U}^* X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}_1^* \hat{U} \quad (117)$$

$$\tilde{B} = \hat{U}^* \bar{B} = \hat{U}^* X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}_2^* \quad (118)$$

which form the reduced-order state equation $\hat{x}_{k+1} = \tilde{A} \hat{x}_k + \tilde{B} u_k$. Eigen- or singular-value decomposition $\tilde{A} W = W \Lambda$ yields the dominant dynamic modes of the network; mapping back to the original space $\Phi = X_2 \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}_1^* \hat{U} W$ recovers the spatial structure of each mode in terms of the original variables, and Λ or $\omega_k = \frac{\ln \lambda_k}{\Delta t}$ explain their growth rates and frequencies.

Repeating the above calculation within a sliding time window updates A and B in real time as the window moves, yielding a directed weighted network with edge weights that evolve over time and that simultaneously distinguishes intrinsic couplings from external driving effects; the entire procedure relies solely on data snapshots and, without prior equations, outputs a visualizable and predictable low-order input-output model [198].

To capture subspace structures that reconfigure rapidly in time, one commonly applies DMD in overlapping sliding windows of length T and step ΔT . In each window, only a small number of dominant modes are retained as a reduced representation of those local dynamics. After all modes have been normalized into a common coordinate system, one can perform unsupervised clustering, for instance using the average inner products between modes as a similarity measure. With a similarity threshold, agglomerative hierarchical clustering assigns modes to a small number of clusters, each representing a reproducible dynamic spatial pattern [200].

Marking the windows with centers that belong to cluster i by an indicator function $I_i(t)$ enables one to define a simultaneous activation index O_{ij} to measure the relative frequency with which clusters i and j are detected together within the same window.

$$O_{ij} = \frac{1}{T_{\text{scan}}} \sum_t I_i(t) I_j(t) \quad (119)$$

where T_{scan} is the total number of scanned windows, and $I_i(t) = 1$ if the center of window t belongs to cluster i (the value is 0 otherwise). The time-delay transition rate T_{ij} describes the probability of transitioning from pattern i to j after a delay τ .

$$T_{ij} = P(I_j(t + \tau) = 1 \mid I_i(t) = 1) = \frac{\sum_t I_i(t) I_j(t + \tau)}{\sum_t I_i(t)} \quad (120)$$

where τ is the chosen time delay. The matrices O and T thus define undirected and directed edge weights between dynamical patterns, organizing DMD modes into a higher-level network on the space of recurrent spatiotemporal patterns.

A second strategy is to apply DMD directly to a time series of adjacency matrices. Graph DMD constructs an analog of X_1, X_2 from a sequence of weighted adjacency matrices $\{A_t\}_{t=1}^M$, uses tensor-train or related decompositions to obtain a reduced one-step operator, and computes graph modes Z_j and spectra ω_j that satisfy an expansion of the form

$$A_t \approx \sum_{j=1}^p Z_j b_{j,0} e^{\omega_j t} \quad (121)$$

Each adjacency matrix is expressed as a linear superposition of graph modes Z_j with modal amplitudes $b_{j,0} e^{\omega_j t}$ [201]. Entries $(Z_j)_{ab}$ encode the joint participation strength of nodes a and b in mode j , and ω_j characterizes the intrinsic time scale of that cooperative structure. Applying Graph DMD in sliding windows yields a time-updated set of graph modes, providing a spectrally organized description of time-varying networks without explicit node-level governing equations.

A third family of approaches combines DMD with delay embedding and kernel methods to define similarity networks between multivariate time series. For a low-dimensional signal, one constructs delay-embedded snapshots $\tilde{y}(t_i) = [x(t_i), \dots, x(t_{i+d})]$ and constructs augmented matrices $X_1^{\text{aug}}, X_2^{\text{aug}}$, and performs low-rank DMD as above. The resulting set of modes for a given series spans an r -dimensional subspace $\Phi \in \mathbb{C}^{n \times r}$. To quantify the similarity between two series i, j in the mode domain, one introduces the projection kernel

$$k(\Phi_i, \Phi_j) = \|\Phi_i^* \Phi_j\|_F^2 \quad (122)$$

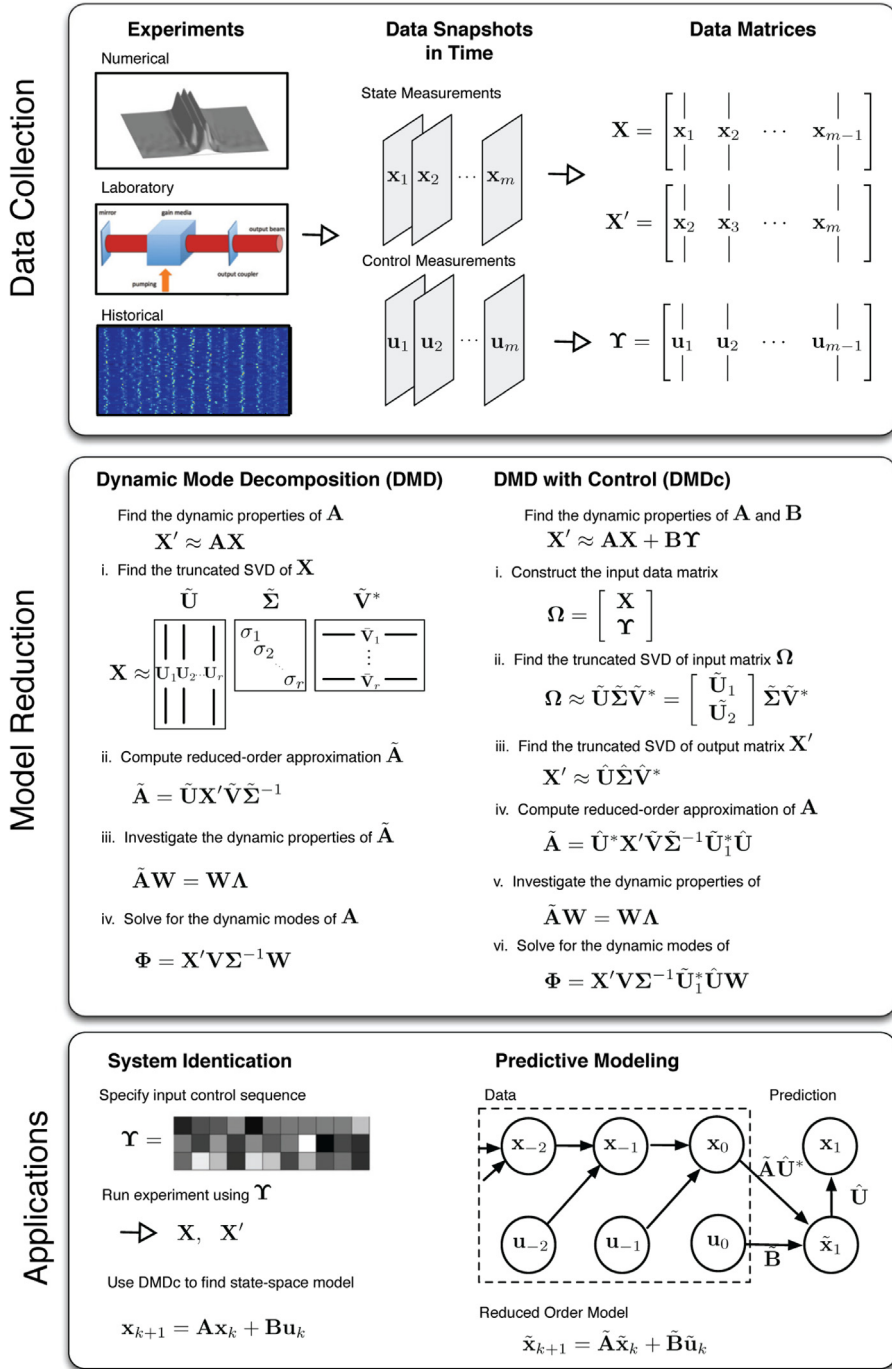


Fig. 9. The illustration outlines the three major components of applying DMDc. The top panel describes the collection of data from numerical, laboratory, or historical data and the curation of the data into matrices for the methods. The middle panel outlines the procedure for DMD and DMDc for comparison. The bottom panel illustrates two practical applications of DMDc.

Source: Reprinted from Ref. [199].

that maps subspaces to the Grassmann manifold and measures the principal-angle similarity, while naturally incorporating mode amplitudes as weights. Evaluating $k(\Phi_s, \Phi_{s'})$ pairwise over a batch of series produces a Gram matrix $K \in \mathbb{R}^{S \times S}$ that can be interpreted as the adjacency matrix of an undirected network with nodes that are time series and whose non-negative edge weights quantifying spectral-subspace similarity [202]. Multi-run or multi-scale data can be handled by averaging such Gram matrices across runs or bands, integrating multiple “views” into a single similarity network.

To separate dynamics operating at different time scales, multi-resolution DMD (mrDMD) recursively applies DMD on nested time intervals [203]. At the top level, DMD is applied to the full time span, and a small number of slow modes with characteristic periods that exceed the window length are removed from the signal. The residual is split into two halves, DMD is applied to each half, and the “remove slow, keep fast” operation is repeated within every segment. After L levels, one obtains a hierarchical expansion

$$\hat{x}(t) = \sum_{l=1}^L \sum_{j=1}^{2^{l-1}} \sum_{k=1}^{m_l} f_{l,j}(t) b_k^{(l,j)} \phi_k^{(l,j)} e^{\omega_k^{(l,j)} t} \quad (123)$$

where l indexes the resolution level, j is the time subinterval, and k is the mode within that subinterval; the indicator $f_{l,j}(t)$ equals 1 only inside the corresponding sub-interval. Thresholds on $|\lambda_k|$ or ω_k are used to select modes with periods that fit within the current window. The resulting modes across all levels can be concatenated into a library $\Psi_r \in \mathbb{R}^{N \times r}$; a column-pivoted QR decomposition of $\Psi_r^\top$ identifies a small set of spatial coordinates (nodes) with the largest mode energy as “optimal sensors”, and the interaction strengths induced by $\Psi_r \Psi_r^\top$ between these nodes can be interpreted as edge weights in a sparse, multiscale network [204].

Finally, if the underlying dynamics are strongly periodic, anti-circulant and sparsity-promoting variants of DMD exploit lifting to a circulant or anti-circulant structure to enforce eigenvalues on (or near) the unit circle, guaranteeing stable, purely periodic modes, and then use ℓ_1 -regularization on modal amplitudes to select a minimal set of distinct frequencies [205]. In such frameworks, the reconstructed signal or network is expressed as a sum of a few periodic DMD modes with automatically chosen frequencies and phases, providing a noise-robust, multiscale decomposition of recurrent network activity.

Across these variants, the interpretability of DMD-type methods has two complementary aspects. First, the fitted linear operators A (and, where applicable, B) can be read directly as directed, weighted adjacency matrices: entries quantify one-step predictive influence between variables or from inputs to states. Second, the DMD modes and spectra furnish a spectral factorization of network dynamics into spatial patterns and characteristic time scales (growth/decay rates and oscillation frequencies), often aligning with physically or functionally meaningful structures (e.g., coherent flow structures, resting-state networks, or recurring behavioral motifs). Moreover, these advantages rest on strong assumptions: the Koopman approximation is linear and global; noise and finite sampling can blur eigenvalues and modes; and the resulting edges represent predictive linear dependence rather than causal influence in the interventionist sense. These limitations motivate combinations of regularization, delay-embedding and kernel methods.

6.2. AI-extended dynamics learning approaches

This section discusses several classic dynamics learning approaches that have undergone extensive development with the integration of machine learning and deep learning techniques. We introduce four representative categories: SINDy extensions, symbolic regression frameworks enhanced by machine learning, neural ODEs and their extensions, and PINNs and their extensions.

6.2.1. SINDy extensions

Recent developments have significantly extended the original SINDy framework to broaden its applicability to noisy, high-dimensional or partially observed systems. Here we review representative extensions: deep SINDy and a two-phase inference approach for PDEs and SDEs, and a machine learning framework combining a symbolic model.

To address the limitation that the proper choice of measurement coordinates is often unknown, Deep SINDy [206] combines deep neural networks with SINDy to learn interpretable dynamical equations from data. As shown in Fig. 10, the core idea is to use a custom deep autoencoder network to discover effective coordinates that make dynamics sparse in the latent space, and then apply SINDy’s sparse regression to identify parsimonious models. Its interpretability mainly arises from the sparse coefficient vectors, the mechanisms of which are selected and their relative contributions.

In real-world systems, observations are often noisy and structurally incomplete. A critical challenge is to autonomously identify the underlying explicit dynamic model instead of merely fitting parameters, without prior knowledge of the mechanisms. A two-phase inference approach [207] was proposed to infer the equations of network dynamics of the general form that $\frac{dx_i(t)}{dt} = \mathbf{F}(\mathbf{x}_i(t)) + \sum_{j=1}^n A_{ij} \mathbf{G}(\mathbf{x}_i(t), \mathbf{x}_j(t))$, where $\mathbf{F}$ is the self-regulation function and $\mathbf{G}$ is the pairwise function. This framework proceeds in two phases. Phase I entails a global regression. It identifies several leading elementary functions for each dimension by assessing the relevance of each elementary function in L_F and L_G to the true network dynamics. This step effectively reduces the search space for finding the potential mechanisms. Phase II is local fine-tuning. It applies topological sampling and the wAIC to select the optimal model. This allows it to reconstruct generative and concise expressions for $\mathbf{F}$ and $\mathbf{G}$.

This framework is interpretable symbolically and mechanistically. It goes beyond a black-box approximation to provide a precise mathematical form of both self-dynamics and pairwise interaction dynamics. These specifically identified terms can correspond to physical processes instead of just being opaque weights.

To infer the hidden stochastic dynamics in coupled SDEs, Gao proposed the Langevin Graph Network Approach (LaGNA) [208]. Here, we specify the stochastic differential equation form as $\mathbf{x}_i(t+dt) = \mathbf{x}_i(t) + (\mathbf{F}(\mathbf{x}_i(t)) + \sum_{j=1}^n A_{ij} \mathbf{G}(\mathbf{x}_i(t), \mathbf{x}_j(t))$,

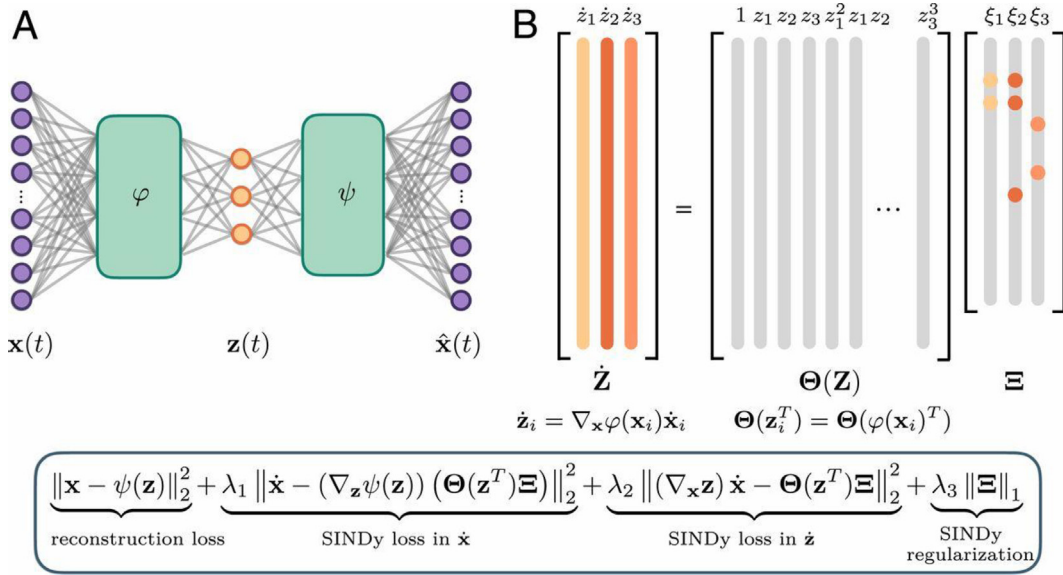


Fig. 10. Schematic of the SINDy autoencoder method for simultaneous discovery of coordinates and parsimonious dynamics. (A) An autoencoder architecture is used to discover intrinsic coordinates $\mathbf{z}$ from high-dimensional input data $\mathbf{x}$. The network consists of 2 components: an encoder $\varphi(\mathbf{x})$, which maps the input data to the intrinsic coordinates $\mathbf{z}$, and a decoder $\psi(\mathbf{z})$, which reconstructs $\mathbf{x}$ from the intrinsic coordinates. (B) A SINDy model captures the dynamics of the intrinsic coordinates. The active terms in the dynamics are identified by the nonzero elements in Ξ , which are learned as part of the NN training. The time derivatives of $\mathbf{z}$ are calculated using the derivatives of $\mathbf{x}$ and the gradient of the encoder φ . Inset shows the pointwise loss function used to train the network. The loss function encourages the network to minimize both the autoencoder reconstruction error and the SINDy loss in $\mathbf{z}$ and $\mathbf{x}$. L_1 regularization on Ξ is also included to encourage parsimonious dynamics. Source: Reprinted from Ref. [206].

$\mathbf{x}_j(t))dt + \Phi(\mathbf{x}_i(t))d\mathbf{W}_t$. The two-phase inference approach is applied to infer the concise forms of own dynamics, and interaction and diffusion parts.

To determine the governing equations of a dynamical system from partial observations, Lu [209] proposed a machine learning framework for both system identification and state reconstruction that combined an encoder with a sparse symbolic model.

Given partially available states $\{\mathbf{x}_v(t_0), \mathbf{x}_v(t_0 + \Delta t), \dots, \mathbf{x}_v(t_N)\}$, an encoder $\mathbf{e}_\eta$ is used to reconstruct these states into hidden states $\{\hat{\mathbf{x}}_h(t_0), \hat{\mathbf{x}}_h(t_0 + \Delta t), \dots, \hat{\mathbf{x}}_h(t_N)\}$. The fully reconstructed state $\hat{\mathbf{x}} = \mathbf{a}(\mathbf{x}_v, \hat{\mathbf{x}}_h)$, obtained by aggregating these states, is then passed into a symbolic model of the governing equations.

$$\frac{d\hat{\mathbf{x}}}{dt} = \hat{\mathbf{F}}_\theta(\hat{\mathbf{x}}) := \theta_1 \mathbf{f}_1(\hat{\mathbf{x}}) + \theta_2 \mathbf{f}_2(\hat{\mathbf{x}}) + \dots + \theta_m \mathbf{f}_m(\hat{\mathbf{x}}) \quad (124)$$

Automatic differentiation is applied to compute higher-order symbolic time derivatives of the reconstructed visible states $\mathbf{g}(\hat{\mathbf{x}})$. These symbolic derivatives can be compared with finite-difference time derivatives $\Delta^p \mathbf{g}(\mathbf{x}) / \Delta t^p = \Delta^p \mathbf{x}_v / \Delta t^p$. The MSE loss is applied to train the encoder and the symbolic model using partial observations.

$$\mathcal{L}(\eta, \theta) = \frac{1}{N} \sum_{i=1}^N \sum_{p=1}^M \frac{\alpha_p}{\sigma_p^2} \left(\frac{d^p \mathbf{g}(\hat{\mathbf{x}}(t_i))}{dt^p} - \frac{\Delta^p \mathbf{x}_v(t_i)}{\Delta t^p} \right)^2 \quad (125)$$

This framework is interpretable mainly due to representing the dynamical system with a sparse symbolic model. By imposing a sparsity prior on the governing equations, it identifies a set of physically meaningful terms that characterize the exact underlying exact dynamics.

6.2.2. Symbolic regression extensions

Here, we present a representative method named the Learning Law of Changes (LLC) [210]. This universal neural symbolic regression tool was proposed to efficiently discover the underlying governing equations of a complex system by combining the excellent fitting ability from deep learning and the equation inference capability from pretrained symbolic regression. LLC introduces a physics prior of decomposability, assuming that a change of system states can be influenced by its own states and its neighbors' states.

Here, $A_{i,j}$ represents the adjacency matrix, and $\mathbf{Q}_i^{(self)}$ and $\mathbf{Q}_{i,j}^{(inter)}$ denote the self-dynamics and the interaction dynamics, respectively. The governing equations of network dynamics can be written as

$$\dot{X}_i(t) = \mathbf{Q}_i^{(self)}(X_i(t)) + \sum_{j=1}^N A_{i,j} \mathbf{Q}_{i,j}^{(inter)}(X_i(t), X_j(t)) \mathbf{Q}_i^{(self)} \mathbf{Q}_{i,j}^{(inter)} \quad (126)$$

The LLC framework has two main steps. By incorporating physical priors, two neural networks $\hat{\mathbf{Q}}_{\theta_1}^{(self)}$ and $\hat{\mathbf{Q}}_{\theta_2}^{(inter)}$ are trained to parameterize $\mathbf{Q}^{(self)}$ and $\mathbf{Q}^{(inter)}$ separately. This effectively achieves the decoupling of the dynamics. Once the neural networks are well trained, a pretrained symbolic regression technique, NSRA, is applied to parse the black-box neural networks into interpretable white-box symbolic equations. The interpretability of LLC is structural and mechanistically explicit. By enforcing the physical prior of dynamic interactions, this framework achieves the clean separation between intrinsic and interaction dynamics. This framework ensures the results' consistency with the underlying dynamics reality.

A deep learning method named Neural Discovery of Network Dynamics (ND²) [211] has recently been proposed to automatically learn network dynamics formulas. ND² can effectively discover such formulas in a reduced search space by designing three operators: source, target and aggregation.

6.2.3. Neural ODEs and extensions

Neural ordinary differential equations (neural ODEs) were first introduced by Chen in 2018 [212]. The central idea is to reinterpret the discrete layer-wise transformations of deep residual networks (ResNets) as the Euler discretization of an underlying continuous-time dynamical system. A continuous-time formulation is provided by representing the evolution of a hidden state h_t through a learnable vector field.

A ResNet updates features via a discrete recursion $h_{t+1} = h_t + f(h_t, \theta_t)$, $t \in \{0 \dots T\}$, $h_t \in \mathbb{R}^D$, which can be viewed as an explicit Euler discretization of an underlying continuous transformation. In the small step-size limit, the ODE form can be described as

$$\frac{dh(t)}{dt} = f(h(t), t, \theta) \quad (127)$$

This formulation effectively transforms the depth of a neural network from a discrete number of layers into a continuous time dimension. This model provides a direct state that can be constrained and regularized.

Neural ODEs are usually applied for causal structure reconstruction in dynamic systems [213,214]. However, they suffer from several practical limitations, including high computational cost and vanishing or exploding gradients. Many extensions have been proposed, such as Augmented Neural ODEs [215], Neural integro-differential equations (Neural IDE) [216], Neural Delay Differential Equations [217], Neural Controlled Differential Equations [218], Stiff Neural Ordinary Differential Equations [219], Modulated Neural ODEs [220], Fourier NODEs [221], its extension to the stochastic setting named Neural SDE [222], and its extension to the latent variable, latent ODEs [223].

To improve interpretability and representation of mechanistic information, researchers have proposed variants that combine neural ODEs with symbolic priors or explicit basis representations. Symbolic neural ODE(SNODE) [224] integrates domain-specific constraints and leverages a symbolic neural network module (SymNet) to represent candidate terms in the governing dynamics, thus enhancing interpretability while preserving expressiveness. ODENet [225] represents dynamics as a linear combination of pre-specified basis functions within the Neural ODE architecture, enabling parameter inference and symbolic recovery of governing equations from observational data. This framework embeds explicit numeric integration schemes into a machine learning framework, providing a general-purpose technique for discovering dynamical equations underlying complex systems with large time series.

Recognizing that neural ODEs may sometimes learn “spurious” or non-identifiable dynamics, particularly if generalizing to unseen interventions, researchers introduced a causal neural ODE (C-ODE) [214] framework. This model imposes causal sparsity regularization to promote structurally meaningful dynamics and has demonstrated accurate recovery of both parameters and causal graphs across synthetic and real-world datasets, including nonlinear, low-dimensional systems and gene regulatory networks.

Furthermore, there is a growing body of research that integrates neural ODEs with graph-based techniques, aiming to recover causal dynamic structures from a predictive perspective. In such models, graph neural networks (GNNs) are combined with continuous-time dynamics to learn the evolution of node states over complex topologies. The models have shown strong performance in predicting system evolution in dynamic networks and offer a promising direction for learning interpretable, network-aware causal dynamics from high-dimensional time series data [226].

The continuous-time counterpart of hybrid modeling is illustrated by Hybrid² Neural ODE Causal Modeling (H²NCM) [227]. In this approach, domain-specific causal information is embedded into a neural ODE through a dedicated causal loss that penalizes trajectories violating known cause–effect constraints. The additional regularization steers the latent dynamics toward structures consistent with the prescribed causal graph, resulting in improved causal validity and enhanced interpretability while maintaining predictive performance.

The PerturbODE model [228] was proposed to explicitly encode the potential GRN structure into the parameters of a neural ODE, thereby endowing the model with biological interpretability. Learning the relevant parameters allows the interactions among genes to be inferred. The model was validated on large-scale datasets and demonstrated strong performance in both GRN inference and cell trajectory prediction.

6.2.4. Physics-Informed Neural Networks and extensions

Physics-Informed Neural Networks (PINNs) were first introduced by Raissi in 2019 [229] as a kind of physics informed machine learning (PIML) [230]. Unlike traditional purely data-driven models that attempt to reconstruct unknown governing equations from observations, PINNs operate on the premise of a priori knowledge of the underlying physical mechanism. They are designed to estimate solutions or identify hidden parameters by simultaneously learning from observational data and enforcing known physical laws. The primary strength of PINNs is in scenarios where observational data is sparse, noisy, or incomplete, but the physical laws are well understood. By integrating fundamental physical laws directly into the neural training process, PINNs act as a regularization mechanism that constrains the hypothesis space to physically valid manifolds. This approach addresses the limitations of traditional black-box learning models, which often suffer from poor generalization and physical inconsistency.

The most crucial aspect of PINNs is in their unique loss function design, which embeds the PDEs into the loss function of a neural network. Consider the Burgers' equation in one spatial dimension as an example.

$$u_t + \lambda_1 u u_x - \lambda_2 u_{xx} = 0 \quad (128)$$

The neural network $u(t, x)$ is trained by minimizing a composite mean square error loss, $MSE = MSE_u + MSE_f$, where

$$MSE_u = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(t_u^i, x_u^i) - u^i|^2 \quad (129)$$

$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_f^i, x_f^i)|^2 \quad (130)$$

and $\{t_u^i, x_u^i, u^i\}_{i=1}^{N_u}$ denote the initial and boundary training data on $u(t, x)$. MSE_u and MSE_f separately correspond to fitting the initial and boundary data and to enforcing the structure. Parameters λ can be learned simultaneously with the solution u , using the gradient-based optimizers such as Adam [231] and L-BFGS [232].

While standard PINNs have been successful, they often struggle with stiffness, and multiscale behaviors, and chaotic systems. A critical failure identified recently is the violation of causality. To address these challenges, recent trends have shifted toward the integration of causal structures into PINNs to enhance interpretability and robustness.

To address the limitation that PINNs may learn spurious correlations, domain causal knowledge has been considered in PINNs. Kang [233] proposed a causal prior-embedded physics-informed neural network, which incorporated quantitative causal relationships that extracted from experimental data via SCMs into the neural architecture. Specifically, the study introduced causal prior regularization and causal weight initialization mechanisms to steer the training toward representations that aligned with known causal structures. This provided mechanistic interpretability by verifying that neural networks logic was consistent with the causal graph. Wang proposed causalPINNs [234], introducing the known causality structure into the training process of PINNs. The core idea was to modify the loss function by considering the time-ordered weighting of residuals; the solution converged sequentially in time. Here, we introduce a reformulation of the PINN training objective that can account for the missing causal structure. A weighted residual loss is defined as

$$\mathcal{L}_r(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \exp\left(-\epsilon \sum_{k=1}^{i-1} \mathcal{L}_r(t_k, \theta)\right) \mathcal{L}_r(t_i, \theta) \quad (131)$$

where θ are all trainable parameters of a neural network, N_t is the number of temporal collocation points, and ϵ is referred to as a causality parameter that controls the steepness of the weights.

The interpretability arises from the convergence trajectory reflecting the actual physical evolution of the system. Many extensions of causalPINNs have also been proposed, such as combining them with transfer learning to enhance generalization [235], a novel adaptive sampling method that considers temporal causality in sampling [236].

Many real-world complex systems exist in which the exact forms of the governing equations are unclear or partially unknown. PINN-SR [237], proposed by Chen in 2021, integrates sparse regularization directly into the PINN framework. This approach effectively discovers dynamic structures such as the Kuramoto–Sivashinsky equation from limited data. It offers symbolic interpretability through a clear mathematical formula output, which can identify exactly which mechanism drives the system.

Katende proposed Counterfactual PINNs [238] for discovery of causal structures in PDEs, combining PINNs and counterfactual perturbations. It formalizes PDEs as SCMs to enable principled reasoning about interventions in dynamical systems. Here we start with a structural PDE model $\mathcal{N}[u; \gamma] = 0$ with solution $u: \Omega \times [0, T] \rightarrow \mathbb{R}$. A counterfactual PDE is defined as a system obtained by an intervention on the parameters γ . The causal response to the intervention $\gamma \rightarrow \gamma'$ is defined as the induced discrepancy between the factual and counterfactual solutions,

$$\delta(\gamma \rightarrow \gamma') := \|u - u^{cf}\|_{L^2(\Omega \times (0, T))} \quad (132)$$

where δ captures the global, nonlinear, and structural changes to the system behavior, and u is the learned solution under the original parameters γ . The total loss of a Causal PINN is as follows:

$$\mathcal{L}(\theta, \alpha) = \frac{1}{N} \sum_{i=1}^N (u_\theta(x_i, t_i) - u_i)^2 + \lambda_r \|\mathcal{R}_\theta\|_{L^2(\Omega \times (0, T))}^2 + \lambda_s \|\alpha\|_1 \quad (133)$$

where λ_r enforces physical consistency, and λ_s induces sparsity in α . Then, the training proceeds by alternating minimization; namely, we first update θ by gradient descent to minimize $\mathcal{L}$, and then update α .

$$\alpha_j \leftarrow \text{ST}_{\lambda_s} \left(\underset{\alpha_j}{\text{argmin}} \|\mathcal{R}_\theta\|^2 \right) \quad (134)$$

By setting $\alpha_j \rightarrow 0$ and evaluating the new solution u_θ^{cf} under the updated residual, the causal effect of the operator can be evaluated.

$$\delta_j := \|u_\theta^{\text{cf}} - u_\theta\|_{L^2(\Omega \times (0, T))} \quad (135)$$

where δ_j represents the influence. Counterfactual evaluation of u_θ^{cf} closes the physical interpretability.

The causal sensitivity index (CSI) has been defined to quantify the causal effect of each operator under a do-intervention on the structural PDE,

$$\text{CSI}_j := \frac{\|\mathcal{R}_\theta - \mathcal{R}_{\theta,(-j)}\|_{L^2(\Omega \times (0, T))}}{\|\mathcal{R}_\theta\|_{L^2(\Omega \times (0, T))} + \delta} \quad (136)$$

This CSI metric provides a continuous, data-driven ranking of operator influence.

7. Applications

7.1. Climate systems

Climate change, as one of today's most urgent global challenges, demands an improved understanding of extreme events, teleconnection structures, and long-term climate variability. Interpretable causal inference provides powerful tools for this purpose, especially in distinguishing linear versus nonlinear influence mechanisms, constructing directed climate networks, analyzing multiscale coupling, and enhancing interpretability in prediction tasks. Compared with traditional correlation-based analysis, causal and information-theoretic methods can uncover genuine drivers in high-dimensional and noisy climate data, thereby deepening the understanding of dynamic climate processes.

A major application concerns the identification of teleconnection patterns and hidden climate linkages. By integrating correlation and normalized mutual information (NMI), recent studies have quantified both linear and nonlinear dependencies between climate variables and have mapped their strength across regimes [239]. The correlation and NMI analysis distinguishes influence regimes via significance testing and reveals nonlinear remote pathways, such as the impact of eastern Montana soil moisture on rainfall in the southern Great Plains. These pathways remain invisible to the correlation analysis alone, and thus the interpretability of drought-onset mechanisms is improved.

Interpretability-oriented causal tools have also advanced the understanding of multiscale climate dynamics. The Synergistic-Unique-Redundant Decomposition (SURD) framework applies partial information decomposition to turbulent flows [94]. It separates causal contributions into synergistic, unique, and redundant components, showing that the forward energy cascade is dominated by unique causality and quantifying causality leak from unobserved variables. Granger causality, CTE or CCM methods cannot solve this problem.

Another line of research focuses on reconstructing global climate networks and their causal structure. The framework of Tsonis et al. connects climate-field grid points using thresholded Pearson correlations [47,240], revealing global connectivity changes during events such as El Niño [241,242]. The global climate network constructed by [243] shows that El Niño events markedly alter the network's linking pattern, reflecting changes in the global teleconnection structure during the event. Mutual-information climate networks extend this approach. Donges et al. showed that MI and correlation analyses produced similar local topologies but differed markedly at global scales, especially in terms of betweenness-centrality patterns [244]. These developments led to causal (directed) climate networks. Graph-based Granger models were used to infer directional pathways between climate indices [245], while probabilistic graph-model methods filtered indirect links to recover direct causal relationships [246].

Recent advances include causal fingerprints for constraining precipitation projections [49] and probabilistic causal networks for assessing subseasonal-to-seasonal predictability [248]. Another study shows that δ -MAPS extracts a compact set of dominant SST regions [247]. The ENSO, Indo-Pacific Warm Pool, western Pacific, and Tropical Atlantic nodes have the greatest strength and thus anchor basin-scale variability (Fig. 11a-c). PCMCi further uncovers directed teleconnections among these key regions. ENSO exhibits the highest Average Causal Effect and acts as the primary driver in the network (Fig. 11d). In addition, the inferred one-month-lag causal links reveal clear propagation pathways, reflecting systematic SST teleconnections at fast timescales. The pathways are that ENSO perturbations transmit to the South Pacific, western Pacific, and Atlantic nodes through positive or negative path coefficients (Fig. 11e).

Mechanism-focused reconstruction of dynamics further contributes to separating external forcing from internal variability. SINDy generates explicit governing equations that expose low-dimensional climate dynamics, with applications to learning latent-coordinate ODE models of climate variability [206]. Dynamic Mode Decomposition (DMD) and DMD with control (DMDc) identify leading modes and teleconnection patterns. In particular, DMDc has been applied to exploring anthropogenic-forcing impacts on global surface-air-temperature's evolution [249].

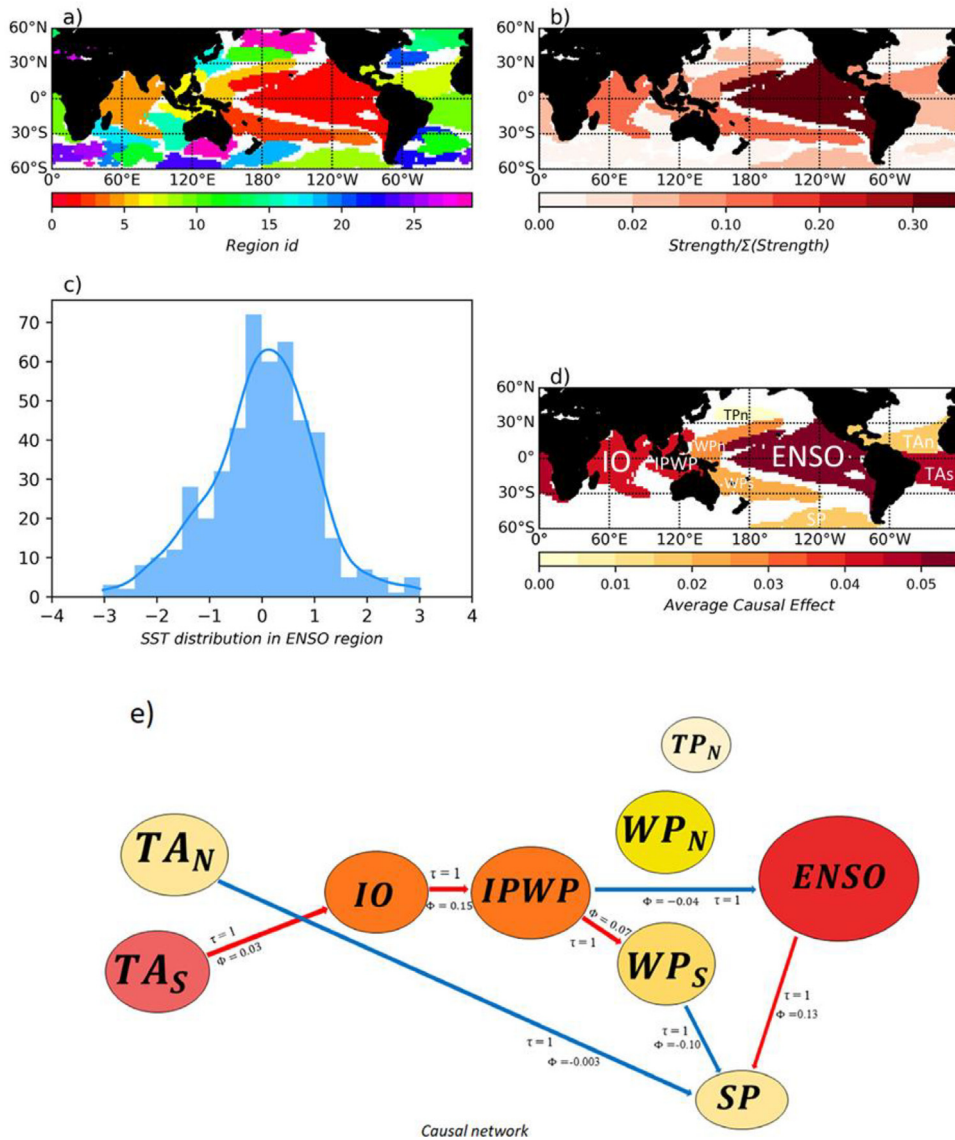


Fig. 11. Regions and links inference with δ -MAPS and PCMCI. a Region map extracted from HadISST dataset with δ -MAPS algorithm. Thirty regions are identified and associated to a signal (cumulative Sea Surface Temperature anomalies). b Corresponding weight map: each region is colored as function of its weight (sum of covariances with all other regions divided by total sum of covariance). The region with biggest weight lies in the central east Pacific Ocean, and is referred to as the El Nino Southern Oscillation (ENSO region). c SST anomalies in ENSO region in HadISST. d Nine regions which serve as nodes for the causal network, labeled with their names: the ENSO region, the Indian Ocean (IO) region, the Indo-Pacific Warm Pool (IPWP), the region in South Pacific (SP) Ocean, the regions in the Western Tropical Pacific Ocean in the north and south hemisphere (WP_N and WP_S), the regions in the Tropical Atlantic Ocean in the north and south hemisphere (TA_N and TA_S) and the region in north of the Tropical Pacific (TP_N). Regions are colored as function of their Average Causal Effect (ACE) value in the reference HadISST dataset. The ENSO region has the largest ACE (e) Causal network reconstructed from nine regions with PCMCI algorithm, and its reference causal effects inferred in HadISST dataset with a lag of one month. Only causal paths at lag one month are displayed here. Link are labeled with the value of the lag τ in months and the value of the path coefficient ϕ which is equal to causal effect for direct causal path. Red and blue link respectively indicate positive and negative causal effects (sign of the path coefficient ϕ). Nodes are colored as function of the ACE displayed on (d). Source: Reprinted from Ref. [247].

Causal diagnostics in the time–frequency domain help resolve scale-dependent interactions between large-scale climate modes and regional hydroclimate. Recurrence plots (CRPs) have been used to analyze the dynamic similarity between ENSO and rainfall in northwestern Argentina [250]. The cited study investigates Tropical Pacific Decadal Variability (TPDV) using scale-dependent analyses such as wavelet power and coherence, which reveal strong relationships between the Niño4 SST and subsurface temperature at decadal timescales [251]. The results in Fig. 12 show that off-equatorial northwestern Pacific subsurface temperature anomalies drive TPDV phase transitions: propagating eastward along the

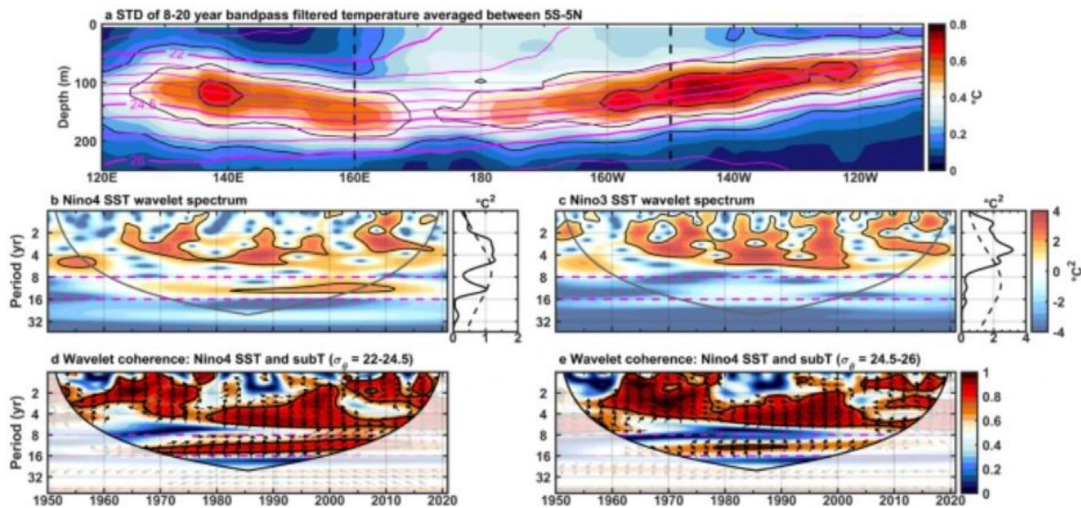


Fig. 12. Niño4 upper ocean decadal variability. a Standard deviation of 8–20 year bandpass filtered temperature anomalies (shading and thin black contours) averaged between 5°S and 5°N. The contour interval is 0.2 °C and zero contours are omitted. Magenta lines denote the mean climatological potential density with 0.5 kg m⁻³ contour intervals. The vertical dashed black lines denote the longitude of 160°E and 150°W. b Bias-rectified wavelet power spectrum (shading) and corresponding global wavelet power spectrum of raw detrended Niño4 SST index. Thin black contours in the wavelet power spectrum and dashed black lines in the global power spectrum indicate statistical significance at the 90% confidence level when tested against a red-noise AR(1) process with lag-1 of 0.91. The parabola regions indicate the “cone of influence” where edge effects become important. The horizontal dashed magenta lines indicate the 8 and 16-year periods. c Same as (b) but for Niño3 SST index. Wavelet coherence between Niño4 SST index and Niño4 subsurface temperature anomalies averaged between (d) 22–24.5 σ_θ and (e) 24.5–26 σ_θ . The 95% significance level against red noise is shown as thick black contours. The vectors indicate the relative phase relationship (pointing right indicates in phase, pointing left indicates out of phase, and pointing vertically upward means the Niño4 SST index lags subsurface temperature by 90°). The horizontal dashed magenta lines are the same as (b).

Source: Reprinted from Ref. [251].

NECC pathway, they reverse the sign of equatorial SST anomalies approximately three years later. Extratropical-tropical interactions, particularly via the Pacific Meridional Mode, amplify these subsurface disturbances 0–12 months before the cold peak. The full phase shift completes in 5–6 years, with combined subsurface and extratropical types of feedback sustaining a preferred 10–20 year oscillation. Additionally, the above study underscores the key role of off-equatorial subsurface thermal processes in initiating TPDV phase reversals.

Recent reviews emphasize integrating physical domain knowledge and temporal structure into causal inference for geosciences [252]. Methods such as conditional independence testing, graph models, and transfer entropy-based measures help disentangle direct drivers from indirect or spurious correlations, addressing challenges such as contemporaneous causality, hidden confounders, and nonstationarity.

Finally, the challenge of interpretable extreme-event prediction under data scarcity has motivated hybrid neural-causal approaches. Spatiotemporal neural architectures learn coherent representations that reveal latent teleconnections, e.g., automatically detecting hurricane spirals and extreme weather events from high-dimensional climate data. Concurrently, hybrid physics-informed neural frameworks incorporate known causal structures and discover additional interactions, such as data-driven corrections to model biases [253]. In 2025, structural causal models were integrated with meta-learning for extreme weather forecasting, enabling causal interventions to generate plausible synthetic extreme weather scenarios and mitigate data scarcity [254].

Overall, this research area integrates diverse interpretable causal methods, including mutual information, partial information decomposition (e.g., SURD), causal graph models, time–frequency causal diagnostics, SINDy, DMD, and hybrid neural-causal frameworks. These approaches uncover the nonlinear, multiscale, and cross-regional causal structures governing the climate system, thereby improving prediction, deepening process understanding, and supporting decision-making.

7.2. Neuroscience

Inferring large-scale interaction networks from neural time series remains a central challenge in systems neuroscience. Beyond simple correlation-based functional connectivity, causal reconstruction in brain networks is essential for distinguishing drivers from responders and for linking dynamics back to anatomy. Specifically, explainable causal reconstruction in neural systems addresses three primary challenges: first, identifying directed, frequency- and time scale-specific influences between regions; second, characterizing how groups of regions share information redundantly or

synergistically rather than only via pairwise links, and third, relating these functional interactions to underlying structural connectivity and known cortical hierarchies [255,256].

Early work on resting-state fMRI showed that spontaneous low-frequency BOLD fluctuations organized into coherent, spatially distributed networks, such as the default-mode and task-positive systems [257]. In the network neuroscience framework, thresholded correlation matrices between regions of interest are interpreted as graphs with small-world topology, modular organization and hubs, providing a compact description of the large-scale co-fluctuation structure [258,259]. These symmetric functional connectivity maps are highly interpretable (a strong undirected edge simply means that two regions tend to co-activate) and have been widely used to define canonical resting-state networks and to relate connectivity patterns to behavior and disease. However, correlation analysis mixes direct and indirect pathways and is blind to direction; it cannot distinguish whether a “hub” region is genuinely driving others, is being driven, or is only passively co-modulated by common input.

To capture directed, time-asymmetric influences, many studies turn to Granger causality-type models in electrophysiology and fMRI. For example, analyzing invasive recordings from macaque sensorimotor cortex, Brovelli et al. used frequency-resolved Granger causality to show that β -band activity flowed from premotor to primary motor to parietal regions, revealing a directed hierarchy that matched known anatomical pathways [258]. In human MEG, Hillebrand et al. applied directed measures to resting-state oscillations and observed that posterior regions predominantly drove frontal areas in the alpha and beta bands, whereas frontal-to-posterior drive was more prominent in the gamma band, suggesting frequency-dependent channels of information flow [259]. Seth et al. reviewed multivariate Granger analysis in the context of fMRI, emphasizing that directed connectivity estimates, when carefully implemented with appropriate preprocessing, model order selection and hemodynamic considerations, could be made robust enough to be meaningfully compared to anatomical connectivity and task manipulations [260]. In all of these cases, edge strengths, lags and frequency spectra provide an interpretable notion of causal influence: one can say not only that region A is connected to region B but also that past activity in A improves the prediction of B at specific time scales and frequencies.

Information-theoretic approaches extend this logic beyond linear models and pairwise links. Multivariate measures such as total correlation, dual total correlation and O-information can quantify whether a set of regions is dominated by sharing redundant information where each region carries overlapping signals, or synergistic information where only joint activity across multiple regions reveals certain features of the system [261]. Building on this, Varley et al. introduced partial entropy decomposition (PED) to decompose the joint entropy of brain activity into nonnegative “atoms” corresponding to redundant, unique and synergistic contributions, thereby directly characterizing a higher-order functional structure that is invisible to any bivariate connectivity measure [95]. This framework provided an explicitly explainable interpretation: redundancy corresponded to distributed, overlapping representations, while synergy reflected integrative processing that only emerged only if multiple regions were considered together.

Fig. 13 illustrates how PED reveals dynamic redundancy-synergy structure in resting-state fMRI. At the spatial level, surface plots show that primary sensory and motor cortices are dominated by redundant information, whereas multimodal association areas, including frontoparietal and default-mode regions, are strongly synergistic. This reproduces and refines earlier findings that “synergy-rich” regions tend to sit at the interface between canonical networks and are well placed to integrate information from multiple sources [261,262]. At the temporal level, Varley et al. computed, for each fMRI frame in a long resting-state scan, the total amount of redundant and synergistic structure present in the whole brain. The resulting time series showed substantial fluctuations in both quantities, but synergy consistently dominated: at no time point did redundancy exceed synergy. Instantaneous redundancy and synergy were strongly anticorrelated, indicating that when the brain temporarily adopts a redundancy-dominated configuration—favoring overlapping, robust representations—synergistic integration is suppressed, and vice versa. In addition, redundancy was positively correlated with the amplitude of conventional bivariate co-fluctuations, while synergy was negatively correlated with it; i.e., “loudest” FC events tended to reflect redundancy rather than integrative coding. At the cellular scale, large-scale two-photon imaging reveals a closely related redundancy-reliability trade-off: in the context of a mouse visual discrimination task, Ebrahimi et al. showed that population codes in the early visual cortex transiently increased their redundancy at stimulus onset, thereby boosting trial-to-trial reliability, while low-dimensional inter-areal communication modes routed sensory- and choice-related signals between visual and higher-order areas along distinct channels [263].

For each fMRI frame, Varley et al. quantified the redundant and synergistic contributions of every region and subtracted the two, yielding a time-region map that indicated whether a given area was currently dominated by redundancy or synergy. Aggregating regions into canonical Yeo networks, they observed clear system-level preferences: visual and sensorimotor systems tended to be redundancy-dominated, whereas frontoparietal and default-mode systems were typically synergy-dominated, yet all networks occasionally flipped their signs. Clustering the time points by their whole-brain redundancy-synergy profiles revealed a small set of recurring “redundancy-synergy states”. When the centroids of these states are projected back onto the cortex, some states closely resemble the average maps, while others show marked departures in which portions of the association cortex temporarily become redundancy-dominated or sensory systems transiently exhibit strong synergy. This analysis makes explicit that higher-order functional organization is dynamic: over time, the brain reallocates which the regions that implement robust, overlapping codes and those that support integrative, synergy-based computations.

PED-based analyses connect naturally to emerging views of brain communication and higher-order connectomes. Large-scale reviews of network communication models emphasize that information flow in brain networks is constrained

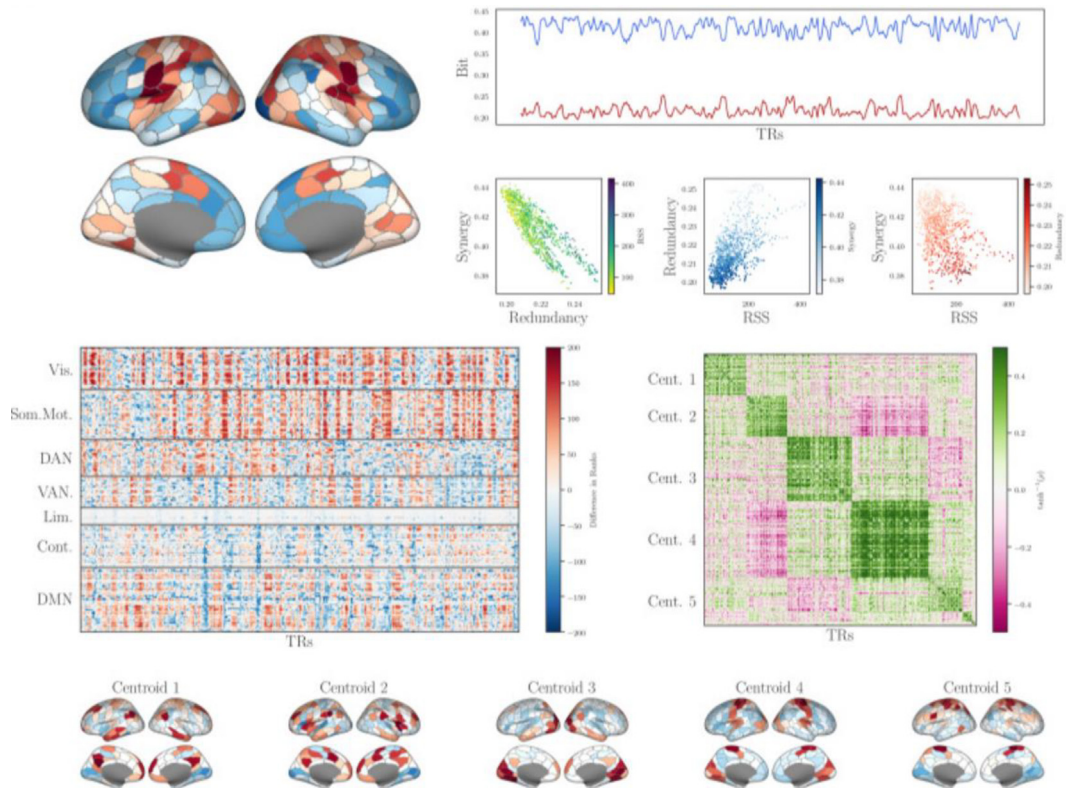


Fig. 13. Time-resolved analysis. (A) Surface plots for the distributions of relative synergies and relative redundancies across the human brain. (B) Over the course of a scan (1100 TRs), the total redundant and synergistic structure varies over time, although never so much that the curves cross (i.e., there is never more redundant structure than synergistic structure present). (C) Instantaneous redundant and synergistic structures are anticorrelated ($r = -0.83, P < 10^{-50}$). (D) Redundancy is positively correlated with the amplitude of bivariate co-fluctuations (Pearson's $r = 0.6, P < 10^{-50}$), and (E) synergy is negatively correlated with co-fluctuation amplitude (Pearson's $r = -0.43, P < 10^{-50}$). (F) For each TR, the rank-redundancy and rank-synergy are shown for each node (red indicates a higher rank-redundancy than rank-synergy and vice versa for blue). When nodes are stratified by system (gray, horizontal lines), different systems alternate between high-redundancy and high-synergy configurations in different ways. (G) For every pair of columns in Panel F, the Pearson correlation is computed to construct a time $\times$ time similarity matrix, which is then clustered using the MRCC algorithm. Rows and columns are reordered to reveal the state-structure of the time series. (H) Five example states (centroids of each community shown in Panel G) projected onto the cortical surface. The instantaneous pattern of relative synergies and redundancies varies from the average structure presented in Panel A.

Source: Reprinted from Ref. [95].

by structural connectivity, transmission delays and metabolic cost, and that different communication strategies (shortest paths, diffusion, or broadcasting) capture complementary aspects of functional interactions [264]. Studies using dynamic models based on intrinsic macroscale oscillatory modes have shown that a small number of coherent spatiotemporal patterns can account for much of resting-state functional connectivity, linking slow fMRI co-fluctuations back to underlying oscillatory dynamics [265]. Moreover, higher-order connectome approaches explicitly infer interactions involving three or more regions and demonstrate that hyperedges or simplex configurations can improve task decoding, subject identification and brain-behavior associations beyond what is possible with pairwise edges alone [197,266]. In this landscape, PED provides a principled way to quantify how much of the functional organization revealed by these models is redundant versus synergistic, and how this balance evolves over time.

Neural-system applications illustrate how explainable causal reconstruction can enrich, rather than replace, traditional functional connectivity analysis. Symmetric correlation-based networks remain useful as transparent baselines that summarize co-fluctuation structure and modular organization. Directed models such as frequency-resolved Granger causality add an explicit notion of “what drives what else, at which frequencies and time scales”, while higher-order information-theoretic tools such as PED expose synergy-rich “workspaces” that support integrative computation. By integrating these approaches with structural connectivity and biophysically grounded communication models, it becomes possible to move beyond static correlation maps toward dynamic and causally interpretable descriptions of how distributed brain regions coordinate perception and behavior, while preserving the interpretability of the resulting networks in terms of redundancy, synergy, anatomical pathways, and oscillatory communication channels.

7.3. Epidemiology

Causal relationship frameworks are being increasingly used to capture data and guide study design in epidemiology. This section reviews some examples in this research field. For instance, Granger causality has been integrated into spatial and mobility-aligned frameworks to better reflect real human movement. One approach uses commuting or GPS data to construct interregional connection networks, and then estimates GC along edges, retaining only corridors with statistically significant predictive power for regional incidence curves. The inferred “functional epidemic regions” often cross administrative boundaries, suggesting that mitigation strategies should target human-network communities rather than political units [267]. Parallel research combines GC with exogenous covariates (meteorological variables, mobility indices, and non-pharmaceutical interventions) in VARX models, enabling explicit tests of directional relationships (e.g., whether mobility reductions Granger-cause incidence declines or vice versa). In all formulations, regression coefficients represent incremental predictive effects after controlling for past processes, making the influence’s direction, timing, and conditional nature explicit.

Complementing empirical GC networks, mechanistic differential equation models transparently encode infection/recovery processes. Classic Kermack–McKendrick SIR/SEIR models describe S/E/I/R population dynamics via coupled ordinary differential equations, with transmission, progression, and recovery rates having direct epidemiological interpretations [268,269]. Parameters correspond to tangible metrics (e.g., effective contacts per infectious individual, or mean latent/infectious periods), and equation terms map to specific biological processes (infection, recovery, or waning immunity), ensuring intrinsic interpretability: interventions (school closures, mask mandates, or vaccination) translate to targeted parameter/term adjustments, not opaque high-dimensional vector changes.

Recent developments have begun to close the loop between mechanistic equations and data-driven inference while retaining this interpretability. Network-based time series models such as the Generalized Network Autoregressive (GNAR) process provide a statistically grounded analog of metapopulation dynamics: weekly COVID-19 incidence across counties is modeled as an autoregression with coefficients along network edges summarizing effective transmission strengths and revealing how dominant pathways reorganize across phases with and without mobility restrictions [270,271]. Additionally, Bayesian calibration and neural-ODE or physics-informed neural network techniques are being used to fit SIR/SEIR-type models with time-varying effective contact rates to incidence data so that each learned trajectory in parameter space can still be read as a changing transmission or recovery rate under interventions [272]. A natural workflow is to use GC networks as a fast, assumption-light tool to screen for candidate transmission corridors and dominant lags, and then embed these corridors into a mechanistic network SIR/SEIR model with parameters being estimated from the same data. In this way, epidemic applications illustrate the central theme of this review: combining GC and mechanistic differential equation approaches yields causal networks where each edge, lag and parameter can be traced back to a concrete transmission route or process, supporting both accurate forecasting and transparent intervention design.

Embedding entropy (EE) was used to detect the dynamical causality of the spread of COVID-19 across 47 Japanese prefectures using 304 days of daily new case data [28]. The analysis revealed significant causal effects from Tokyo to neighboring Kanagawa, Saitama, and Chiba, as well as the major hubs Osaka and Aichi. These results were both statistically significant and epidemiologically interpretable, aligning with known interregional mobility, an established driver of infection spread. By integrating domain knowledge, EE transcends mere correlation analysis to provide explainable causal relationships, confirming that reducing human movement is an effective intervention. The DyMES framework was also applied to model pandemic spread [4]. By incorporating total disease incidence (I) as a state variable into the transition function at the microscale of social clusters, DyMES revealed that epidemic deceleration is driven by downward causation: macroscale reported incidence influences individual behavior, reducing effective contact rates. This cross-scale interplay provides an explanation for how public health warnings modify epidemic trajectories, a mechanism that is not captured by traditional SIR models because they lack explicit bidirectional causal interactions across scales.

A causal graph was used to investigate the causal effects of weather and climate on infectious diseases [28]. The core central step was to represent the cause-and-effect relationship using a directed acyclic graph (DAG). This framework could not only help clarify hypotheses, assumptions, and potential bias sources for investigators but also allow causal inference theory to prescribe rules for determining the valid model form (if any) to answer the causal question. The above study demonstrates the practical implementation of the causal inference framework by means of a simplified mathematical model, which characterizes the population-level transmission dynamics of acute infectious diseases transmitted via direct contact between susceptible and infected hosts. While the subsequent analysis does not focus on any specific pathogen, this model, referred to as the Susceptible–Infected–Recovered–Susceptible (SIRS) model in the realm of infectious disease modeling, is highly applicable to respiratory viruses with short generation times, including influenza viruses and respiratory syncytial viruses.

A novel AI-driven Bayesian causal deep learning framework was proposed to explore the time-varying causal effects of PM_{2.5} concentrations and public health interventions on COVID-19 infection rates across the UK [28]. The key results reveal that short-term outdoor PM_{2.5} pollution exerted a significantly influenced on increasing infection rates, especially during the early phase of the pandemic. School closures proved most effective during the initial waves, whereas the suspension of public transport grew critical in the later stages. These findings provide novel implications for public health policy formulation. Early interventions aimed at reducing air pollution and improving indoor ventilation can play a critical

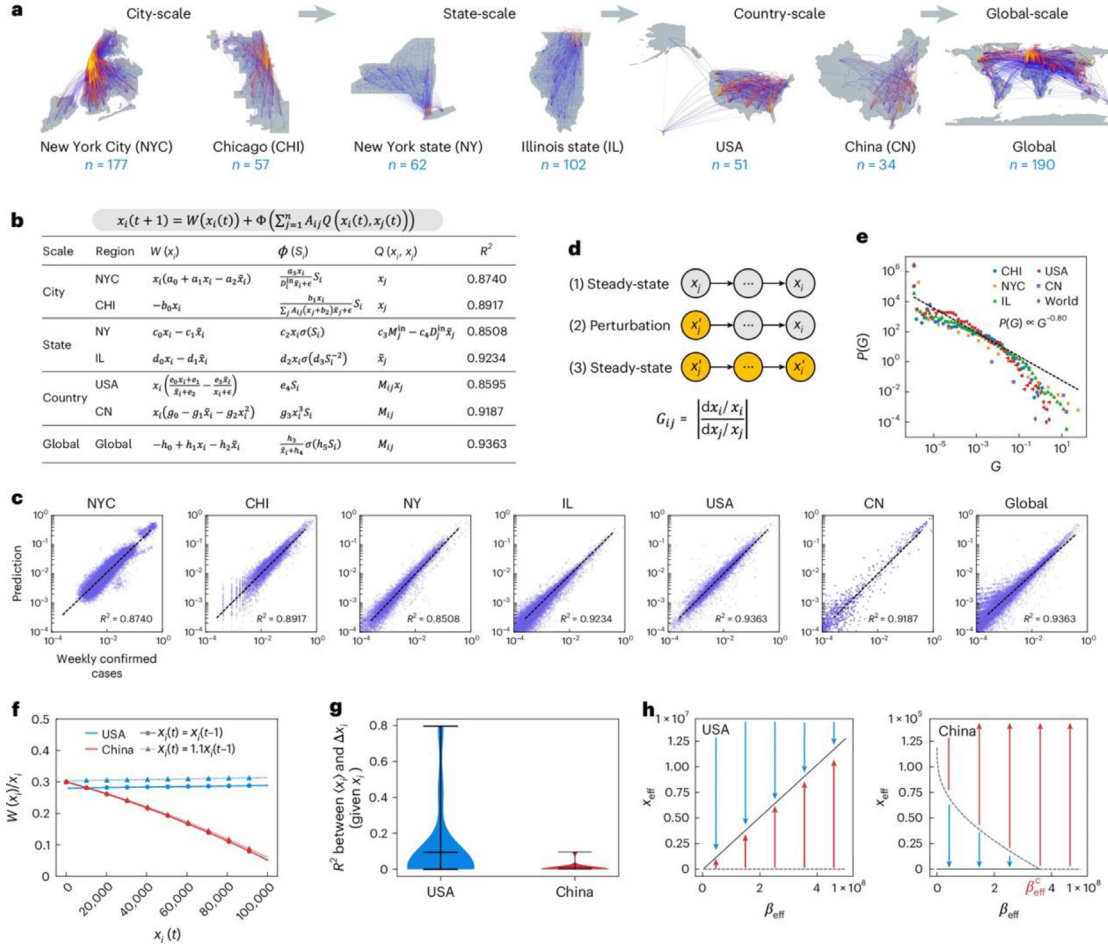


Fig. 14. Epidemic transmission dynamics. a, The dataset consists of seven different regions at four spatial scales (city, state, country and global) worldwide, where the color of curves represents the traffic flow between different areas. b, Discovered epidemic dynamics in all seven regions, where $x_i(t)$ denotes the confirmed cases in the i th area at the t th week, $\bar{x}_i(t) = x_i(t) + x_i(t-1)$ denotes the confirmed cases in history 14 days, denotes the network structure, $D_i^{in} = \sum_j A_{ij}$ denotes the in-degree of i , M_{ij} denotes the flow of people from j to i , $M_i^{in} = \sum_j M_{ij}$ denotes the total-in flow of i , $\sigma(x) = 1/(1 + \exp(-x))$ is the sigmoid function, ϵ is a small nonzero constant, and $a_i \sim h_i$ are parameters. c, Discovered dynamics fit well with the ground-truth results. d, e, Perturbation experiments (d) show the same power-law distribution properties for all discovered dynamics $P(G) \propto G^{-0.80}$ (e). f–h, Comparison of the discovered COVID-19 dynamics in the USA and China in terms of self-evolution dynamics (f), infection dynamics (g) and steady-state analysis (h). The center line and whiskers in g mark the mean and minimum/maximum values ($n = 15$). Source: Reprinted from Ref. [211].

role in effective pandemic preparedness. Furthermore, adaptive public health policies that evolve based on pandemic phases can optimize infection control efforts.

ND² was applied to discover the COVID-19 transmission dynamics in seven regions at four spatial scales: city, state, country, and global [211]. The partial results comparing the dynamics in the USA and China are shown in Fig. 14. In particular, the interaction dynamics in China were weaker than those in the U.S. because of the aggressive interventions in China.

7.4. Economic and financial systems

Macroeconomic and financial systems are highly nonlinear, nonstationary, and tightly interconnected, which makes causal inference essential for understanding systemic risk, contagion, and policy transmission. Interpretable causal methods help reconstruct hidden interactions, uncover multiscale propagation channels, and improve the robustness of predictions under structural breaks. By moving beyond correlation and static econometric models, modern information-theoretic and graph-based approaches provide deeper insights into direct causality, cross-market linkages, and regime-dependent dynamics.

The challenge of inferring causality from nonstationary and heavy-tailed financial time series has motivated the development of robust information-theoretic measures. Partial Symbolic Transfer Entropy (PSTE) extends symbolic TE

to isolate direct nonlinear effects using ranked delay vectors, which makes it resistant to mean/variance shifts and outliers [273]. When evaluated in simulation settings, PSTE reliably identified true causal directions even under severe nonlinear couplings, long-tailed noise, and structural breaks, i.e., conditions under which CGCI produced spurious or unstable links. Applied to real U.S. interest-rate and volatility data, PSTE consistently detected a nonlinear driving effect from the 10-year Treasury bond rate to the 3-month T-bill rate across raw price, return, and VAR-filtered series, and additionally uncovered a causal influence from the VIX index to short-term rates in residual data, revealing how market uncertainty shaped monetary-policy expectations. Fractional Transfer Entropy (FTE) was introduced to address the difficulty of capturing multiscale memory in market dynamics by embedding fractional calculus operators into TE [274]. Empirical analysis across major financial indices showed that short-memory configurations of FTE highlighted transient shocks, while long-memory settings revealed persistent structural dependencies that reflected slow-moving regime transitions and contagion channels.

Another research direction focuses on uncovering the nature of causal interactions rather than only their existence. Pattern Causality (PC) detects not only positive and negative but also dark causality, reflecting neither same-direction nor opposite-direction temporal pattern propagation [5]. In a 69-country sovereign CDS network, positive and negative causal links both collapsed rapidly if weak interactions were filtered out, leaving only 3–15 nodes at a threshold of 0.6 (Fig. 15 A–F). In contrast, dark causality remained extremely robust, persisting in 65 of 69 countries even under the strictest thresholding (Fig. 15 G–I). This dominance of complex, neither-mutualistic-nor-competitive interactions indicates that sovereign CDS markets operate in a regime of opaque interdependence, which is consistent with high systemic fragility and the difficulty of assessing counterparty risk.

High-dimensional financial interactions pose another challenge, motivating neural architectures that encode the causal structure explicitly. The CauGNN framework embeds a neural Granger-causal graph into a hybrid convolutional-GNN architecture, improving forecasting accuracy while yielding interpretable multivariate dependencies in financial markets [275]. Extending this idea, the Hurst-Exponent Effective Transfer-Entropy Graph Neural Network (H-ETE-GNN) integrates effective TE with Hurst exponent-based regime detection to track asymmetric directional flows and adapt to evolving market conditions [276]. Experiments on a 19-year, 10-country ETF dataset showed that H-ETE-GNN achieved the lowest RMSE (0.1623) and MAE (0.1024) among TE-, Granger-, correlation- and RNN-based baselines, and achieved the highest correlation (0.9605) with realized volatility.

Complementing market-level predictions, MagicNet offers a hierarchical causal representation of financial textual information by integrating temporal-spatial dependencies in documents with causal relations among stocks [277]. The model's dynamic GNN captures news-driven market interactions, while a causality-informed differential-equation module imposes interpretable structural constraints. Empirical evaluations demonstrated that MagicNet improved prediction stability and interpretability by revealing how information flowed through financial texts and how these flows shaped cross-stock causality. The results indicated that combining document-level signals with causal graph dynamics enhanced both the predictive performance and the mechanistic understanding of market reactions.

A different set of problems involves understanding how macro-financial structures evolve and how economic mechanisms co-move across cycles. The DyMES framework formalizes the dynamic coupling between income distribution and gross domestic income through a transition function $f(n, I)$, illustrating how macroeconomic growth trajectories coevolve with inequality dynamics and generate feedback loops influencing long-term development [4]. Recurrence plot-based analyses further revealed synchronization and convergence between U.S. financial and business cycles [278], demonstrating interpretable cyclical coordination relevant for early-warning systems and macro-financial surveillance.

Systemic risk propagation remains a central problem, and early network-based approaches laid empirical foundations for understanding contagion. Mantegna's minimum-spanning-tree construction revealed hierarchical clustering and industry-based communities in stock-return networks [39], while the partial-correlation filtering approach of Kenett et al. exposed a dominant, star-shaped organizational structure centered on major investment banks [279]. Building on these foundations, directed approaches based on Granger causality have mapped spillovers among major institutions [280], and quantified bidirectional return/volatility transmissions across global markets [281]. Sparse-group LASSO-GN uncovered high-dimensional causal links and showed that the financial crisis news topic persistently drives VIX movements, with no reverse effect from VIX to the news [282]. Studies have also focused on nonlinear contagion. During the COVID-19 pandemic, stock and bond markets acted as major risk exporters, while FX markets became the most globally connected nodes [283]. Crypto assets further amplified contagion during episodes of extreme volatility [284]. Entropy-based and time-varying networks added another layer of insight. They revealed that the connections between commodity and equity markets strengthened sharply only during crisis periods [285]. Effective TE further identified copper, silver, and crude oil as key information transmitters to equity markets, helping delineate distinct contagion phases and the broader systemic reconfiguration under financial stress [285].

Overall, contemporary approaches to causal reconstruction in macroeconomics and finance now integrates transfer-entropy methods (PSTE and FTE), Granger-based causal networks, partial-correlation filtering, DyMES, recurrence analysis, and causal graph neural networks (CauGNN, H-ETE-GNN, MagicNet). These approaches jointly advance the interpretability and reliability of macro-financial analysis, improving our capacity to identify contagion channels, diagnose systemic fragility, and support policy-oriented decision-making.

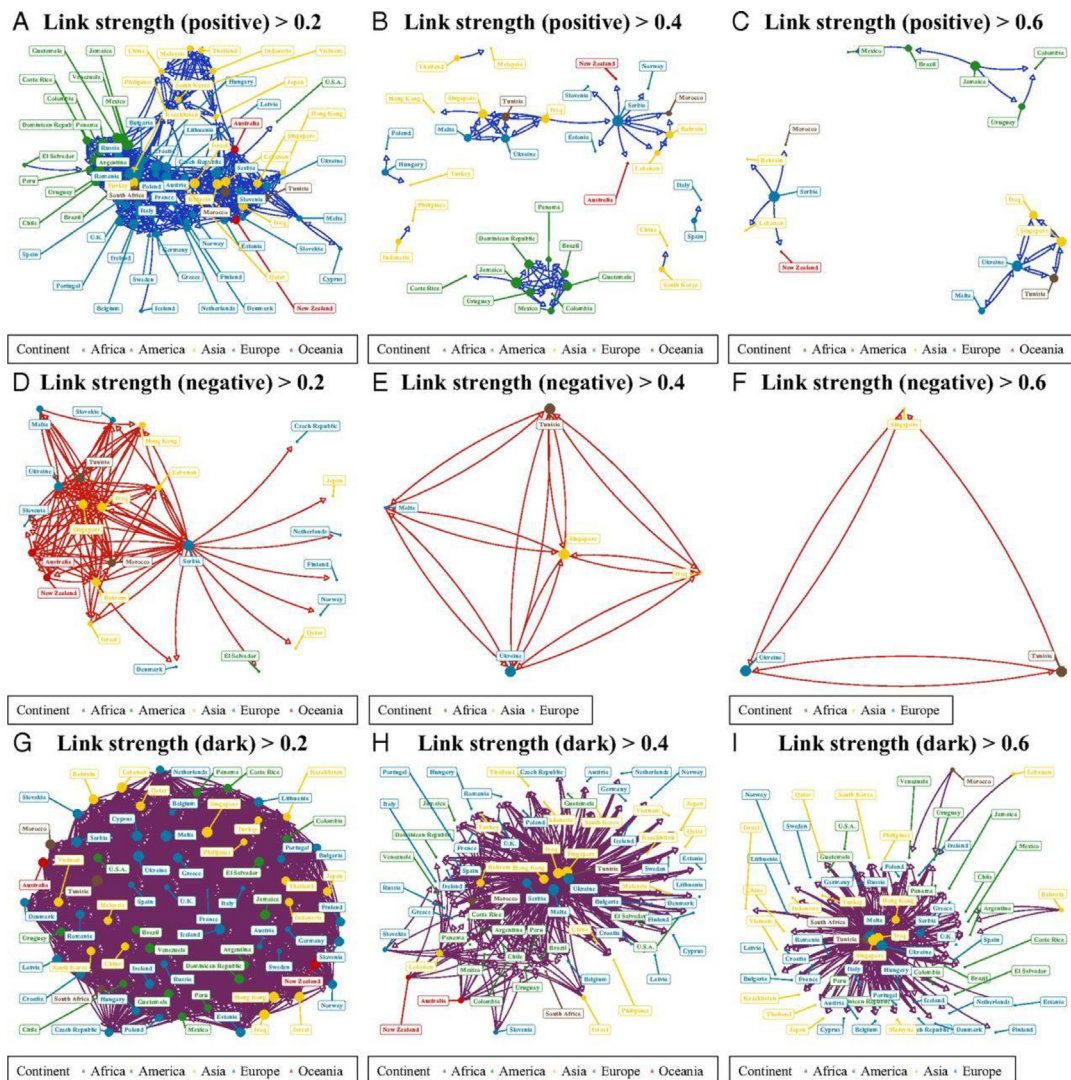


Fig. 15. Nature of causality in CDS network. (A–C) Positive aspect of the network, gradual elimination of links below 0.2 (A), 0.4 (B), and 0.6 (C) PC. (D–F) Negative aspect of the network, gradual elimination of links below 0.2 (D), 0.4 (E), and 0.6 (F) PC. (G–I) Dark aspect of the network, gradual elimination of links below 0.2 (G), 0.4 (H), and 0.6 (I) PC. Overall dark causality is the most persistent type of PC. Link color scheme: Blue is used for positive causality. Red is used for negative causality. Purple is used for dark causality. Source: Reprinted from Ref. [5].

7.5. Social science

Social systems are a paradigmatic example of human systems. A central challenge is to understand how local interactions on networks give rise to large-scale information diffusion, collective coordination and the evolution of social norms, and how platform algorithms and policy interventions reshape these processes. Recent research has converged on a toolkit based on large-scale online experiments, quantitative measures of information flows and causal inference for analyzing and redesigning social systems.

In the study of diffusion and coordination, online randomized controlled trials have clarified the role of complex contagion in social networks. Centola used online health behavior interventions on networks with different topologies and showed that when adoption required reinforcement from multiple neighbors, clustered networks with local redundancy supported wider and more durable spread than random small world networks [6]. Borge Holthoefer et al. used Twitter data and transfer entropy to construct cross regional and cross group information flow networks that mapped directional “who drives whom” relationships in information-driven coordination such as protest mobilization [286]. Fig. 16 shows the information flow balance of each specific geographical area. Across the 15M, Brazilian, and Higgs cases, diffusion was consistently kick-started by a small set of early driver urban areas, but once the first major announcement occurred,

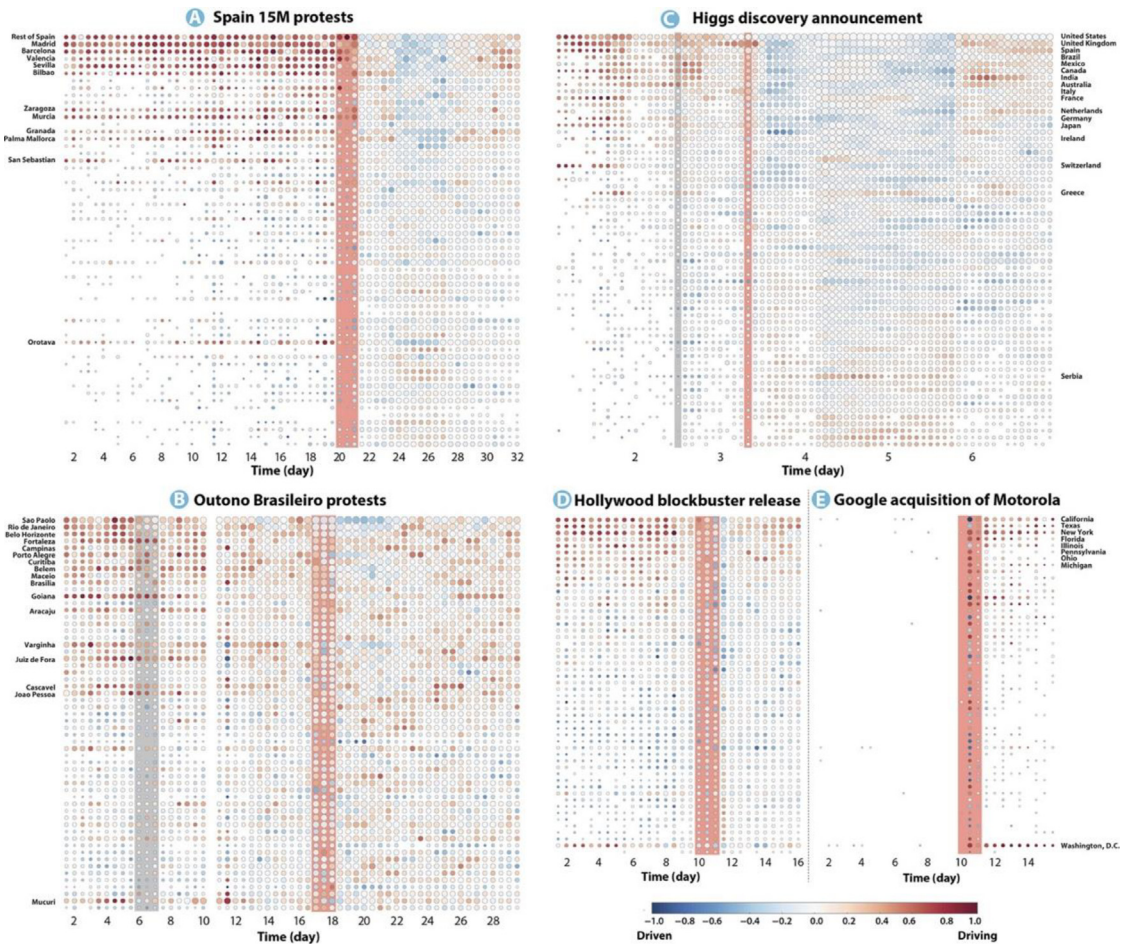


Fig. 16. Evolution of information flow balance between geographical locations for the analyzed events. (A to E) The color goes from dark blue to dark red (white corresponds to null driving), with the former standing for negative values of $\sum_y T_{x,y}^S$ (that is, driven locations) and the latter corresponding to positive information flow balances (that is, drivers). The size of the circles is log-proportional to the number of messages sent from the location at that time, and the vertical bars mark the day of the main event. The geographical locations are ordered according to population size, except for (C), in which countries are ranked with the amount of Higgs-related tweets produced. Source: Reprinted from Ref. [286].

influence rapidly spreads out, and the driver structure became increasingly diffuse or indistinguishable. Together, these studies illustrate how information theory and controlled interventions can link local contacts to collective outcomes.

Content dynamics, exemplified by the spread of misinformation and emotionally charged messages, show the coupling between content, emotion and structure in social systems. Using propagation records for 126,000 fact-checked news stories on Twitter, Vosoughi et al. showed that false news spread more widely than true news across most domains and more often evoked negative emotions such as fear and disgust [287]. Centola et al. ran online coordination experiments in which a committed minority promoting an alternative naming convention was embedded in groups that had already converged on a common label and observed a tipping point at approximately one quarter of the population at which the incumbent convention was overturned in favor of the new norm [288]. These findings identified critical masses and network structures that shaped norm change and information spread.

Beyond diffusion and content dynamics, a second line of research focuses on the systemic effects of platform algorithms and information environments. Guess et al. collaborated with major social media platforms during the 2020 U.S. election and randomly assigned some Facebook and Instagram users to strictly chronological feeds instead of algorithmically ranked feeds, showing that ranking changed the time spent and the mix of political and cross-partisan content but left issue positions and affective polarization unchanged over three months [289]. Falkenberg et al. analyzed Twitter discussions around United Nations climate conferences and observed that climate change conversations were only weakly polarized between COP20 and COP25 but showed a strong increase in ideological polarization by COP26 [290]. Cross-national studies reported that political Twitter interactions in nine countries shared a structure of partisan segregation in which cross-camp interactions were more toxic than within-camp interactions and characterized by lower engagement, revealing a similar ally-enemy pattern in online political communication [291].

On the intervention and governance side, attention has shifted from describing polarization to evaluating depolarization strategies in real-world settings. Combs et al. built an anonymous mobile chat platform that randomly paired Democratic and Republican voters for repeated conversations and observed that cross-partisan anonymous exchanges reduced standard attitude polarization measures relative to a control group [292], while Voelkel et al. showed in large-N randomized trials that several classic depolarization strategies reduced affective polarization but had little impact on anti-democratic attitudes, such as support for undemocratic candidates and tolerance of partisan violence [293]. Altay et al. using multi-country data, showed that individuals who actively followed news on social media tended to possess higher levels of factual political knowledge and belief accuracy and greater trust in media and science without stronger polarization [294], and Ecker et al. reviewed motivational, cognitive reflective and affective mechanisms that drove belief in misinformation and resistance to correction, arguing that these mechanisms helped explain why many interventions had asymmetric or short-lived effects [295].

Causal questions in such high-dimensional and strongly coupled social systems cannot be addressed solely by traditional regression or simple experiments; this has stimulated advances in causal structure learning and high-dimensional causal inference. Lagemann et al. proposed a framework for causal discovery that used deep learning and differentiable graph models to recover latent causal graphs from combinations of observational and interventional data under severe sample size constraints [296], and Martínez Sánchez et al. introduced the SURD framework, which decomposed causal influence into redundant, unique and synergistic components and distinguished the contributions of single-variable direct effects and multivariable synergistic effects on system behavior [94]. Nathaniel et al. combined deep Koopman operators with causal discovery and showed that the resulting framework could identify causal structure in nonlinear complex dynamic systems and recover key causal signals in high-dimensional climate time series [297], suggesting that these tools might also identify structural driver variables and synergistic causal clusters in social systems.

The emergence of generative artificial intelligence has created new opportunities for large-scale personalized social interventions. Costello et al. used GPT 4 Turbo to run three one-to-one conversational sessions with more than 2000 individuals who endorsed conspiracy theories and observed an approximately 20% reduction in belief in the targeted conspiracy theory that persisted for at least two months and generalized to other conspiracy beliefs and related behavioral intentions [298]. This study showed that large language models could act as programmable social actors embedded in social systems while also raising concerns about privacy, manipulation and long-term evaluation.

Overall, research on social systems is advancing along a trajectory that combines digital trace data, large-scale experimentation and causal structure learning and is moving from descriptive correlations toward mechanistic causal explanations. Online experiments and platform-level collaborations provide opportunities to identify the structural conditions under which behaviors diffuse, norms shift and polarization emerges [286–295], and new information-theoretic and causal inference methods offer tools for uncovering latent causal patterns in high-dimensional social data [94,296,297]. Moreover, AI-driven personalized interventions show that it is feasible to implement micro-level dialog-scale manipulations within macro level social systems [298], which makes normative and governance questions more pressing, especially when these approaches are extended beyond Western contexts and over longer-term trajectories of political and institutional change under ethical and data access constraints.

7.6. Ecology

Disentangling causal relationships in complex ecological systems is essential for addressing climate change impacts, biodiversity loss, and adaptive environmental management. Traditional correlation-based approaches often conflate spurious associations with genuine causality and fail to capture nonlinear, state-dependent dynamics, limiting reliable predictions. Interpretable causality that quantifies causal direction, strength, and spatiotemporal variability has emerged as a critical need in the context of mechanistically explaining ecosystem interactions. This review focuses on three leading causal inference methods in ecology: empirical dynamic modeling (EDM), quasi-experimental designs, and SCM with fixed-effects regression. For each method, we synthesize its core principles, ecological applications, key findings, and the challenges it resolves, highlighting how these tools advance beyond correlation to deliver actionable mechanistic insights for ecosystem science and management.

EDM is a nonparametric framework rooted in nonlinear dynamic systems theory, designed to detect causal relationships without restrictive distributional assumptions [12]. It leverages state-space reconstruction to infer causal links, encompassing co-prediction (identifying shared dynamics), convergent cross-mapping (CCM, quantifying directionality), and S-map analysis (estimating time-varying interaction strengths). EDM has been transformative for resolving nonlinear, weakly connected processes: in the California Current System, Kuriyama [7] used co-prediction to show that temperature, salinity, and ichthyoplankton diversity share nonlinear dynamics, improving the hindcast skill by 23% and clarifying the role of temperature as a key community driver. Exploring terrestrial carbon cycling, Ping [299] applied CCM to European heatwave data, detecting bidirectional causality between gross primary productivity (GPP) and ecosystem respiration (Re) and observing that heatwaves enhanced GPP's causal effect on Re, explaining why ecosystems maintained carbon sink resilience. Ushio [300] used an S-map to uncover seasonal fluctuations in fish community interactions where weak competition in the summer stabilized communities, while strong predation dominated in the winter, resolving static models' failure to capture temporal variability. The results are shown in Fig. 17. Hybrid EDM-mechanistic models further address nonstationarity: Deyle [301] integrated EDM with hydrodynamic simulations for Lake Geneva, showing that a 3 °C

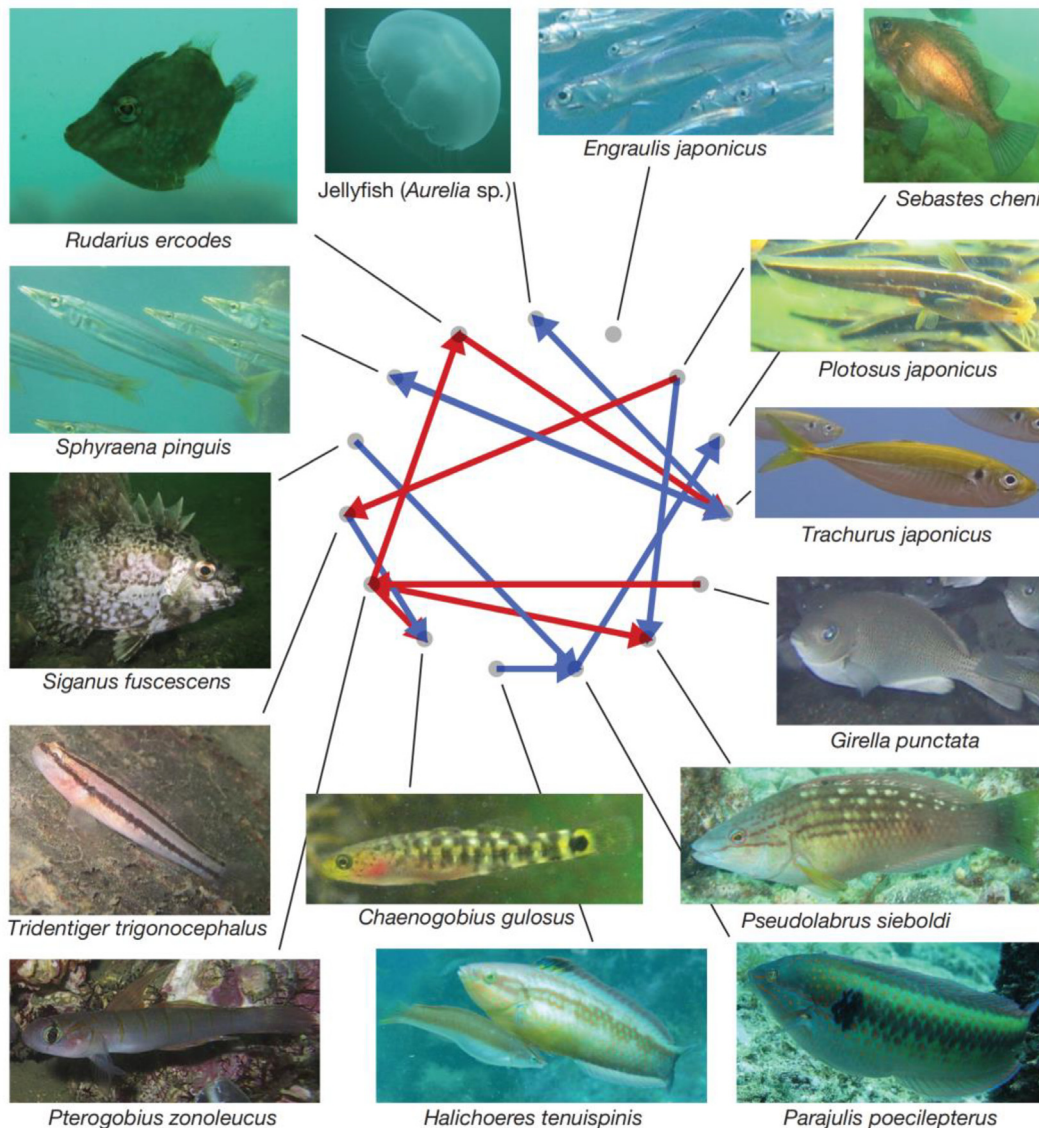


Fig. 17. Reconstructed interaction network of a subset of the Maizuru Bay fish community. Arrows indicating interspecific interactions are assigned on the basis of the results of convergent cross mapping (CCM). Blue and red colors indicate positive and negative interactions, respectively, calculated by the S-map method based on the 12-year average.

Source: Reprinted from Ref. [300].

warming and eutrophication had comparable water quality impacts, challenging single-factor management and advocating for integrated policies.

Quasi-experimental designs exploit natural or policy-driven variations to isolate causal effects, critical for large-scale or long-term processes where controlled experiments are unfeasible. Key approaches include difference-in-differences (DiD, comparing pre-/post-treatment changes between affected and control groups) and regression discontinuity design (RDD, leveraging sharp thresholds to compare adjacent units). These methods resolve the confounding problem associated with observational studies: Schrodtt [302] used DiD to analyze European grasslands, observing that agricultural expansion indirectly reduced mammal diversity indirectly via forest fragmentation rather than direct habitat loss, guiding targeted reforestation. Xu [303] applied RDD to protected areas, showing that they buffered land surface temperature by 0.8 °C relative to unprotected forests; the sharp boundary of reserves minimized confounding from topography, confirming that human disturbance was the driver. In wildfire ecology, Grace [304] used DiD to estimate the Glacier National Park's forest canopy recovery, taking 16 years, with sensitivity tests validating robustness to trend violations and informing fire recovery timelines.

SCMs paired with DAGs formalize causal assumptions by mapping confounders, mediators, and colliders, while fixed-effects panel regression controls for time-invariant (e.g., soil type) confounders and those varying with site or time (e.g., regional climate) confounders, isolating within-unit effects. Together, they link causal effects to mechanistic processes: Dee [305] used DAGs to identify precipitation as a critical confounder in grassland biodiversity–productivity relationships, showing omitting it biased richness effect estimates by 35%. Fixed-effects regression further showed that, in natural systems, greater plant richness was linked to lower productivity, likely due to resource competition from rare species, contrasting with experimental findings and refining the “biodiversity-enhances-productivity” paradigm. In fire ecology, Grace [304] combined SCMs with nonlinear regression to identify a fire severity threshold: sites above it failed to recover (remaining effect: -0.2 density units after 22 years), while those below it regained pre-fire levels. DAGs confirmed that fire severity directly drove recovery, providing a mechanistic basis for prioritizing low-severity fire regimes.

Machine learning/AI has become pivotal in ecology, addressing the limitations of traditional methods. In particular, Double Machine Learning (DML) and Targeted Maximum Likelihood Estimation (TMLE) handle high-dimensional confounders and nonlinearity to estimate causal effects, e.g., by disentangling honeybee impacts on wild bees [306]. The Deep Instrumental Variables (Deep IV) resolves unmeasured confounding or bidirectional relationships via deep neural networks and valid instruments. Causal forests uncover heterogeneous effects, such as depth’s context-dependent influence on kelp abundance [306,307]. These methods outperform correlational approaches, aiding large-scale observational data analysis. Some scholars highlight generative AI’s potential: it augments scarce data (e.g., rare species in camera traps) via GANs and diffusion models, extends ecological pattern observations (e.g., simulating animal movements), integrates with mechanistic models (e.g., generative agent-based models for animal decision-making), and enhances data accessibility (e.g., extracting information from non-English texts) [308,309]. However, challenges persist, including model biases, high energy consumption, and privacy risks. Interdisciplinary collaboration is key to leveraging AI for robust ecological insights [310].

7.7. Molecular biology

The inference of gene regulatory networks (GRNs) [311] remains a central challenge in biological systems. In advancing beyond simple correlation-based models, causal reconstruction in gene regulation is essential for understanding intricate mechanisms and distinguishing drivers from responders. Specifically, explainable causal reconstruction addresses primary challenges in gene regulation. Recent advances in single-cell RNA sequencing (scRNA-seq) have spurred the development of computational frameworks capable of analyzing GRNs at single-cell resolution, enabling a deeper understanding of cell type-specific and dynamic regulatory mechanisms.

Chen [312] proposed using a single cell-specific causal network (SiCNet) to construct molecular regulatory networks at the single-cell level. The study identified several potential high-activity regulators, such as CEBPB, TRAF3, and BATF3, during the lineage of monocyte cell development. Zhou [313] introduced a Single-cell Network Using Mutual Information (SINUM) to estimate pairwise gene dependencies within each cell, which effectively tracked temporal changes in gene-gene associations during embryonic development. The cited study identified cell-type marker genes and gene pairs that exhibited differential connectivity but no differential gene expression, such as MTHFD1 and MTHFD1-IFI6 in HFF cells. Li [116] proposed scNME to infer directed GRNs from single-cell RNA-seq data. The study identified cells from F_c3 that were highly enriched in the PD state, suggesting F_c3 might be a novel cancer-associated fibroblast related to drug resistance. Deshpande et al. [314] proposed Single-cell Inference of Networks using Granger Ensembles (SINGE), demonstrating that *Asxl2* and *Tet1* might play critical regulatory roles in hematopoiesis regulation and its associated diseases. Lee [315] developed TENET to successfully identify key regulators such as *Nme2* in embryonic stem cells and during cardiomyocyte reprogramming. TE scores were significantly elevated for validated target genes and sensitive to regulator perturbations, underscoring TE’s utility in revealing functional gene hierarchies.

Gao [316] proposed CausCell to combine SCM with diffusion models to increase the interpretability of single-cell data. The scholars observed that the age-upregulated genes in Micro were enriched in the striatum domain. *Lilrb4a* was also identified as the gene with the most enriched pathways, including T-cell activation, leucocyte adhesion and type 2 immune response. *Lilrb4a* might be a common regulator of aging across various cell types in the brains of mice. Shu [317] introduced the DeepSEM method to identify 12 potential upstream transcription factors regulating *Rorb* in L4 cells, including genes *Mef2c*, *Nr2f1* and *Pknox2*.

CauFinder [318] was proposed to integrate the delineated causal features with prior network knowledge using a novel weighted feedback vertex set (FVS) method. It showed a striking directional asymmetry in LUAD–LUSC transdifferentiation: LUSC bifurcates into two trajectories from a common LUAD origin, yet robust reversibility (transition from LUSC to LUAD) was largely confined to only one branch. This pattern implied a controllable intermediate state and highlighted that state accessibility was inherently direction dependent. CellOracle [319] has recently been proposed to simulate the transcription factor perturbations. The study supported the existence of this relatively mild Gata1 phenotype. In addition, it accurately recapitulated the previously reported role of *Tal1* in regulating cell fate specifically at the early stages of erythropoiesis. Furthermore, CellOracle was also used to reconstruct the underlying GRN for brassinosteroids [320]. The study demonstrated that *HAT7* and *GTL1* were brassinosteroid-responsive regulators along cortex trajectories that controlled cell elongation. CIMLA [321], considering counterfactual inference and machine learning, revealed the potential key regulators of AD, *CREB3* and *NEUROD6*, and additional potential regulators such as *LYL1* and *HOXD1*.

Zhang [50] applied a statistics-based CVP method to identify the regulatory targets of the functional driver genes SNRNPP200 and RALGAPB in the liver cancer. Meng [322] performed ancestry-specific two-sample MR analyses, observing that MD genetic liability increased the odds of a higher BMI, with the effect estimate of 0.092 (95%CI 0.024–0.161, $P = 0.00812$), which had not been previously reported. Importantly, the positive and bidirectional relationship between MD genetic liability and BMI was only observed in samples from individuals of European ancestry. In samples of individuals of East Asian ancestry, decreasing TGs appeared to increase the odds of MD, suggesting that the direction of causal relationships between MD and metabolic traits might differ between European and East Asian ancestry groups. Justin [63] introduced SWING to simultaneously evaluate multiple upstream regulators at several potential time delays. The study identified 10- and 20-minute time delays between *crp* and target genes in the *E. coli* *tdcABC* operon.

Most of these advances try to detect the true interventional causality without requiring real-world experimental manipulations. KOCMI [133], aligned with the do-calculus-based causal analysis, was proposed to detect interventional direct causality and quantify causal strength by leveraging the distributional invariance observed before and after this virtual intervention. Specifically, the cited study used KOCMI to explore the causal relationship between microbial phylum levels and physiological skin indicators, and 265 forehead samples and 277 cheek samples were gathered (Fig. 18A), where each sample included measurements of both the skin microbiome and physiological skin indicators. Permutation test analyses demonstrated that physiological skin indicators, transepidermal water loss (TEWL), sebum, pH and porphyrin, which are exhibited significant variations between the forehead and cheek regions (Fig. 18B), and the difference in microbiomes might be the cause or the effect. The KOCMI framework was established for the dataset, which could help infer causal regulatory relationships exerted by the microbiota on forehead and cheek skin physiological indicators. The directionality of causality (positive or negative) was determined based on the Pearson correlation coefficient (PCC) (Fig. 18C).

In an analysis of breast cancer data with 2184 genes, KOCMI identifies 9600 regulatory relationships. Among the inferred GRNs, several hub genes, such as HLA-E, VIM, and ACTA2, and their downstream genes and shared genes were identified and are shown in Fig. 18D. Notably, HLA-E is widely observed to be overexpressed in breast cancer cells and is implicated in tumor immune escape. VIM expression can be used as a biomarker for the prognosis of triple-negative breast cancer (TNBC). ACTA2 overexpression accelerated the invasiveness and metastasis of breast cancer cells. Moreover, the shared upstream regulators of these hub genes may be associated with breast cancer's occurrence and progression. The top 10 enriched potential drugs based on DSigDB infer several compounds that have already been shown to have anti-cancer properties; examples are 5-Azacytidine, fenretinide, Pyrvinium, and Pentoxifylline. Hence, these genes may serve as promising drug targets for further cancer treatment. The top 10 enriched potential pathways (Fig. 18F) show that one of the most crucial pathways was the phosphatidylinositol 3-kinase–Akt. Mutations in PIK3CA promotes tumor survival, growth, and metastasis.

These studies illustrate how causal reconstruction has evolved from simple correlation-based networks into a multifaceted toolbox that integrates cellular heterogeneity, temporal dynamics and causal directionality. These advanced methodologies originate from information theory, SCM, and interventional proxies. However, challenges remain. Many predicted regulators still lack experimental validation. Scalability to whole-transcriptome or full multiomics data is nontrivial. Future advances should focus on integrating causal inference, representation learning, and interventional experiments, to construct GRNs that are not only interpretable but also mechanistically valid and experimentally testable.

The empirical application of causality detection approaches is strongly domain-dependent, with different fields favoring different methodological families according to their characteristic data structures, modeling assumptions, and interpretability requirements. To make these cross-domain patterns more explicit, Fig. 19 provides a synthesized view of how representative causality detection approaches have evolved over time and how they have been adopted across empirical domains. The map shows that Granger causality and its extensions remain broadly used across disciplines and continue to be particularly influential in economic and financial systems, where temporal predictability and lagged dependence are central concerns. Information-theoretic approaches are widely adopted in climate systems, neuroscience, epidemiology, and social science because of their ability to characterize nonlinear and asymmetric information transfer. Nonlinear state–space approaches have developed rapidly in ecology and molecular biological systems, where feedback, nonlinear coupling, and state-dependent interactions are common. In molecular biology, graphical causal approaches and AI-enhanced approaches are increasingly used to recover structured causal relationships. This may be because these data usually combine high dimensionality, nonlinear regulation, strong heterogeneity, measurement noise, and partial observability. In such settings, AI-extended approaches offer greater flexibility in learning nonlinear causal dependencies.

8. Outlook and future directions

8.1. Classic vs. AI-driven approaches

Both classic and AI-driven approaches to interpretable causal network reconstruction from time series possess distinct strengths and limitations, with classic methods excelling in transparency and theoretical rigor, while AI-driven methods offer superior scalability and flexibility in handling complex data; their complementary natures drive the emerging trend of hybrid frameworks that integrate mechanistic insights with data-driven learning. Nevertheless, some differences exist between the two approaches in terms of foundational assumptions, mechanisms enabling interpretability, robustness and performance, computational efficiency, and domain suitability.

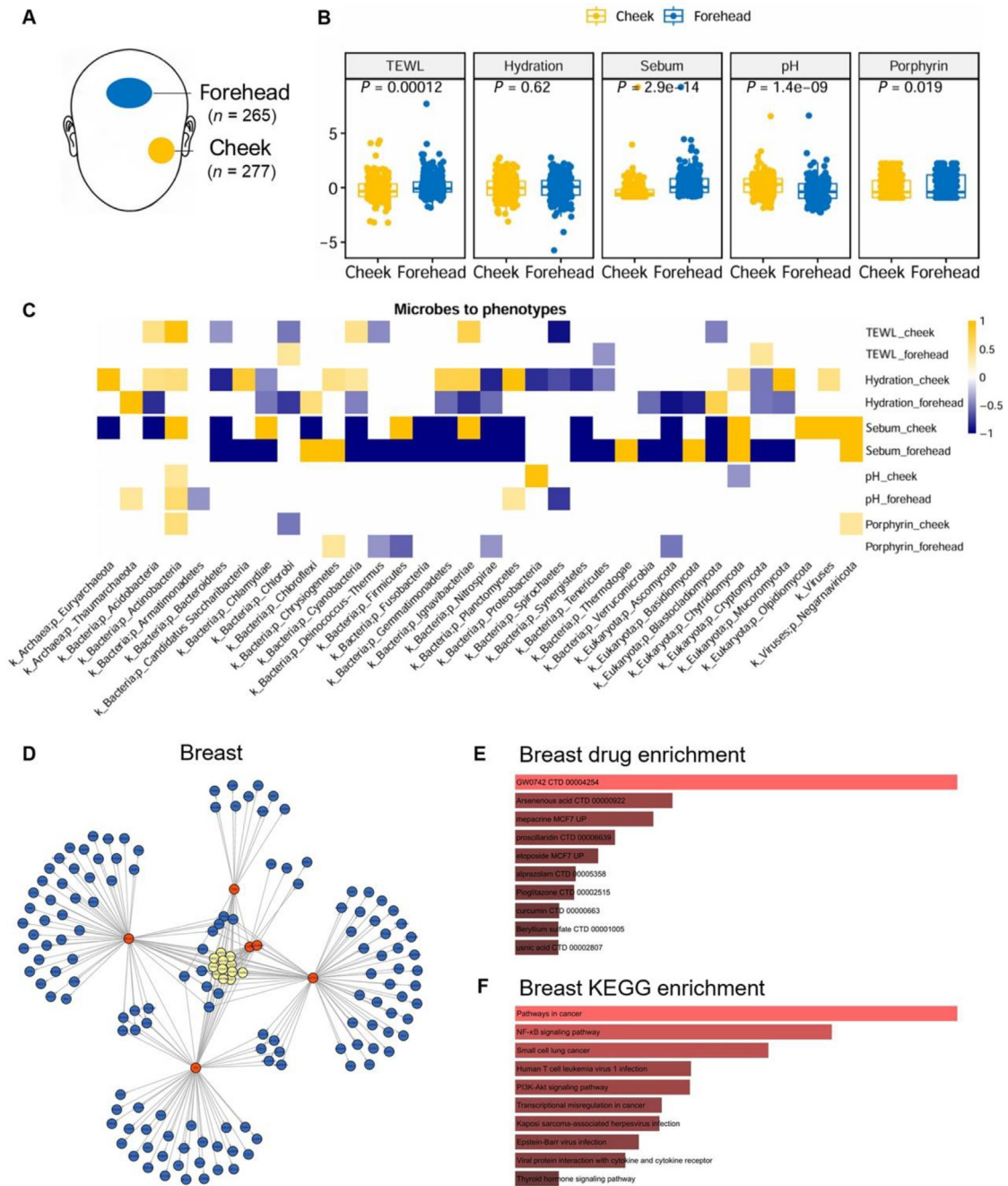


Fig. 18. Validations of KOCMI using metagenomic data and breast cancer data. (A) A total of 265 forehead samples and 277 cheek samples were collected and measured for skin microbiome and skin physiological indicators. (B) The difference of skin physiological indicators between cheek and forehead. (C) The causality identified by KOCMI from microbiota to skin physiological indicators. A positive value represents positive regulation, while a negative value represents negative regulation. The absolute value represents the causal strength. (D) A subnetwork of the network reconstructed by KOCMI using breast cancer data, composing of hub genes and their downstream and shared upstream genes. (E and F) Drug enrichment analysis and Kyoto Encyclopedia of Genes and Genome (KEGG) pathway enrichment analysis for the hub genes and their shared upstream genes. NF- κ B, nuclear factor κ B; PI3K, phosphatidylinositol 3-kinase. Source: Reprinted from Ref. [133].

(1) The foundational assumptions of many classic methods include explicit, often restrictive assumptions to ensure causal identifiability. Statistical association approaches (e.g., Granger causality) assume linearity and stationarity [10], while nonlinear state-space methods (e.g., convergent cross mapping) require low-dimensional deterministic dynamics [12]. Graph models (e.g., PCMC) depend on causal sufficiency, the Markov property, and faithfulness [48,323],

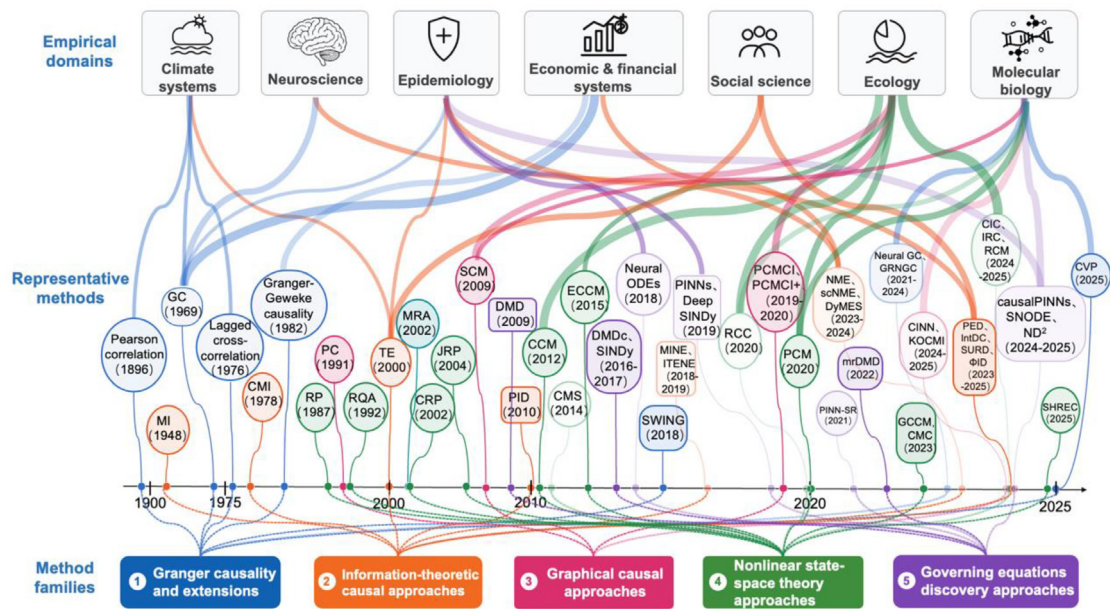


Fig. 19. Evolution and empirical domains of representative causal reconstruction approaches. The timeline summarizes the historical development of selected representative methods. Curves connect these methods to the empirical domains in which they have been widely applied. The bottom labels group these methods into five broad methodological families.

which may not hold in the presence of latent confounders or nonstationarity. These assumptions enable rigorous theoretical guarantees but limit the applicability to real-world data with complex characteristics. On the other hand, AI-driven approaches often relax parametric assumptions, leveraging neural networks to model nonlinear, nonstationary relationships without explicit structural priors [71,121]. However, AI-driven approaches introduce implicit requirements: large amounts of high-quality data to avoid overfitting, and a stable data distribution for generalization. Methods such as neural Granger causality and PINNs bypass linearity constraints but still depend on indirect assumptions such as feature relevance and network architecture suitability. The trade-off shifts from strict modeling assumptions to data quality and computational resources.

(2) Classic methods are inherently interpretable by design, producing transparent outputs consistent with domain knowledge. Statistical and information-theoretic approaches (e.g., transfer entropy) yield quantifiable metrics for causal strength and direction. Graph models output explicit directed acyclic graphs where edges directly represent causal links [48,120]. Mechanistic methods (e.g., SINDy and DMD) recover symbolic equations or modal structures that can be directly mapped to physical or biological mechanisms, facilitating actionable scientific insights. AI-driven methods typically require additional efforts to achieve interpretability, as deep neural networks are often regarded as “black boxes”. Post hoc explanation techniques (e.g., attention weights, and feature attribution) help identify influential variables, while structured architectures (e.g., causal graph neural networks, and neural ODEs) embed interpretability through design. Hybrid models (e.g., physics-informed neural networks) integrate domain constraints into loss functions, ensuring that outputs adhere to known physical laws [233,324,325]. However, the interpretability of AI-driven methods remains partial, and is often limited to local explanations or structural approximations rather than a full mechanistic understanding [32,34].

(3) Classic methods demonstrate robust performance under ideal conditions but degrade if assumptions are violated. They handle low-dimensional data effectively but struggle with high-dimensional systems due to computational complexity and multiple-testing issues. Classic methods’ robustness is affected not only by noise but also by latent confounders and temporal nonstationarity (e.g., regime shifts in time series). Noise tolerance varies: information-theoretic methods (e.g., symbolic transfer entropy) are relatively robust to noise [99,273], while linear Granger causality suffers from spurious correlations in noisy environments [10,56]. Such methods’ performance is highly dependent on data stationarity and completeness. In contrast, AI-driven approaches excel in high-dimensional, noisy, and partially observed systems, leveraging representation learning to extract meaningful features from raw data [71,206]. Regularization techniques enhance resilience to noise, and deep architectures naturally handle nonlinear dependencies. However, AI-driven methods require sufficient data to avoid overfitting and may fail to generalize to distribution shifts. Recent advances (e.g., latent convergent cross mapping) address data scarcity issues, while causal regularization improves robustness to confounding variables [233,234]. AI methods outperform classic approaches in complex real-world scenarios but lack the consistent reliability of classic methods in well-constrained problems [15,192].

(4) Classic methods are computationally efficient for problems with scales ranging from small to moderate (dozens of variables), with linear time complexity for pairwise measures (e.g., correlation, and Granger causality). However, their

Table 1
Suitable conditions of interpretable causal reconstruction approaches.

Method	Data length	Dimension	Noise tolerance	Missing data	Nonstationarity	Nonlinearity	Interpretability
Granger causality and extensions	Long	Low-moderate	Moderate	Complete observations	Stationarity required	Nonlinear variants	Temporal predictability
Information-theoretic approaches	Long	Low-moderate	Moderate	Complete observations	No requirement	Nonlinear	Information flow
Graphical causal approaches	Long	High	Moderate	Partially tolerant	No requirement	Linear and nonlinear	Causal graph
Nonlinear state-space theory approaches	Short to medium	Low-moderate	Low	Complete observations	Nonstationarity tolerated	Nonlinear	Manifold reconstruction
Governing equations discovery approaches	Dense	Low-moderate	Low	Complete observations	Nonstationary variants	Nonlinear	Explicit equations

complexity increases combinatorially with the number of variables, making them impractical for high-dimensional data due to repeated conditional independence tests or matrix operations [10,38,120]. AI-driven approaches leverage parallel computing and GPU acceleration to handle large-scale data efficiently [212]. End-to-end optimization frameworks enable simultaneous evaluation of all variable interactions, avoiding the combinatorial explosion of classic methods. However, training deep models requires significant computational resources and multiple passes over the dataset. Memory overhead is also a concern for large networks, though sparse architectures and model compression techniques mitigate this issue. For extremely high-dimensional systems, AI-driven methods are superior in terms of scalability compared to classic approaches [324,325].

(5) Classic methods are well-suited for domains with established theoretical frameworks (e.g., physics and engineering) where mechanistic insights are prioritized. They are preferred in fields with limited data (e.g., ecology and small-scale neuroscience) and if interpretability is critical for validation [12,258]. Methods such as SINDy and DMD are widely used in engineering and climate science because of their ability to recover governing equations [15,192,199]. On the other hand, AI-driven approaches dominate in data-rich, complex domains (e.g., systems biology, finance, and social networks), where nonlinear interactions and high dimensionality are prevalent [5,116,319]. They are ideal for exploratory analysis in emerging fields with incomplete theoretical models. PINNs and neural ODEs are being increasingly used in climate science and biology, while graph neural networks excel in social network analysis and brain connectivity mapping [253,319]. Hybrid methods are particularly valuable in interdisciplinary research, as they combine classic domain knowledge with AI’s predictive power [206,233].

In summary, both classic and AI-driven approaches offer complementary strengths for causal network reconstruction. Classic methods provide transparency, theoretical rigor, and reliability in well-constrained problems, while AI-driven methods deliver scalability, flexibility, and performance in complex, data-rich environments [8,32,206]. The future of causal reconstruction is in hybrid frameworks that integrate classic mechanistic insights with AI’s learning capabilities, addressing the limitations of both approaches. Such integration will enable more robust, interpretable, and generalizable causal inference across diverse scientific domains [206,229].

8.2. Practical approach selection

While the integration of AI significantly enhances scalability and representation power, it does not alter the foundational assumptions or boundary conditions of the underlying causal paradigms. Practical approach selection also depends on the intrinsic properties of the data, including data length, dimensionality, noise level, missingness, nonstationarity and nonlinearity, as well as the desired form of causal interpretation. To serve as a comprehensive navigation map for practical application, Table 1 therefore summarizes the suitable empirical conditions for major interpretable causal reconstruction approaches.

Granger causality and extensions remain attractive for long, low- to moderate-dimensional systems because they provide a clear interpretation in terms of temporal predictability. However, they usually require nonlinear variants and stationarity treatment for strongly nonlinear or nonstationary systems. Information-theoretic approaches are well suited to capturing nonlinear and asymmetric information flow, but their performance depends on sample size, estimator choice, and high-dimensional conditioning. Local estimation is often required for nonstationarity. Graphical causal approaches are useful when the objective is to infer multivariate causal structures, especially in high-dimensional systems where dynamic graph variants are available. Nonlinear state–space approaches are particularly relevant for systems governed by nonlinear dynamical coupling, but they generally require relatively complete trajectories for reliable local reconstruction. Governing-equation discovery approaches provide explicit mechanistic interpretability by seeking dynamic equations, but they generally require dense observations and appropriate equation structures. Overall, this comparison emphasizes that method selection should be matched to both empirical data conditions and the desired form of causal interpretation.

8.3. Challenges

Reconstructing causal network dynamics from time series data has become a cornerstone of scientific inquiry across physics, genomics, neuroscience, and economics, offering profound insights into the underlying mechanisms of complex systems [1,2,8]. While traditional statistical methods and emerging AI-driven approaches have made significant strides, numerous challenges persist in translating theoretical advances into practical applications [22,23]. We highlight five critical bottlenecks in the field and propose three promising research directions to address these limitations, aiming to guide future efforts toward more robust, reliable, and interpretable causal network reconstruction.

First, enhancing the robustness of causal inference methods against low-quality time series data remains an urgent priority. Real-world time series are often plagued by missing values, class imbalance, and structured noise, which can severely distort causal estimates and lead to spurious connections [48]. Conventional methods, such as Granger causality and dynamic Bayesian networks [326], typically assume ideal data conditions, which makes these methods vulnerable to such imperfections [327,328]. Although some imputation techniques or noise reduction algorithms have been integrated into existing frameworks, they often treat data preprocessing and causal inference as separate stages, failing to account for the interdependencies between data quality and causal structure [207,208]. Future studies can develop end-to-end robust methods that explicitly model data imperfections during causal inference. This could involve incorporating probabilistic models to quantify uncertainty from missing values [122,124], designing imbalance-aware loss functions for AI-driven approaches, or leveraging adaptive regularization techniques to suppress noise-induced spurious correlations [42,69].

Second, the lack of comprehensive benchmark datasets hinders the rigorous evaluation of causal network reconstruction methods [29,252]. Current validation techniques primarily rely on synthetic data or small-scale real-world datasets, which often fail to capture the complexity of real-world systems (e.g., high dimensionality, non-stationarity, and latent variables) [8,49]. Moreover, existing benchmarks rarely provide standardized metrics for assessing robustness across different data quality scenarios, which makes it difficult to compare the performance of traditional and AI-driven methods fairly [296,297]. To address this gap, there is a pressing need to establish large-scale, multi-domain benchmark datasets that include diverse data quality challenges and provide ground-truth causal structures [50,121]. Such benchmarks should also incorporate domain-specific characteristics, such as the dynamic causal relationships in gene regulatory networks [311,314] or brain activity data [258,259], enabling more accurate assessments of method validity and generalizability.

Third, identifying high-order causal relationships represents a major frontier in the field [93,95]. Most existing methods focus on pairwise or low-order causal interactions, overlooking the complex high-order dependencies prevalent in many systems [94]. For example, in multivariate time series, the causal effect of one variable on another may depend on the state of multiple intermediate variables or exhibit time-lagged higher-order interactions [329]. Traditional methods struggle to model such complex relationships due to their linearity assumptions of the methods, while AI-driven approaches often lack effective mechanisms for extracting and interpreting high-order causal structures [94,266]. It is promising to develop scalable high-order causal inference methods, particularly for high-dimensional time series. This could involve integrating tensor decomposition techniques with AI models to efficiently capture high-order interactions [197], or extending information-theoretic frameworks to quantify high-order causal entropy [93].

Fourth, improving domain-specific interpretability is crucial for the practical adoption of causal network reconstruction methods [32]. While traditional methods offer inherent interpretability through explicit graphical structures, they often lack the ability to model complex nonlinear dynamics [12]. In contrast, AI-driven methods excel at capturing nonlinear relationships but are often criticized as “black boxes”, with their causal inferences being difficult to validate in domain-specific applications [35]. To enhance interpretability, future efforts could focus on developing domain-adaptive interpretability frameworks. It would be helpful to incorporate domain knowledge into model design (e.g., using clinical guidelines to constrain causal structures in medical data [228]) or develop post hoc interpretation tools that translate model outputs into domain-specific insights (e.g., highlighting key causal paths in gene regulatory networks [311,319]).

Fifth, rigorous uncertainty quantification in causal inference remains an underdeveloped area, posing risks to the reliability of reconstructed causal networks [122]. A key limitation of most existing methods is their failure to systematically capture and propagate uncertainty throughout the causal reconstruction process [248]. Such uncertainty originates from multiple interconnected sources: low-quality time series data (e.g., missing values, noise, and imbalance) [207,208], model misspecification (e.g., incorrect assumptions about linearity or stationarity) [212,330], and inherent ambiguity in complex causal relationships (e.g., overlapping causal paths or latent confounders) [124,181]. To address this gap, it is possible to prioritize integrating uncertainty quantification (UQ) into the core of causal inference frameworks [247]. This involves moving beyond point estimates of causal links to generating probabilistic causal structures, where each potential causal interaction is accompanied by a measure of confidence or uncertainty [247]. Such advances would not only enhance the robustness of causal inferences but also enable domain experts to make more informed decisions by distinguishing between high-confidence and low-confidence causal relationships [133,321].

8.4. Perspectives

We summarize four promising directions. The first goal is to unite rigorous statistical discovery with data-rich dynamic modeling in a single framework that spans the range from identifying network structure to capturing the underlying

mechanistic equations. A central trajectory for the field of causal network reconstruction is a deep integration of classic causal inference techniques with modern AI-driven modeling [331]. Traditional methods provide strong theoretical guarantees and interpretability, while AI models offer powerful nonlinear modeling and scalability [332]. For example, the Granger Causality Kolmogorov–Arnold Networks (GC-KAN) [333] combines the interpretability of Granger causality [10] with the nonlinear mapping capabilities of Kolmogorov–Arnold Networks [334], achieving competitive performance in inferring causal relationships from nonlinear time series [79]. Similarly, Temporal Causal Attention frameworks integrate causal inference principles with attention mechanisms, filtering out spurious correlations while maintaining interpretability [335]. Future integration efforts could involve using traditional methods to constrain the structure of AI models (e.g., using Bayesian network priors to regularize neural networks [327]) or using AI models to enhance traditional methods (e.g., using neural ODEs to model dynamic causal structures in Bayesian networks [212]).

The second possible direction is leveraging neural networks to simulate digital twins for intervention experiments [208]. Intervention is the gold standard for validating causal relationships, but it is often impractical in real-world systems (e.g., medical treatments or ecological systems [299]). Digital twins, i.e., virtual replicas of physical systems, offer a solution by enabling simulated interventions to test causal hypotheses [228]. Neural networks can be used to construct high-fidelity digital twins of complex systems, capturing their dynamic causal relationships from time series data [210]. Extending this idea, AI-driven digital twins could be used to simulate the effects of perturbing different variables, allowing researchers to validate causal inferences and identify high-order causal relationships that are difficult to detect from observational data alone [228]. This approach also enhances interpretability by providing tangible evidence of causal effects through simulated interventions, making it easier for domain experts to validate and act on the results [133,321].

The third direction focuses on advancing uncertainty-aware causal inference frameworks that integrate probabilistic modeling with causal discovery [122]. Building on the challenges highlighted earlier, this direction aims to address the lack of systematic uncertainty quantification by designing end-to-end frameworks that model uncertainty explicitly during causal reconstruction [296,297]. This could involve integrating Bayesian probabilistic models (e.g., Bayesian neural networks) with causality discovery algorithms to generate probabilistic causal graphs, where each edge is associated with a confidence interval or a probability distribution reflecting the uncertainty of the causal relationship [122]. Additionally, uncertainty-aware frameworks can enhance robustness against low-quality data by explicitly modeling noise and missing data as probabilistic variables instead of treating them as nuisances to be preprocessed away [207,208]. In practice, such frameworks would enable domain experts to make more informed decisions, for example, in clinical settings, prioritizing interventions based on causal relationships with low uncertainty [321]. Moreover, uncertainty quantification can facilitate more rigorous validation of causal models on benchmark datasets, as it provides a richer metric than simple accuracy, capturing both the validity and the reliability of inferred causal structures [296].

The fourth direction is ensuring that the outputs of these advanced methods are transparent and trustworthy. As causal models become central to high-stakes decisions, for example in financial markets or medical interventions, simply mapping out a network of links is no longer sufficient [336]. Decision-makers also need to understand how and why those links exist [336], and how confident one can be in them [337]. Accordingly, researchers are focusing on interpretability (providing human-understandable explanations of causal relationships) and on rigorous uncertainty quantification to assess the reliability of each inferred link, including its robustness remains under perturbations or changing conditions. This push for trustworthy inference goes hand-in-hand with efforts to systematically evaluate and compare new methods. In particular, evaluation criteria are evolving beyond simple structural accuracy, and methods are now judged by how well they elucidate the underlying mechanisms and by the stability of methods' predictions under uncertainty. The community is also building a shared benchmarking infrastructure, including standardized datasets and evaluation metrics for both structural accuracy and dynamic fidelity, to test algorithms across multiple domains and ensure that progress is grounded in reproducible empirical evidence. By validating approaches on common datasets in fields such as climate science, biology, neuroscience, and economics, researchers can identify the strengths and weaknesses of different techniques and observe how well methods generalize across disparate domains, driving continual improvement while also building the confidence needed for such tools to be adopted in practice. Together, these developments are converging into a new paradigm of causal discovery that fuses data-driven learning with mechanistic insight. This paradigm is poised to shift the discipline from merely identifying statistical associations toward uncovering causal mechanisms that can directly inform interventions. In the long term, such an approach could establish a robust foundation for quantitative decision support in complex problem domains.

CRedit authorship contribution statement

Xiangyun Gao: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Xiaotian Sun:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Tao Wu:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Sufang An:** Writing – review & editing, Writing – original draft. **Feng An:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Siyang Leng:** Writing – review & editing, Funding acquisition. **Hongyu Wei:** Writing – review & editing, Writing – original draft. **Yupeng Zhang:** Writing – review & editing, Writing – original draft. **Norbert Marwan:** Writing – review & editing, Supervision, Conceptualization. **Jürgen Kurths:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] M. Schlüter, C. Brelsford, P.J. Ferraro, K. Orach, M. Qiu, M.D. Smith, Unraveling complex causal processes that affect sustainability requires more integration between empirical and modeling approaches, *Proc. Natl. Acad. Sci.* 120 (2023) e2215676120.
- [2] J. Runge, V. Petoukhov, J.F. Donges, J. Hlinka, N. Jajcay, M. Vejmelka, D. Hartman, N. Marwan, M. Paluš, J. Kurths, Identifying causal gateways and mediators in complex spatio-temporal systems, *Nat. Commun.* 6 (2015) 8502.
- [3] Z. q.K. Tian, K. Chen, S. Li, D.W. McLaughlin, D. Zhou, Causal connectivity measures for pulse-output network reconstruction: Analysis and applications, *Proc. Natl. Acad. Sci.* 121 (2024) e2305297121.
- [4] J. Harte, M. Brush, K. Umemura, P. Muralikrishnan, E.A. Newman, Dynamical theory of complex systems with two-way micro-macro causation, *Proc. Natl. Acad. Sci. USA* 121 (2024) e2408676121.
- [5] S.K. Stavrogliou, A.A. Pantelous, H.E. Stanley, K.M. Zuev, Hidden interactions in financial markets, *Proc. Natl. Acad. Sci.* 116 (2019) 10646–10651.
- [6] D. Centola, The spread of behavior in an online social network experiment, *Science* 329 (2010) 1194–1197.
- [7] P.T. Kuriyama, G. Sugihara, A.R. Thompson, B.X. Semmens, Identification of shared spatial dynamics in temperature, salinity, and ichthyoplankton community diversity in the California current system with empirical dynamic modeling, *Front. Mar. Sci.* 7 (2020) 557940.
- [8] J. Runge, S. Bathiany, E. Bollt, G. Camps-Valls, D. Coumou, E. Deyle, C. Glymour, M. Kretschmer, M.D. Mahecha, J. Muñoz-Marí, E.H. van Nes, J. Peters, R. Quax, M. Reichstein, M. Scheffer, B. Schölkopf, P. Spirtes, G. Sugihara, J. Sun, K. Zhang, J. Zscheischler, Inferring causation from time series in earth system sciences, *Nat. Commun.* 10 (2019) 2553.
- [9] T. Wu, X. Gao, F. An, X. Sun, H. An, Z. Su, S. Gupta, J. Gao, J. Kurths, Predicting multiple observations in complex systems through low-dimensional embeddings, *Nat. Commun.* 15 (2024) 2242.
- [10] C.W. Granger, Investigating causal relations by econometric models and cross-spectral methods, *Econ.: J. Econ. Soc.* (1969) 424–438.
- [11] T. Schreiber, Measuring information transfer, *Phys. Rev. Lett.* 85 (2000) 461–464.
- [12] G. Sugihara, R. May, H. Ye, C.H. Hsieh, E. Deyle, M. Fogarty, S. Munch, Detecting causality in complex ecosystems, *Science* 338 (2012) 496–500.
- [13] N.M. Mangan, S.L. Bruntton, J.L. Proctor, J.N. Kutz, Inferring biological networks by sparse identification of nonlinear dynamics, *IEEE Trans. Mol. Biol. Multi-Scale Commun.* 2 (2016) 52–63.
- [14] C.W. Rowley, I. Mezić, S. Bagheri, P. Schlatter, D.S. Henningson, Spectral analysis of nonlinear flows, *J. Fluid Mech.* 641 (2009) 115–127.
- [15] P.J. Schmid, Dynamic mode decomposition of numerical and experimental data, *J. Fluid Mech.* 656 (2010) 5–28.
- [16] M. Bunge, *Causality and Modern Science: Third Revised Edition*, Dover Publications, 2012.
- [17] P. Suppes, *A Probabilistic Theory of Causality*, North-Holland Publishing Company, 1970.
- [18] E. Eells, *Probabilistic Causality*, Cambridge University Press, 1991.
- [19] D. Lewis, Causation, *J. Philos.* 70 (1973) 556–567.
- [20] J.Y. Halpern, J. Pearl, Causes and explanations: A structural-model approach. Part I: Causes, *British J. Philos. Sci.* (2005).
- [21] J. Woodward, *Making Things Happen: A Theory of Causal Explanation*, Oxford University Press, 2005.
- [22] J. Pearl, *Causality*, Cambridge University Press, 2009.
- [23] J. Pearl, M. Glymour, N.P. Jewell, *Causal Inference in Statistics: A Primer*, John Wiley & Sons, 2016.
- [24] N. Wiener, The theory of prediction, *Mod. Math. Eng.* (1956).
- [25] C.E. Shannon, A mathematical theory of communication, *Bell Syst. Tech. J.* 27 (1948) 379–423.
- [26] K. Hlaváčková-Schindler, M. Paluš, M. Vejmelka, J. Bhattacharya, Causality detection based on information-theoretic approaches in time series analysis, *Phys. Rep.* 441 (2007) 1–46.
- [27] A. Lozano-Durán, G. Arranz, Information-theoretic formulation of dynamical systems: Causality, modeling, and control, *Phys. Rev. Res.* 4 (2022) 023195.
- [28] J. Shi, L. Chen, K. Aihara, Embedding entropy: a nonlinear measure of dynamical causality, *J. R. Soc. Interface* 19 (2022) 20210766.
- [29] I. Brugere, B. Gallagher, T.Y. Berger-Wolf, Network structure inference, A survey: Motivations, methods, and applications, *ACM Comput. Surv.* 51 (2018) Article 24.
- [30] B. Barzel, A.-L. Barabási, Universality in network dynamics, *Nat. Phys.* 9 (2013) 673–681.
- [31] S. An, X. Gao, X. Sun, T. Wu, H. An, H. Huang, X. Liu, Forecasting early warning signals for abrupt transitions in oil systems: A deep learning-based network and regime switching model, *Eng. Manag.* (2026) <http://dx.doi.org/10.1007/s42524-026-5377-y>.
- [32] W.J. Murdoch, C. Singh, K. Kumbier, R. Abbasi-Asl, B. Yu, Definitions, methods, and applications in interpretable machine learning, *Proc. Natl. Acad. Sci.* 116 (2019) 22071–22080.
- [33] H. Hagras, Toward human-understandable, explainable AI, *Computer* 51 (2018) 28–36.
- [34] T. Miller, Explanation in artificial intelligence: Insights from the social sciences, *Artificial Intelligence* 267 (2019) 1–38.
- [35] S. Bassan, G. Amir, G. Katz, Local vs. global interpretability: A computational complexity perspective, 2024, arXiv preprint arXiv:2406.02981.
- [36] D.D. Nauck, Measuring interpretability in rule-based classification systems, in: *The 12th IEEE International Conference on Fuzzy Systems*, 2003. FUZZ '03, vol. 191, 2003, pp. 196–201.
- [37] E. Aguirre-Zapata, H. Alvarez, L. Lema-Perez, F. di Sciascio, A.N. Amicarelli, Interpretability in the modeling spectrum: A conceptual framework and a quantification index, *Ecol. Model.* 498 (2024) 110882.
- [38] K. Pearson, Mathematical contributions to the theory of evolution. III. Regression, heredity, and panmixia, *Philos. Trans. R. Soc. Lond. Ser. A, Contain. Pap. A Math. Or Phys. Character* 187 (1896) 253–318.
- [39] R.N. Mantegna, Hierarchical structure in financial markets, *Eur. Phys. J. B - Condens. Matter Complex Syst.* 11 (1999) 193–197.

- [40] A. Grinsted, J.C. Moore, S. Jevrejeva, Application of the cross wavelet transform and wavelet coherence to geophysical time series, *Nonlin. Process. Geophys.* 11 (2004) 561–566.
- [41] L.D. Haugh, Checking the independence of two covariance-stationary time series: A univariate residual cross-correlation approach, *J. Amer. Statist. Assoc.* 71 (1976) 378–385.
- [42] S. Epskamp, E.I. Fried, A tutorial on regularized partial correlation networks, *Psychol. Methods* 23 (2018) 617–634.
- [43] N. Boers, B. Goswami, A. Rheinwalt, B. Bookhagen, B. Hoskins, J. Kurths, Complex networks reveal global pattern of extreme-rainfall teleconnections, *Nature* 566 (2019) 373–377.
- [44] N. Marwan, M.C. Romano, M. Thiel, J. Kurths, Recurrence plots for the analysis of complex systems, *Phys. Rep.-Rev. Sect. Phys. Lett.* 438 (2007) 237–329.
- [45] Z. Cao, Y. Zhang, H. Chen, C. Li, Y. Luo, Understanding the influencing factors and evolving trends of the yellow river water-sediment regulation system from a system perspective, *Front. Eng. Manag.* 11 (2024) 528–541.
- [46] G. Nolte, A. Ziehe, V.V. Nikulin, A. Schlögl, N. Krämer, T. Brismar, K.R. Müller, Robustly estimating the flow direction of information in complex physical systems, *Phys. Rev. Lett.* 100 (2008) 234101.
- [47] A.A. Tsonis, P.J. Roebber, The architecture of the climate network, *Phys. A* 333 (2004) 497–504.
- [48] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, D. Sejdinovic, Detecting and quantifying causal associations in large nonlinear time series datasets, *Sci. Adv.* 5 (2019) eaau4996.
- [49] P. Nowack, J. Runge, V. Eyring, J.D. Haigh, Causal networks for climate model evaluation and constrained projections, *Nat. Commun.* 11 (2020) 1415.
- [50] Y. Zhang, Q. Li, J. Wang, X. Chang, L. Chen, X. Liu, Causal network inference based on cross-validation predictability, *Commun. Phys.* 8 (2025) 173.
- [51] A. Roebroeck, E. Formisano, R. Goebel, Mapping directed influence over the brain using granger causality and fMRI, *NeuroImage* 25 (2005) 230–242.
- [52] H. Yahyaie, V. Kitsios, L. De Mello, Risk transference between climate variability and financial derivatives: Implications for global food security, *J. Clim. Financ.* 7 (2024) 100038.
- [53] A.A. Salisu, A.E. Ogbonna, R. Gupta, Y. Zhang, Climate policy uncertainty and the forecastability of inflation, *J. Clim. Financ.* 14 (2026) 100080.
- [54] H. Iltkepohl, *New Introduction to Multiple Time Series Analysis*, Springer Publishing Company, Incorporated, 2007.
- [55] L. Barnett, A.K. Seth, The MVGC multivariate granger causality toolbox: A new approach to granger-causal inference, *J. Neurosci. Methods* 223 (2014) 50–68.
- [56] F. Diebold, K. Yilmaz, On the network topology of variance decompositions: Measuring the connectedness of financial firms, *J. Econometrics* 182 (2014) 119–134.
- [57] J. Geweke, Measurement of linear dependence and feedback between multiple time series, *J. Amer. Statist. Assoc.* 77 (1982) 304–313.
- [58] M. Dhamala, G. Rangarajan, M. Ding, Estimating granger causality from Fourier and wavelet transforms of time series data, *Phys. Rev. Lett.* 100 (2008) 018701.
- [59] M.J. Kamiński, K.J. Blinowska, A new method of the description of the information flow in the brain structures, *Biol. Cybernet.* 65 (1991) 203–210.
- [60] L.A. Baccalá, K. Sameshima, Partial directed coherence: a new concept in neural structure determination, *Biol. Cybernet.* 84 (2001) 463–474.
- [61] D.Y. Takahashi, L.A. Baccalá, K. Sameshima, Information theoretic interpretation of frequency domain connectivity measures, *Biol. Cybernet.* 103 (2010) 463–469.
- [62] T. Tian, D. Liu, G. Zhang, J. Wang, C. Wan, J. Fang, D. Wu, Y. Zhou, Y. Qin, H. Zhu, Y. Li, C. Liu, J. Zhang, J. Li, W. Zhu, Dynamic effective connectivity among large-scale brain networks mediates risk of anxiety, *Hum. Brain Mapp.* 44 (2023) 3730–3743.
- [63] J.D. Finkle, J.J. Wu, N. Bagheri, Windowed granger causal inference strategy improves discovery of gene regulatory networks, *Proc. Natl. Acad. Sci.* 115 (2018) 2252–2257.
- [64] G.E. Primiceri, Time varying structural vector autoregressions and monetary policy, *Rev. Econ. Stud.* 72 (2005) 821–852.
- [65] A. Bitto, S. Frühwirth-Schnatter, Achieving shrinkage in a time-varying parameter model framework, *J. Econometrics* 210 (2019) 75–97.
- [66] M. Caporin, M. Costola, J.-C. Garibal, B. Mailet, Systemic risk and severe economic downturns: A targeted and sparse analysis, *J. Bank. Financ.* 134 (2022) 106339.
- [67] N. Ancona, D. Marinazzo, S. Stramaglia, Radial basis function approach to nonlinear granger causality of time series, *Phys. Rev. E* 70 (2004) 056221.
- [68] D. Marinazzo, M. Pellicoro, S. Stramaglia, Kernel-granger causality and the analysis of dynamical networks, *Phys. Rev. E* 77 (2008) 056215.
- [69] N.-J. Hsu, H.-L. Hung, Y.-M. Chang, Subset selection for vector autoregressive processes using lasso, *Comput. Statist. Data Anal.* 52 (2008) 3645–3657.
- [70] A. Shojaie, G. Michailidis, Discovering graphical granger causality using the truncating lasso penalty, *Bioinformatics* 26 (2010) 517–523.
- [71] A. Tank, I. Covert, N. Foti, A. Shojaie, E.B. Fox, Neural granger causality, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (2022) 4267–4279.
- [72] A. Duggento, M. Guerrisi, N. Toschi, Echo state network models for nonlinear granger causality, *Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 379 (2021) 20200256.
- [73] A. Montalto, S. Stramaglia, L. Faes, G. Tessitore, R. Prevete, D. Marinazzo, Neural networks with non-uniform embedding and explicit validation phase to assess granger causality, *Neural Netw.* 71 (2015) 159–171.
- [74] Y. Antonacci, L. Minati, L. Faes, R. Pernice, G. Nollo, J. Toppi, A. Pietrabissa, L. Astolfi, Estimation of granger causality through artificial neural networks: applications to physiological systems and chaotic electronic oscillators, *PeerJ Comput. Sci.* 7 (2021) e429.
- [75] M. Rosol, M. Młyńczak, G. Cybulski, Granger causality test with nonlinear neural-network-based methods: Python package and simulation study, *Comput. Methods Programs Biomed.* 216 (2022) 106669.
- [76] W. Zhou, S. Bai, S. Yu, Q. Zhao, B. Chen, Jacobian regularizer-based neural granger causality, in: *Proceedings of the 41st International Conference on Machine Learning, JMLR.org, Vienna, Austria, 2024*, Article 2555.
- [77] S. Khanna, V.Y.F. Tan, Economy statistical recurrent units for inferring nonlinear granger causality, in: *International Conference on Learning Representations*, 2020.
- [78] B. Bussmann, J. Nys, S. Latré, Neural additive vector autoregression models for causal discovery in time series, in: *International Conference on Discovery Science*, Springer, 2021, pp. 446–460.
- [79] R. Marcinkevičs, J. Vogt, Interpretable models for granger causality using self-explaining neural networks, in: *International Conference on Learning Representations*, 2021.
- [80] Z. Yin, P. Barucca, Deep recurrent modelling of granger causality with latent confounding, *Expert Syst. Appl.* 207 (2022) 118036.
- [81] Suryadi, L.Y. Chew, Y.-S. Ong, Granger causality using Jacobian in neural networks, *Chaos: Interdiscip. J. Nonlinear Sci.* 33 (2023).
- [82] M. Liu, H. Dong, X. Yang, Y. Xu, Z. Li, Z. Si, X. Yang, Z. Zhao, Gradient regularization-based neural granger causality, 2025, *arXiv preprint arXiv:2507.11178*.
- [83] M. Wang, Z. Fu, A new method of nonlinear causality detection: Reservoir computing granger causality, *Chaos Solitons Fractals* 154 (2022) 111675.

- [84] X. Li, Q. Zhu, C. Zhao, X. Duan, B. Zhao, X. Zhang, H. Ma, J. Sun, W. Lin, Higher-order granger reservoir computing: simultaneously achieving scalable complex structures inference and accurate dynamics prediction, *Nat. Commun.* 15 (2024) 2506.
- [85] T.M. Cover, *Elements of Information Theory*, John Wiley & Sons, 1999.
- [86] A. Kraskov, H. Stögbauer, P. Grassberger, Estimating mutual information, *Phys. Rev. E* 69 (2004) 066138.
- [87] Y.I. Moon, B. Rajagopalan, U. Lall, Estimation of mutual information using kernel density estimators, *Phys. Rev. E Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.* 52 (1995) 2318–2321.
- [88] A.D. Wyner, A definition of conditional mutual information for arbitrary ensembles, *Inf. Control* 38 (1978) 51–59.
- [89] D.J. Albers, G. Hripcsak, Using time-delayed mutual information to discover and interpret temporal correlation structure in complex populations, *Chaos* 22 (2012).
- [90] A.M. Fraser, H.L. Swinney, Independent coordinates for strange attractors from mutual information, *Phys. Rev. A* 33 (1986) 1134–1140.
- [91] P.L. Williams, R.D. Beer, Nonnegative decomposition of multivariate information, *Physics* 1004 (2012).
- [92] S.E. Chiang, Z. Liu, R.L. Peach, M. Barahona, Information-theoretic signatures of causality in Bayesian networks and hypergraphs, 2025, arXiv preprint [arXiv:2512.10552](https://arxiv.org/abs/2512.10552).
- [93] P.A. Mediano, F.E. Rosas, A.I. Luppi, R.L. Carhart-Harris, D. Bor, A.K. Seth, A.B. Barrett, Toward a unified taxonomy of information dynamics via integrated information decomposition, *Proc. Natl. Acad. Sci.* 122 (2025) e2423297122.
- [94] A. Martínez-Sánchez, G. Arranz, A. Lozano-Durán, Decomposing causality into its synergistic, unique, and redundant components, *Nat. Commun.* 15 (2024).
- [95] T.F. Varley, M. Pope, M.G. Puxeddu, J. Faskowitz, O. Sporns, Partial entropy decomposition reveals higher-order information structures in human brain activity, *Proc. Natl. Acad. Sci. USA* 120 (2023).
- [96] D. Massaro, S. Rezaeiravesh, P. Schlatter, On the potential of transfer entropy in turbulent dynamical systems, *Sci. Rep.* 13 (2023).
- [97] L. Barnett, A.B. Barrett, A.K. Seth, Granger causality and transfer entropy are equivalent for Gaussian variables, *Phys. Rev. Lett.* 103 (2009) 238701.
- [98] V.A. Vakorin, O.A. Krakovska, A.R. McIntosh, Confounding effects of indirect connections on causality estimation, *J. Neurosci. Methods* 184 (2009) 152–160.
- [99] M. Staniek, K. Lehnertz, Symbolic transfer entropy, *Phys. Rev. Lett.* 100 (2008).
- [100] K. Keller, M. Sinn, Ordinal analysis of time series, *Phys. A-Stat. Mech. Appl.* 356 (2005) 114–120.
- [101] D. Kugiumtzis, Partial transfer entropy on rank vectors, *Eur. Phys. J.-Special Top.* 222 (2013) 401–420.
- [102] D. Kugiumtzis, Transfer entropy on rank vectors, *Eur. Phys. J. Spec. Top.* 222 (2010).
- [103] J. Shi, Y. Li, J. Zhao, S. Leng, K. Aihara, L. Chen, W. Lin, Deciphering interventional dynamical causality from non-intervention systems, 2024, [arXiv:2407.01621](https://arxiv.org/abs/2407.01621).
- [104] M.I. Belghazi, A. Baratin, S. Rajeswar, S. Ozair, Y. Bengio, A. Courville, R.D. Hjelm, MINE: Mutual information neural estimation, 2018.
- [105] M. Donsker, Asymptotic evaluation of certain Markov process for large time, I, *Comm. Pure Appl. Math.* 28 (1975).
- [106] A. Keziou, Dual representation of φ -divergences and applications, *C. R. Math.* 336 (2003) 857–862.
- [107] X.L. Nguyen, M.J. Wainwright, M.I. Jordan, Estimating divergence functionals and the likelihood ratio by convex risk minimization, *IEEE Trans. Inform. Theory* 56 (2010) 5847–5861.
- [108] S. Nowozin, B. Cseke, R. Tomioka, F-gan: Training generative neural samplers using variational divergence minimization, *Adv. Neural Inf. Process. Syst.* 29 (2016).
- [109] A.V.D. Oord, Y. Li, O. Vinyals, Representation learning with contrastive predictive coding, 2018.
- [110] J. Zhang, O. Simeone, Z. Cvetkovic, E. Abela, M.P. Richardson, ITENE: Intrinsic transfer entropy neural estimator, in: *International Zurich Seminar on Information and Communication (IZS 2020)*. Proceedings, 2020.
- [111] R.G. James, B.D.M. Ayala, B. Zakirov, J.P. Crutchfield, Modes of information flow, 2018, arXiv preprint [arXiv:1808.06723](https://arxiv.org/abs/1808.06723).
- [112] O. Luxembourg, D. Tsur, H. Permuter, TREET: Transfer entropy estimation via transformer, 2024, [arXiv:2402.06919](https://arxiv.org/abs/2402.06919).
- [113] K.A. Murphy, D.S. Bassett, Information decomposition in complex systems via machine learning, *Proc. Natl. Acad. Sci. USA* 121 (2024) e2312988121.
- [114] I. Estella Aguerri, A. Zaidi, Distributed information bottleneck method for discrete and Gaussian sources, in: *International Zurich Seminar on Information and Communication (IZS 2018)*. Proceedings, ETH Zurich, 2018, pp. 35–39.
- [115] I.E. Aguerri, A. Zaidi, Distributed variational representation learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (2019) 120–138.
- [116] L. Li, R. Xia, W. Chen, Q. Zhao, P. Tao, L.A. Chen, Single-cell causal network inferred by cross-mapping entropy, *Brief. Bioinform.* 24 (2023) bbad281.
- [117] M. Eichler, Causal inference with multiple time series: principles and problems, *Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 371 (2013) 20110613.
- [118] J. Pearl, Causal inference, causality: objectives assessment, 2010, pp. 39–58.
- [119] J. Peters, D. Janzing, B. Schölkopf, *Elements of Causal Inference: Foundations and Learning Algorithms*, The MIT Press, 2017.
- [120] P. Spirtes, C. Glymour, An algorithm for fast recovery of sparse causal graphs, *Soc. Sci. Comput. Rev.* 9 (1991) 62–72.
- [121] Z. Wang, X. Shi, L. Shu, X. Yin, K. Zhou, P. Xu, Quantifying climate factor contributions to groundwater level changes under different soil freezing-thawing states with the WT-PCMC model, *J. Hydrol.* 656 (2025) 132997.
- [122] J. Runge, Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets, in: *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence, PMLR, Virtual*, 2020, pp. 1388–1397.
- [123] E. Saggioro, J. de Wiljes, M. Kretschmer, J. Runge, Reconstructing regime-dependent causal relationships from observational time series, *Chaos: Interdiscip. J. Nonlinear Sci.* 30 (2020) 113–115.
- [124] A. Gerhardus, J. Runge, High-recall causal discovery for autocorrelated time series with latent confounders, 2020, arXiv e-prints, [arXiv:2007.01884](https://arxiv.org/abs/2007.01884).
- [125] S. Arif, M.A. MacNeil, Applying the structural causal model framework for observational causal inference in ecology, *Ecol. Monograph.* 93 (2023) e1554.
- [126] J. Splawa-Neyman, D.M. Dabrowska, T.P. Speed, On the application of probability theory to agricultural experiments. Essay on principles. Section 9, *Statist. Sci.* (1990) 465–472.
- [127] D.B. Rubin, Estimating causal effects of treatments in randomized and nonrandomized studies, *J. Educ. Psychol.* 66 (1974) 688–701.
- [128] H. Pu, Z. Chen, J. Liu, X. Yang, C. Ren, H. Liu, Y. Jian, Research on decision-level fusion method based on structural causal model in system-level fault detection and diagnosis, *Eng. Appl. Artif. Intell.* 126 (2023) 107095.
- [129] J. Pearl, D. Mackenzie, *The Book of Why: The New Science of Cause and Effect*, Basic books, 2018.
- [130] F. Iglesias-Suarez, P. Gentine, B. Solino-Fernandez, T. Beucler, M. Pritchard, J. Runge, V. Eyring, Causally-informed deep learning to improve climate models and projections, *J. Geophys. Res.: Atmospheres* 129 (2024) e2023JD039202.
- [131] S. Rasp, M.S. Pritchard, P. Gentine, Deep learning to represent subgrid processes in climate models, 115, 2018, pp. 9684–9689.
- [132] T. Beucler, P. Gentine, J. Yuval, A. Gupta, L. Peng, J. Lin, S. Yu, S. Rasp, F. Ahmed, P.A. O’Gorman, Climate-invariant machine learning, *Sci. Adv.* 10 (2024) eadj7250.

- [133] X. Zhang, L. Chen, Quantifying interventional causality by knockoff operation, *Sci. Adv.* 11 (2025) eadu6464.
- [134] R.F. Barber, E.J. Candès, Controlling the false discovery rate via knockoffs, *Ann. Stat.* (2015) 2055–2085.
- [135] D. Harnack, E. Laminaki, M. Schunemann, K.R. Pawelzik, Topological causality in dynamical systems, *Phys. Rev. Lett.* 119 (2017) 098301.
- [136] N.F. Rulkov, M.M. Sushchik, L.S. Tsimring, H.D. Abarbanel, Generalized synchronization of chaos in directionally coupled chaotic systems, *Phys. Rev. E* 51 (1995) 980.
- [137] S.J. Schiff, P. So, T. Chang, R.E. Burke, T. Sauer, Detecting dynamical interdependence and generalized synchrony through mutual prediction in a neural ensemble, *Phys. Rev. E* 54 (1996) 6708.
- [138] J. Arnhold, P. Grassberger, K. Lehnertz, C.E. Elger, A robust method for detecting interdependences: application to intracranially recorded EEG, *Phys. D: Nonlinear Phenom.* 134 (1999) 419–430.
- [139] E. Saberski, T. Lorimer, D. Carpenter, E. Deyle, E. Merz, J. Park, G.M. Pao, G. Sugihara, The impact of data resolution on dynamic causal inference in multiscale ecological networks, *Commun. Biol.* 7 (2024) 1442.
- [140] H. Ma, S. Leng, C. Tao, X. Ying, J. Kurths, Y.-C. Lai, W. Lin, Detection of time delays and directional interactions based on time series from complex dynamical systems, *Phys. Rev. E* 96 (2017) 012221.
- [141] H. Ye, E.R. Deyle, L.J. Gilaranz, G. Sugihara, Distinguishing time-delayed causal interactions using convergent cross mapping, *Sci. Rep.* 5 (2015) 14750.
- [142] H. Ye, G. Sugihara, Information leverage in interconnected ecosystems: Overcoming the curse of dimensionality, *Science* 353 (2016) 922–925.
- [143] Y. Duan, Y. Guo, J. Yang, M. Yin, Causal discovery in symmetric dynamic systems with convergent cross mapping, *Chaos: Interdiscip. J. Nonlinear Sci.* 35 (2025).
- [144] K. Sawada, Y. Shimada, T. Ikeguchi, Detecting causality based on state space reconstruction from interspike intervals for neural spike trains, *Phys. Rev. E* 112 (2025) 014222.
- [145] S. Leng, H. Ma, J. Kurths, Y.-C. Lai, W. Lin, K. Aihara, L. Chen, Partial cross mapping eliminates indirect causal influences, *Nat. Commun.* 11 (2020) 2632.
- [146] A.T. Clark, H. Ye, F. Isbell, E.R. Deyle, J. Cowles, G.D. Tilman, G. Sugihara, Spatial convergent cross mapping to detect causal relationships from short time series, *Ecology* 96 (2015) 1174–1181.
- [147] B. Gao, J. Yang, Z. Chen, G. Sugihara, M. Li, A. Stein, M.-P. Kwan, J. Wang, Causal inference from cross-sectional earth system data with geographical convergent cross mapping, *Nat. Commun.* 14 (2023) 5875.
- [148] S. Avvaru, K.K. Parhi, Effective brain connectivity extraction by frequency-domain convergent cross-mapping (FDCCM) and its application in parkinson's disease classification, *IEEE Trans. Biomed. Eng.* 70 (2023) 2475–2485.
- [149] Z. Benko, B. Varga, M. Stippinger, Z. Somogyvari, Detecting causality in the frequency domain with cross-mapping coherence, *Phys. D: Nonlinear Phenom.* 477 (2025) 134708.
- [150] D. Gu, A. Lin, G. Lin, Detection of attention deficit hyperactivity disorder in children using CEEMDAN-based cross frequency symbolic convergent cross mapping, *Expert Syst. Appl.* 226 (2023) 120105.
- [151] P. Tao, Q. Wang, J. Shi, X. Hao, X. Liu, B. Min, Y. Zhang, C. Li, H. Cui, L. Chen, Detecting dynamical causality by intersection cardinal concavity, *Fundam. Res.* (2023).
- [152] X. Ying, S.-Y. Leng, H.-F. Ma, Q. Nie, Y.-C. Lai, W. Lin, Continuity scaling: A rigorous framework for detecting and quantifying causality accurately, *Research* (2022).
- [153] H. Poincaré, Sur le problème des trois corps et les équations de la dynamique, *Acta Math.* 13 (1890) A3–A270.
- [154] J.P. Eckmann, S.O. Kamphorst, D. Ruelle, Recurrence plots of dynamic-systems, *Europhys. Lett.* 4 (1987) 973–977.
- [155] J.S. Iwanski, E. Bradley, Recurrence plots of experimental data: To embed or not to embed? *Chaos* 8 (1998) 861–871.
- [156] J.M. Choi, B.H. Bae, S.Y. Kim, Divergence in perpendicular recurrence plot; quantification of dynamical divergence from short chaotic time series, *Phys. Lett. A* 263 (1999) 299–306.
- [157] S. Horai, T. Yamada, K. Aihara, Determinism analysis with iso-directional recurrence plots, *IEEJ Trans. Electron. Inf. Syst.* 122 (2002) 141–147.
- [158] T.D. Pham, Fuzzy recurrence plots, *Epl* 116 (2016).
- [159] C.L. Webber, J.P. Zbilut, Dynamical assessment of physiological systems and states using recurrence plot strategies, *J. Appl. Physiol.* 76 (1994) 965–973.
- [160] J.P. Zbilut, C.L. Webber, Embeddings and delays as derived from quantification of recurrence plots, *Phys. Lett. A* 171 (1992) 199–203.
- [161] N. Marwan, J. Kurths, Nonlinear analysis of bivariate data with cross recurrence plots, *Phys. Lett. A* 302 (2002) 299–307.
- [162] N. Marwan, M. Thiel, N.R. Nowaczyk, Cross recurrence plot based synchronization of time series, *Nonlinear Process. Geophys.* 9 (2002) 325–331.
- [163] J.H. Feldhoff, R.V. Donner, J.F. Donges, N. Marwan, J. Kurths, Geometric detection of coupling directions by means of inter-system recurrence networks, *Phys. Lett. A* 376 (2012) 3504–3513.
- [164] A. Groth, Visualization of coupling in time series by order recurrence plots, *Phys. Rev. E* 72 (2005).
- [165] M.C. Romano, M. Thiel, J. Kurths, W. von Bloh, Multivariate recurrence plots, *Phys. Lett. A* 330 (2004) 214–223.
- [166] E. Peluso, T. Craciunescu, A. Murari, A refinement of recurrence analysis to determine the time delay of causality in presence of external perturbations, *Entropy* 22 (2020).
- [167] E. Peluso, A. Murari, T. Craciunescu, E. Lerche, P. Gaudio, M. Gelfusa, D. Gallart, D. Taylor, Conditional recurrence plots for the investigation of sawteeth pacing with RF modulation, *Plasma Phys. Control. Fusion* 64 (2022).
- [168] X.D. Yang, M.H. Chen, Y.L. Ren, B.Y. Hong, A.J. He, J. Wang, Multivariate joint order recurrence networks for characterization of multi-lead ECG time series from healthy and pathological heartbeat dynamics, *Chaos* 33 (2023).
- [169] M.C. Romano, M. Thiel, J. Kurths, C. Grebogi, Estimation of the direction of the coupling by conditional probabilities of recurrence, *Phys. Rev. E-Stat., Nonlinear, Soft Matter Phys.* 76 (2007) 036211.
- [170] A.M.T. Ramos, A. Builes-Jaramillo, G. Poveda, B. Goswami, E.E.N. Macau, J. Kurths, N. Marwan, Recurrence measure of conditional dependence and applications, *Phys. Rev. E* 95 (2017).
- [171] M. Porfiri, M.R. Marin, Transfer entropy on symbolic recurrences, *Chaos* 29 (2019).
- [172] W. Gilpin, Recurrences reveal shared causal drivers of complex time series, *Phys. Rev. X* 15 (2025) 011005.
- [173] D. Chicharro, R.G. Andrzejak, Reliable detection of directional couplings using rank statistics, *Phys. Rev. E* 80 (2009) 026217.
- [174] S.K. Stavroglou, A.A. Pantelous, H.E. Stanley, K.M. Zuev, Unveiling causal interactions in complex systems, *Proc. Natl. Acad. Sci.* 117 (2020) 7599–7605.
- [175] H. Ma, K. Aihara, L. Chen, Detecting causality from nonlinear dynamics with short-term time series, *Sci. Rep.* 4 (2014).
- [176] J. Yan, S.-W. Zhang, C. Zhang, W. Huang, J. Shi, L. Chen, Dynamical causality under invisible confounders, 2024, arXiv:2408.05584.
- [177] X.-Y. Duan, X. Ying, S.-Y. Leng, J. Kurths, W. Lin, H.-F. Ma, Embedding theory of reservoir computing and reducing reservoir network using time delays, *Phys. Rev. Res.* 5 (2023) L022041.
- [178] Y. Huang, Z. Fu, C.L. Franzke, Detecting causality from time series in a machine learning framework, *Chaos: Interdiscip. J. Nonlinear Sci.* 30 (2020).
- [179] R. Cao, J. Zhao, C. Guan, H. Ma, J. Shi, S. Leng, Reservoir cross mapping as a nonlinear framework for detecting dynamical causality, *Cell Rep. Phys. Sci.* 6 (2025) 102683.

- [180] J. Zhao, Z. Gan, R. Huang, C. Guan, J. Shi, S. Leng, Detecting dynamical causality via intervened reservoir computing, *Commun. Phys.* 7 (2024) 232.
- [181] E. De Brouwer, A. Arany, J. Simm, Y. Moreau, Latent convergent cross mapping, in: *International Conference on Learning Representations*, 2020.
- [182] C.J. Oates, F. Dondelinger, N. Bayani, J. Korkola, J.W. Gray, S. Mukherjee, Causal network inference using biochemical kinetics, *Bioinformatics* 30 (2014) i468–474.
- [183] F. Dondelinger, D. Husmeier, S. Rogers, M. Filippone, ODE parameter inference using adaptive gradient matching with Gaussian processes, in: *Artificial Intelligence and Statistics*, PMLR, 2013, pp. 216–228.
- [184] O.S. Venturelli, A.C. Carr, G. Fisher, R.H. Hsu, R. Lau, B.P. Bowen, S. Hromada, T. Northen, A.P. Arkin, Deciphering microbial interactions in synthetic human gut microbiome communities, *Mol. Syst. Biol.* 14 (2018) e8157.
- [185] R.L. Clark, B.M. Connors, D.M. Stevenson, S.E. Hromada, J.J. Hamilton, D. Amador-Noguez, O.S. Venturelli, Design of synthetic human gut microbiome assembly and butyrate production, *Nat. Commun.* 12 (2021) 3254.
- [186] C.H. Remien, M.J. Eckwright, B.J. Ridenhour, Structural identifiability of the generalized Lotka–Volterra model for microbiome studies, *R. Soc. Open Sci.* 8 (2021) 201378.
- [187] J.O. Ramsay, G. Hooker, D. Campbell, J. Cao, Parameter estimation for differential equations: a generalized smoothing approach, *J. R. Stat. Soc. Ser. B Stat. Methodol.* 69 (2007) 741–796.
- [188] B. Calderhead, M. Girolami, N.D. Lawrence, Accelerating Bayesian inference over nonlinear differential equations with Gaussian processes, in: *Proceedings of the 22nd International Conference on Neural Information Processing Systems*, Curran Associates Inc. Vancouver, British Columbia, Canada, 2008, pp. 217–224.
- [189] L. Dony, F. He, M.P.H. Stumpf, Parametric and non-parametric gradient matching for network inference: a comparison, *BMC Bioinformatics* 20 (2019) 52.
- [190] F.J. Bruggeman, H.V. Westerhoff, J.B. Hoek, B.N. Kholodenko, Modular response analysis of cellular regulatory networks, *J. Theoret. Biol.* 218 (2002) 507–520.
- [191] D. Lill, O.S. Rukhlenko, A.J. Mc Elwee, E. Kashdan, J. Timmer, B.N. Kholodenko, Mapping connections in signaling networks with ambiguous modularity, *Npj Syst. Biol. Appl.* 5 (2019) 19.
- [192] S.L. Brunton, J.L. Proctor, J.N. Kutz, Discovering governing equations from data by sparse identification of nonlinear dynamical systems, *Proc. Natl. Acad. Sci.* 113 (2016) 3932–3937.
- [193] J.A. Nabuco Leva Ferreira de Freitas, O. Bischof, Dynamic modeling of the cellular senescence gene regulatory network, *Heliyon* 9 (2023) e14007.
- [194] M. Hoffmann, C. Fröhner, F. Noé, Reactive SINDy: Discovering governing reactions from concentration data, *J. Chem. Phys.* 150 (2019) 025101.
- [195] J. Huang, Y. Zhou, W.-A. Yong, Data-driven discovery of multiscale chemical reactions governed by the law of mass action, *J. Comput. Phys.* 448 (2022) 110743.
- [196] R. Jiang, P. Singh, F. Wrede, A. Hellander, L. Petzold, Identification of dynamic mass-action biochemical reaction networks using sparse Bayesian methods, *PLoS Comput. Biol.* 18 (2022) e1009830.
- [197] R. Delabays, G. De Pasquale, F. Dörfler, Y. Zhang, Hypergraph reconstruction from dynamics, *Nat. Commun.* 16 (2025) 2691.
- [198] J. Mann, J.N. Kutz, Dynamic mode decomposition for financial trading strategies, *Quant. Finance* 16 (2016) 1643–1655.
- [199] J.L. Proctor, S.L. Brunton, J.N. Kutz, Dynamic mode decomposition with control, *SIAM J. Appl. Dyn. Syst.* 15 (2016) 142–161.
- [200] R. Fukuma, K. Majima, Y. Kawahara, O. Yamashita, Y. Shiraishi, H. Kishima, T. Yanagisawa, Fast, accurate, and interpretable decoding of electrocorticographic signals using dynamic mode decomposition, *Commun. Biol.* 7 (2024) 595.
- [201] K. Fujii, N. Takeishi, M. Hojo, Y. Inaba, Y. Kawahara, Physically-interpretable classification of biological network dynamics for complex collective motions, *Sci. Rep.* 10 (2020) 3005.
- [202] S. Ikeda, K. Kawano, S. Watanabe, O. Yamashita, Y. Kawahara, Predicting behavior through dynamic modes in resting-state fMRI data, *NeuroImage* 247 (2022) 118801.
- [203] M.M. Kelp, S. Lin, J.N. Kutz, L.J. Mickley, A new approach for determining optimal placement of PM2.5 air quality sensors: case study for the contiguous United States, *Environ. Res. Lett.* 17 (2022) 034034.
- [204] P. Climaco, J. Garcke, R. Iza-Teran, Multi-resolution dynamic mode decomposition for damage detection in wind turbine gearboxes, *Data-Centric Eng.* 4 (2023) e1.
- [205] X. Wang, L. Sun, Anti-circulant dynamic mode decomposition with sparsity-promoting for highway traffic dynamics analysis, *Transp. Res. Part C: Emerg. Technol.* 153 (2023) 104178.
- [206] K. Champion, B. Lusch, J.N. Kutz, S.L. Brunton, Data-driven discovery of coordinates and governing equations, *Proc. Natl. Acad. Sci.* 116 (2019) 22445–22451.
- [207] T.-T. Gao, G. Yan, Autonomous inference of complex network dynamics from incomplete and noisy data, *Nat. Comput. Sci.* 2 (2022) 160–168.
- [208] T.-T. Gao, B. Barzel, G. Yan, Learning interpretable dynamics of stochastic complex systems from experimental data, *Nat. Commun.* 15 (2024) 6029.
- [209] P.Y. Lu, J. Ariño Bernad, M. Soljačić, Discovering sparse interpretable dynamics from partial observations, *Commun. Phys.* 5 (2022) 206.
- [210] J. Hu, J. Cui, B. Yang, Learning interpretable network dynamics via universal neural symbolic regression, *Nat. Commun.* 16 (2025) 6226.
- [211] Z. Yu, J. Ding, Y. Li, Discovering network dynamics with neural symbolic regression, *Nat. Comput. Sci.* (2025) <http://dx.doi.org/10.1038/s43588-025-00893-8>.
- [212] R.T. Chen, Y. Rubanova, J. Bettencourt, D.K. Duvenaud, Neural ordinary differential equations, *Adv. Neural Inf. Process. Syst.* 31 (2018).
- [213] A. Bellot, K. Branson, M. van der Schaar, Neural graphical modelling in continuous-time: consistency guarantees and algorithms, 2021, arXiv preprint [arXiv:2105.02522](https://arxiv.org/abs/2105.02522).
- [214] H. Aliee, F.J. Theis, N. Kilbertus, Beyond predictions in neural ODEs: identification and interventions, 2021, arXiv preprint [arXiv:2106.12430](https://arxiv.org/abs/2106.12430).
- [215] E. Dupont, A. Doucet, Y.W. Teh, Augmented neural odes, *Adv. Neural Inf. Process. Syst.* 32 (2019).
- [216] E. Zappala, A.H.d.O. Fonseca, A.H. Moberly, M.J. Higley, C. Abdallah, J.A. Cardin, D. van Dijk, Neural integro-differential equations, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023, pp. 11104–11112.
- [217] Q. Zhu, Y. Guo, W. Lin, Neural delay differential equations, 2021, arXiv preprint [arXiv:2102.10801](https://arxiv.org/abs/2102.10801).
- [218] J. Morrill, P. Kidger, L. Yang, T. Lyons, Neural controlled differential equations for online prediction tasks, 2021, arXiv preprint [arXiv:2106.11028](https://arxiv.org/abs/2106.11028).
- [219] S. Kim, W. Ji, S. Deng, Y. Ma, C. Rackauckas, Stiff neural ordinary differential equations, *Chaos: Interdiscip. J. Nonlinear Sci.* 31 (2021).
- [220] I.A. Auzina, Ç. Yildiz, S. Magliacane, M. Bethge, E. Gavves, Modulated neural odes, *Adv. Neural Inf. Process. Syst.* 36 (2023) 44572–44594.
- [221] X. Li, J. Zhang, Q. Zhu, C. Zhao, X. Zhang, X. Duan, W. Lin, From fourier to neural odes: Flow matching for modeling complex systems, 2024, arXiv preprint [arXiv:2405.11542](https://arxiv.org/abs/2405.11542).
- [222] X. Liu, T. Xiao, S. Si, Q. Cao, S. Kumar, C.-J. Hsieh, Neural sde: Stabilizing neural ode networks with stochastic noise, 2019, arXiv preprint [arXiv:1906.02355](https://arxiv.org/abs/1906.02355).
- [223] Y. Rubanova, R.T. Chen, D.K. Duvenaud, Latent ordinary differential equations for irregularly-sampled time series, *Adv. Neural Inf. Process. Syst.* 32 (2019).

- [224] X. Li, C. Zhao, X. Zhang, X. Duan, Symbolic neural ordinary differential equations, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025, pp. 18511–18519.
- [225] P. Hu, W. Yang, Y. Zhu, L. Hong, Revealing hidden dynamics from time-series data by odenet, *J. Comput. Phys.* 461 (2022) 111203.
- [226] C. Zang, F. Wang, Neural dynamics on complex networks, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2020, pp. 892–902.
- [227] B.J. Zou, M.E. Levine, D.P. Zaharieva, R. Johari, E.B. Fox, Hybrid $\mathcal{H}^2$ neural ODE causal modeling and an application to glycemic response, 2024, arXiv preprint arXiv:2402.17233.
- [228] Z. Lin, S. Chang, A. Zweig, M. Kang, E. Azizi, D.A. Knowles, Interpretable neural odes for gene regulatory network discovery under perturbations, 2025, arXiv preprint arXiv:2501.02409.
- [229] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *J. Comput. Phys.* 378 (2019) 686–707.
- [230] G.E. Karniadakis, I.G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, *Nat. Rev. Phys.* 3 (2021) 422–440.
- [231] K. Diederik, Adam: A method for stochastic optimization, 2014.
- [232] R.H. Byrd, P. Lu, J. Nocedal, C. Zhu, A limited memory algorithm for bound constrained optimization, *SIAM J. Sci. Comput.* 16 (1995) 1190–1208.
- [233] Q. Kang, B. Zhang, Y. Cao, X. Song, X. Ye, X. Li, H. Wu, Y. Chen, B. Chen, Causal prior-embedded physics-informed neural networks and a case study on metformin transport in porous media, *Water Res.* 261 (2024) 121985.
- [234] S. Wang, S. Sankaran, P. Perdikaris, Respecting causality for training physics-informed neural networks, *Comput. Methods Appl. Mech. Engrg.* 421 (2024) 116813.
- [235] T. Kapoor, H. Wang, A. Núñez, R. Dollevoet, Transfer learning for improved generalizability in causal physics-informed neural networks for beam simulations, *Eng. Appl. Artif. Intell.* 133 (2024) 108085.
- [236] J. Guo, H. Wang, S. Gu, C. Hou, TCAS-PINN: Physics-informed neural networks with a novel temporal causality-based adaptive sampling method, *Chin. Phys. B* 33 (2024) 050701.
- [237] Z. Chen, Y. Liu, H. Sun, Physics-informed learning of governing equations from scarce data, *Nat. Commun.* 12 (2021) 6136.
- [238] R. Katende, Causal operator discovery in partial differential equations via counterfactual physics-informed neural networks, 2025, arXiv preprint arXiv:2506.20181.
- [239] C.S. Shin, P.A. Dirmeyer, B.H. Huang, A joint approach combining correlation and mutual information to study land and ocean drivers of US droughts: Methodology, *J. Clim.* 36 (2023) 2795–2814.
- [240] A.A. Tsonis, K.L. Swanson, P.J. Roebber, What do networks have to do with climate? *Bull. Am. Meteorol. Soc.* 87 (2006) 585–596.
- [241] A.A. Tsonis, K. Swanson, S. Kravtsov, A new dynamical mechanism for major climate shifts, *Geophys. Res. Lett.* 34 (2007).
- [242] A.A. Tsonis, K.L. Swanson, G. Wang, On the role of atmospheric teleconnections in climate, *J. Clim.* 21 (2008) 2990–3001.
- [243] K. Yamasaki, A. Gozolchiani, S. Havlin, Climate networks around the globe are significantly affected by El Niño, *Phys. Rev. Lett.* 100 (2008) 228501.
- [244] J.F. Donges, Y. Zou, N. Marwan, J. Kurths, Complex networks in climate dynamics, *Eur. Phys. J. Spec. Top.* 174 (2009) 157–179.
- [245] T. Chu, D. Danks, C. Glymour, Data driven methods for nonlinear granger causality: Climate teleconnection mechanisms, 2005.
- [246] I. Ebert-Uphoff, Y. Deng, Causal discovery for climate research using graphical models, *J. Clim.* 25 (2012) 5648–5665.
- [247] L. Ricard, F. Falasca, J. Runge, A. Nenes, Network-based constraint to evaluate climate sensitivity, *Nat. Commun.* 15 (2024) 6942.
- [248] E. Saggioro, T.G. Shepherd, J. Knight, Probabilistic causal network modeling of southern hemisphere jet subseasonal to seasonal predictability, *J. Clim.* 37 (2024) 3055–3071.
- [249] N. Mankovich, S. Bouabid, P. Nowack, D. Bassotto, G. Camps-Valls, Analyzing climate scenarios using dynamic mode decomposition with control, *Environ. Data Sci.* 4 (2025) e16.
- [250] N. Marwan, M.H. Trauth, M. Vuille, J. Kurths, Comparing modern and pleistocene ENSO-like influences in NW Argentina using nonlinear time series analysis methods, *Clim. Dyn.* 21 (2003) 317–326.
- [251] S.-C. San, Y.-H. Tseng, R. Ding, E. Di Lorenzo, A key role of off-equatorial subsurface temperature anomalies in tropical Pacific decadal variability, *Npj Clim. Atmospher. Sci.* 7 (2024) 109.
- [252] J. Runge, A. Gerhardus, G. Varando, V. Eyring, G. Camps-Valls, Causal inference for time series, *Nat. Rev. Earth Environ.* 4 (2023) 487–505.
- [253] M. Reichstein, G. Camps-Valls, B. Stevens, M. Jung, J. Denzler, N. Carvalhais, Prabhat, Deep learning and process understanding for data-driven earth system science, *Nature* 566 (2019) 195–204.
- [254] J. Wang, X. Zhang, Y. Wang, J. Liu, H. Wang, J. Lin, C. Xu, S. Hua, A zero-shot load forecasting method for extreme weather integrating causal learning and meta-learning, *Energy* (2025) 137666.
- [255] E. Bullmore, O. Sporns, Complex brain networks: graph theoretical analysis of structural and functional systems, *Nature Rev. Neurosci.* 10 (2009) 186–198.
- [256] D.S. Bassett, O. Sporns, Network neuroscience, *Nature Neurosci.* 20 (2017) 353–364.
- [257] M.D. Fox, A.Z. Snyder, J.L. Vincent, M. Corbetta, D.C. Van Essen, M.E. Raichle, The human brain is intrinsically organized into dynamic, anticorrelated functional networks, *Proc. Natl. Acad. Sci. USA* 102 (2005) 9673–9678.
- [258] A. Brovelli, M. Ding, A. Ledberg, Y. Chen, R. Nakamura, S.L. Bressler, Beta oscillations in a large-scale sensorimotor cortical network: directional influences revealed by granger causality, *Proc. Natl. Acad. Sci. USA* 101 (2004) 9849–9854.
- [259] A. Hillebrand, P. Tewarie, E. van Dellen, M. Yu, E.W. Carbo, L. Douw, A.A. Gouw, E.C. van Straaten, C.J. Stam, Direction of information flow in large-scale resting-state networks is frequency-dependent, *Proc. Natl. Acad. Sci. USA* 113 (2016) 3867–3872.
- [260] A.K. Seth, A.B. Barrett, L. Barnett, Granger causality analysis in neuroscience and neuroimaging, *J. Neurosci.* 35 (2015) 3293–3297.
- [261] T.F. Varley, M. Pope, J. Faskowitz, O. Sporns, Multivariate information theory uncovers synergistic subsystems of the human cerebral cortex, *Commun. Biol.* 6 (2023) 451.
- [262] A.I. Luppi, P.A.M. Mediano, F.E. Rosas, J. Allanson, J. Pickard, R.L. Carhart-Harris, G.B. Williams, M.M. Craig, P. Finoia, A.M. Owen, L. Naci, D.K. Menon, D. Bor, E.A. Stamatakis, A synergistic workspace for human consciousness revealed by integrated information decomposition, *Elife* 12 (2024).
- [263] S. Ebrahimi, J. Lecoq, O. Romyantsev, T. Tasci, Y. Zhang, C. Irimia, J. Li, S. Ganguli, M.J. Schnitzer, Emergent reliability in sensory cortical coding and inter-area communication, *Nature* 605 (2022) 713–721.
- [264] C. Seguin, O. Sporns, A. Zalesky, Brain network communication: concepts, models and applications, *Nat. Rev. Neurosci.* 24 (2023) 557–574.
- [265] J. Cabral, F.F. Fernandes, N. Shemesh, Intrinsic macroscale oscillatory modes driving long range functional connectivity in female rat brains detected by ultrafast fMRI, *Nat. Commun.* 14 (2023) 375.
- [266] A. Santoro, F. Battiston, M. Lucas, G. Petri, E. Amico, Higher-order connectomics of human brain function reveals local topological signatures of task decoding, individual identification, and behavior, *Nat. Commun.* 15 (2024) 10244.
- [267] C. Andris, C. Koylu, M.A. Porter, Human-network regions as effective geographic units for disease mitigation, *EPJ Data Sci.* 12 (2023) 60.
- [268] H.W. Hethcote, The mathematics of infectious diseases, *SIAM Rev.* 42 (2000) 599–653.
- [269] R.M. Anderson, R.M. May, *Infectious Diseases of Humans: Dynamics and Control*, Oxford University Press, 1991.

- [270] M. Knight, K. Leeming, G. Nason, M. Nunes, Generalized network autoregressive processes and the GNAR package, *J. Stat. Softw.* 96 (2020) 1–36.
- [271] S. Armbruster, G. Reinert, Network-based time series modeling for COVID-19 incidence in the Republic of Ireland, *Appl. Netw. Sci.* 9 (2024) 23.
- [272] J.P. Gleeson, T. Brendan Murphy, J.D. O'Brien, N. Friel, N. Bargary, D.J.P. O'Sullivan, Calibrating COVID-19 susceptible-exposed-infected-removed models with time-varying effective contact rates, *Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 380 (2022) 20210120.
- [273] A. Papan, C. Kyrtou, D. Kugiumtzis, C. Diks, Detecting causality in non-stationary time series using partial symbolic transfer entropy: Evidence in financial data, *Comput. Econ.* 47 (2016) 341–365.
- [274] O. Akgüller, M.A. Balci, L.M. Batrancea, L. Gaban, Fractional transfer entropy networks: Short- and long-memory perspectives on global stock market interactions, *Fractal Fract.* 9 (2025).
- [275] Z.H. Duan, H.Y. Xu, Y.D. Huang, J. Feng, Y.Y. Wang, Multivariate time series forecasting with transfer entropy graph, *Tsinghua Sci. Technol.* 28 (2023) 141–149.
- [276] S. Lee, P. Cho, Graph-based stock volatility forecasting with effective transfer entropy and hurst-based regime adaptation, *Fractal Fract.* 9 (2025) 339.
- [277] D. Luo, S. Li, W. Liao, R. Yan, MagicNet: Memory-aware graph interactive causal network for multivariate stock price movement prediction, *IEEE Trans. Knowl. Data Eng.* 37 (2025) 1989–2000.
- [278] S. Ashe, P. Egan, Examining financial and business cycle interaction using cross recurrence plot analysis, *Financ. Res. Lett.* 51 (2023).
- [279] D.Y. Kenett, M. Tumminello, A. Madi, G. Gur-Gershoren, R.N. Mantegna, E. Ben-Jacob, Dominating clasp of the financial sector revealed by partial correlation analysis of the stock market, *PLoS One* 5 (2010) e15032.
- [280] M. Billio, M. Getmansky, A.W. Lo, L. Pelizzon, Econometric measures of connectedness and systemic risk in the finance and insurance sectors, *J. Financ. Econ.* 104 (2012) 535–559.
- [281] F.X. Diebold, K. Yilmaz, Measuring financial asset return and volatility spillovers, with application to global equity markets, *Econ. J.* 119 (2008) 158–171.
- [282] A. Babii, E. Ghysels, J. Striaukas, High-dimensional granger causality tests with an application to VIX and news*, *J. Financ. Econ.* 22 (2022) 605–635.
- [283] P. Zhang, S. Yin, Y. Sha, Global systemic risk dynamic network connectedness during the COVID-19: Evidence from nonlinear granger causality, *J. Int. Financ. Mark. Institutions Money* 85 (2023) 101783.
- [284] F. Zhou, Cryptocurrency: A new player or a new crisis in financial markets? — evolutionary analysis of association and risk spillover based on network science, *Phys. A* 648 (2024) 129955.
- [285] S. Bekiros, D.K. Nguyen, L. Sandoval, G.S. Uddin, Information diffusion, cluster formation and entropy-based network dynamics in equity and commodity markets, *European J. Oper. Res.* 256 (2017) 945–961.
- [286] J. Borge-Holthoefer, N. Perra, B. Gonçalves, S. González-Bailón, A. Arenas, Y. Moreno, A. Vespignani, The dynamics of information-driven coordination phenomena: A transfer entropy analysis, *Sci. Adv.* 2 (2016).
- [287] S. Vosoughi, D. Roy, S. Aral, The spread of true and false news online, *Science* 359 (2018) 1146–1151.
- [288] D. Centola, J. Becker, D. Brackbill, A. Baronchelli, Experimental evidence for tipping points in social convention, *Science* 360 (2018) 1116–1119.
- [289] A.M. Guess, N. Malhotra, J. Pan, P. Barberá, H. Allcott, T. Brown, A. Crespo-Tenorio, D. Dimmery, D. Freelon, M. Gentzkow, How do social media feed algorithms affect attitudes and behavior in an election campaign? *Science* 381 (2023) 398–404.
- [290] M. Falkenberg, A. Galeazzi, M. Torricelli, N. Di Marco, F. Larosa, M. Sas, A. Mekacher, W. Pearce, F. Zollo, W. Quattrociocchi, Growing polarization around climate change on social media, *Nat. Clim. Chang.* 12 (2022) 1114–1121.
- [291] M. Falkenberg, F. Zollo, W. Quattrociocchi, J. Pfeffer, A. Baronchelli, Patterns of partisan toxicity and engagement reveal the common structure of online political communication across countries, *Nat. Commun.* 15 (2024) 9560.
- [292] A. Combs, G. Tierney, B. Guay, F. Merhout, C.A. Bail, D.S. Hillygus, A. Volfovsky, Reducing political polarization in the United States with a mobile chat platform, *Nat. Hum. Behav.* 7 (2023) 1454–1461.
- [293] J.G. Voelkel, J. Chu, M.N. Stagnaro, J.S. Mernyk, C. Redekopp, S.L. Pink, J.N. Druckman, D.G. Rand, R. Willer, Interventions reducing affective polarization do not necessarily improve anti-democratic attitudes, *Nat. Hum. Behav.* 7 (2023) 55–64.
- [294] S. Altay, E. Hoes, M. Wojcieszak, Following news on social media boosts knowledge, belief accuracy and trust, *Nat. Hum. Behav.* (2025) 1–10.
- [295] U.K. Ecker, S. Lewandowsky, J. Cook, P. Schmid, L.K. Fazio, N. Brashier, P. Kendeou, E.K. Vraga, M.A. Amazeen, The psychological drivers of misinformation belief and its resistance to correction, *Nat. Rev. Psychol.* 1 (2022) 13–29.
- [296] K. Lagemann, C. Lagemann, B. Taschler, S. Mukherjee, Deep learning of causal structures in high dimensions under data limitations, *Nat. Mach. Intell.* 5 (2023) 1306–1316.
- [297] J. Nathaniel, C. Roesch, J. Buch, D. DeSantis, A. Rupe, K.D. Lamb, P. Gentine, Deep koopman operators for causal discovery, *Commun. Phys.* (2025).
- [298] T.H. Costello, G. Pennycook, D.G. Rand, Durably reducing conspiracy beliefs through dialogues with AI, *Science* 385 (2024) eadq1814.
- [299] J. Ping, E. Cui, Y. Du, N. Wei, J. Zhou, J. Wang, S. Niu, Y. Luo, J. Xia, Enhanced causal effect of ecosystem photosynthesis on respiration during heatwaves, *Sci. Adv.* 9 (2023) eadi6395.
- [300] M. Ushio, C. h. Hsieh, R. Masuda, E.R. Deyle, H. Ye, C.-W. Chang, G. Sugihara, M. Kondoh, Fluctuating interaction network and time-varying stability of a natural fish community, *Nature* 554 (2018) 360–363.
- [301] E.R. Deyle, D. Bouffard, V. Frossard, R. Schwefel, J. Melack, G. Sugihara, A hybrid empirical and parametric approach for managing ecosystem complexity: Water quality in lake geneva under nonstationary futures, *Proc. Natl. Acad. Sci.* 119 (2022) e2102466119.
- [302] F. Schrodtt, M. Beck, J. Estopinan, D.E. Bowler, C. Fontaine, P. Gaüzère, R. Goury, M. Grenié, I.S. Martins, N. Morueta-Holme, Advancing causal inference in ecology: Pathways for biodiversity change detection and attribution, *Methods Ecol. Evol.* 16 (2025) 2276–2304.
- [303] X. Xu, A. Huang, E. Belle, P. De Frenne, G. Jia, Protected areas provide thermal buffer against climate change, *Sci. Adv.* 8 (2022) eabo0119.
- [304] J.B. Grace, N. Huntington-Klein, E.W. Schweiger, M. Martinez, M.J. Osland, L.C. Feher, G.R. Guntenspergen, K.M. Thorne, Causal effects versus causal mechanisms: Two traditions with different requirements and contributions towards causal understanding, *Ecol. Lett.* 28 (2025) e70029.
- [305] L.E. Dee, P.J. Ferraro, C.N. Severen, K.A. Kimmel, E.T. Borer, J.E. Byrnes, A.T. Clark, Y. Hautier, A. Hector, X. Raynaud, Clarifying the effect of biodiversity on productivity in natural ecosystems with longitudinal data and methods for causal inference, *Nat. Commun.* 14 (2023) 2607.
- [306] S. Arif, Estimating causal effects with machine learning: A guide for ecologists, *Methods Ecol. Evol.* (2025).
- [307] J.T. Delaney, D.M. Larson, Using explainable machine learning methods to evaluate vulnerability and restoration potential of ecosystem state transitions, *Conserv. Biol.* 38 (2024) e14203.
- [308] K. Rafiq, S. Beery, M.S. Palmer, Z. Harchaoui, B. Abrahms, Generative AI as a tool to accelerate the field of ecology, *Nat. Ecol. Evol.* 9 (2025) 378–385.
- [309] D. Tuia, B. Kellenberger, S. Beery, B.R. Costelloe, S. Zuffi, B. Risse, A. Mathis, M.W. Mathis, F. Van Langevelde, T. Burghardt, Perspectives in machine learning for wildlife conservation, *Nat. Commun.* 13 (2022) 792.
- [310] M.S. Norouzzadeh, A. Nguyen, M. Kosmala, A. Swanson, M.S. Palmer, C. Packer, J. Clune, Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning, *Proc. Natl. Acad. Sci.* 115 (2018) E5716–E5725.

- [311] P. Badia-i Mompel, L. Wessels, S. Müller-Dott, R. Trimbou, R.O. Ramirez Flores, R. Argelaguet, J. Saez-Rodriguez, Gene regulatory network inference in the era of single-cell multi-omics, *Nature Rev. Genet.* 24 (2023) 739–754.
- [312] X. Huang, L. Chen, X. Liu, Constructing cell-specific causal networks of individual cells for depicting dynamical biological processes, *Research* 8 (2025) 0743.
- [313] L.Y. Chang, T.Y. Hao, W.J. Wang, C.Y. Lin, Inference of single-cell network using mutual information for scRNA-seq data analysis, *Bmc Bioinform.* 25 (2024).
- [314] A. Deshpande, L.F. Chu, R. Stewart, A. Gitter, Network inference with granger causality ensembles on single-cell transcriptomics, *Cell Rep.* 38 (2022) 110333.
- [315] J. Kim, S.T. Jakobsen, K.N. Natarajan, K.J. Won, TENET: gene network reconstruction using transfer entropy reveals key regulatory factors from single cell transcriptomic data, *Nucleic Acids Res.* 49 (2021).
- [316] Y. Gao, K. Dong, C. Shan, D. Li, Q. Liu, Causal disentanglement for single-cell representations and controllable counterfactual generation, *Nat. Commun.* 16 (2025) 6775.
- [317] H. Shu, J. Zhou, Q. Lian, H. Li, D. Zhao, J. Zeng, J. Ma, Modeling gene regulatory networks using neural network architectures, *Nat. Comput. Sci.* 1 (2021) 491–501.
- [318] C. Zhang, Z. Chen, Y. Miao, Y. Xue, D. Cai, W. Guo, H. Ji, K. Aihara, L. Chen, Steering cell-state and phenotype transitions by causal disentanglement learning, 2024, *bioRxiv*, 2024.2008. 2016.607277.
- [319] K. Kamimoto, B. Stringa, C.M. Hoffmann, K. Jindal, L. Solnica-Krezel, S.A. Morris, Dissecting cell identity via network inference and in silico gene perturbation, *Nature* 614 (2023) 742–751.
- [320] T.M. Nolan, N. Vukašinović, C.-W. Hsu, J. Zhang, I. Vanhoutte, R. Shahan, I.W. Taylor, L. Greenstreet, M. Heitz, A. Afanassiev, P. Wang, P. Szekely, A. Brosnan, Y. Yin, G. Schiebinger, U. Ohler, E. Russinova, P.N. Benfey, Brassinosteroid gene regulatory networks at cellular resolution in the *Arabidopsis* root, *Science* 379 (2023) eadf4721.
- [321] P. Dibaeinia, A. Ojha, S. Sinha, Interpretable AI for inference of causal molecular relationships from omics data, *Sci. Adv.* 11 (2025) eadk0837.
- [322] X. Meng, G. Navoly, O. Giannakopoulou, D.F. Levey, D. Koller, G.A. Pathak, N. Koen, K. Lin, M.J. Adams, M.E. Rentería, Y. Feng, J.M. Gaziano, D.J. Stein, H.J. Zar, M.L. Campbell, D.A. van Heel, B. Trivedi, S. Finer, A. McQuillin, N. Bass, V.K. Chundru, H.C. Martin, Q.Q. Huang, M. Valkovskaya, C.-Y. Chu, S. Kanjira, P.-H. Kuo, H.-C. Chen, S.-J. Tsai, Y.-L. Liu, K.S. Kendler, R.E. Peterson, N. Cai, Y. Fang, S. Sen, L.J. Scott, M. Burmeister, R.J.F. Loos, M.H. Preuss, K.E.V. Actkins, L.K. Davis, M. Uddin, A.H. Wani, D.E. Wildman, A.E. Aiello, R.J. Ursano, R.C. Kessler, M. Kanai, Y. Okada, S. Sakaue, J.A. Rabinowitz, B.S. Maher, G. Uhl, W. Eaton, C.S. Cruz-Fuentes, G.A. Martinez-Levy, A.I. Campos, I.Y. Millwood, Z. Chen, L. Li, S. Wassertheil-Smoller, Y. Jiang, C. Tian, N.G. Martin, B.L. Mitchell, E.M. Byrne, S. Awasthi, J.R.I. Coleman, S. Ripke, T. Sofer, R.G. Walters, A.M. McIntosh, R. Polimanti, E.C. Dunn, M.B. Stein, J. Gelernter, C.M. Lewis, K. Kuchenbaecker, P.-M.W. Group, G. China Kadoorie Biobank Collaborative, T. the 23andMe Research, T. Genes, P. Health Research, BioBank Japan, Multi-ancestry genome-wide association study of major depression aids locus discovery, fine mapping, gene prioritization and causal inference, *Nature Genet.* 56 (2024) 222–233.
- [323] P. Spirtes, C.N. Glymour, R. Scheines, *Causation, Prediction, and Search*, MIT Press, 2000.
- [324] S. Job, X. Tao, T. Cai, H. Xie, L. Li, Q. Li, J. Yong, Exploring causal learning through graph neural networks: An in-depth review, *Wiley Interdiscip. Rev.: Data Min. Knowl. Discov.* 15 (2025) e70024.
- [325] L. Wang, A. Adiga, J. Chen, A. Sadilek, S. Venkatramanan, M. Marathe, Causalgnn: Causal-based graph neural networks for spatio-temporal epidemic forecasting, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022, pp. 12191–12199.
- [326] F. Wang, K. Li, X. Chen, W. Zhang, Assessing supply chain risks for chip industry with LDA and multi-layer Bayesian network method, *Front. Eng. Manag.* 12 (2025) 1037–1057.
- [327] S.L. Lauritzen, *Graphical Models*, Clarendon Press, 1996.
- [328] M. Eichler, Graphical modelling of multivariate time series, *Probab. Theory Related Fields* 153 (2012) 233–268.
- [329] F. Battiston, E. Amico, A. Barrat, G. Bianconi, G. Ferraz de Arruda, B. Franceschiello, I. Iacopini, S. Kéfi, V. Latora, Y. Moreno, The physics of higher-order interactions in complex systems, *Nat. Phys.* 17 (2021) 1093–1098.
- [330] X. Zheng, B. Aragam, P.K. Ravikumar, E.P. Xing, Dags with no tears: Continuous optimization for structure learning, *Adv. Neural Inf. Process. Syst.* 31 (2018).
- [331] Y. Luo, J. Peng, J. Ma, When causal inference meets deep learning, *Nat. Mach. Intell.* 2 (2020) 426–427.
- [332] L. Wu, P. Cui, J. Pei, L. Zhao, X. Guo, Graph neural networks: foundation, frontiers and applications, in: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 4840–4841.
- [333] H. Lin, M. Ren, P. Barucca, T. Aste, Granger causality detection with kolmogorov-arnold networks, 2024, *arXiv preprint arXiv:2412.15373*.
- [334] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T.Y. Hou, M. Tegmark, Kan: Kolmogorov-arnold networks, 2024, *arXiv preprint arXiv:2404.19756*.
- [335] H. Zhu, H. Huang, K. Yin, Z. Fan, H. Jin, B. Liu, Causalnet: Unveiling causal structures on event sequences by topology-informed causal attention, in: *Proceedings of the IJCAI*, 2024, pp. 7144–7152.
- [336] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, *Nat. Mach. Intell.* 1 (2019) 206–215.
- [337] E. Begoli, T. Bhattacharya, D. Kusnezov, The need for uncertainty quantification in machine-assisted medical decision making, *Nat. Mach. Intell.* 1 (2019) 20–23.